1. GENERAL

1.1. ATIS

ATIS Arrival	131.850
	125.875 (Russian)
ATIS Departure	124.450
	127.8 (Russian)

1.2. LOW VISIBILITY PROCEDURES (LVP)

1.2.1. GENERAL

LVP shall be implemented when RVR is below 550m at least at one of three observation sites. Flight crews will be informed by ATIS or ATS unit when LVP are in progress: "Low visibility procedures in progress, check your minimum".

Taxiing when visibility is less than 2000m shall be carried out with air navigation lights and landing lights switched on.

When visibility is less than 350m ACFT taxiing shall be carried out along TWY only with TWY centerline lights switched on or after Follow-me car.

1.2.2. ARRIVAL

Flight crew shall report RWY vacation to VNUKOVO Tower only after complete crossing of RWY holding position marking painted on TWY or passing last yellow light of exit TWY centerline indicating boundary of ILS critical area.

ACFT must vacate ILS critical area as soon as possible.

Arriving ACFT landing on RWY 06/24 and RWY 19 shall taxi via established routes.

Flight crew shall report the presence of Follow-me car in front of the ACFT as following: "VNUKOVO Ground, TTF 9075, on Mike - 1, Follow-me vehicle in sight".

Thereafter ACFT shall taxi under control of VNUKOVO Ground.

Flight crew shall report to VNUKOVO Ground about ACFT arrival at the stand as following: "TTF 9075, on stand 12".

1.2.3. DEPARTURE

When visibility is less than 350m, the flight crew of the departing ACFT shall carry out taxiing only along the TWYs equipped with green TWY centerline lights. In case of failure of TWY centerline lights or stop bar lights, when visibility is less than 350m, the flight crew must carry out taxiing after the Follow-me car only.

When visibility is less than 350m, ACFT taxiing along the apron area shall be carried out after the Follow-me car only under control of VNUKOVO Ground.

During taxiing alon g the apron area and maneuvering area the flight crew must constantly check the established marking, especially at TWY intersections, to ensure safety of ta xi operation. In case of difficulty or doubt, flight crew must stop taxiing and report to VNUKOVO Ground or Tower.

When visibility is below 350m, ACFT intended to take off from RWY 06 shall taxi under assistance of the Follow-me car to the first illuminated on TWY centerline light on TWY A4, TWY M1 and MAIN TWY M2.

Flight crew shall continue taxiing along green lights by the instruction of VNUKOVO Ground controller to stop bars on TWY A10, TWY A11, TWY A13, stop and hold. The transfer of control from VNUKOVO Ground to Tower shall be carried out at stop bars.

When visibility is below 350m, ACFT intended to take off from RWY 24 shall taxi under assistance of the Follow-me car to the first illuminated centerline light on TWYs A1 thru A5 and MAIN TWY M2. Flight crew shall continue taxiing along green lights following the instruction of VNUKOVO Ground controller to stop bars on TWYs A1 thru A5, stop and hold. The transfer of control from VNUKOVO Ground to Tower shall be carried out at stop bars.

1. GENERAL

When visibility is 350-550m ACFT shall taxi for take-off from RWY 01/19 along established routes. The transfer of control from VNUKOVO Ground to Tower shall be carried out before the marking of RWY holding position.

It is prohibited to cross the RWY holding position line (ILS critical area) marked by stop bars and by DAY without the permission of VNUKOVO Tower.

Holding short of the RWY, t he flight crew should read back all instructions of VNUKOVO Tower and Ground.

Having obtained line-up clearance, flight crew must start taxiing only after stop bars are switched off. It is prohibited to cross the illuminated stop bars.

1.3. TAXI PROCEDURES

Taxiing along TWYs B1, B3, B6, B8 and M4 (from TWY B1 to TWY B4) at reduced speed, strictly along taxi guideline with increased CAUTION.

Taxiing on segments where TWY centerline lights and marking sign of position and movement direction are absent or on TWY C2 thru C5, TWY C9 or TWYs A13 and M2 (from TWYs A13 to C6): At NIGHT or when VIS is 2000m or below taxiing on TWY segments where TWY centerline lights and location and direction signs are not provided shall be carried out under assistance of Follow-me car.

Taxi route from TWY C8 along hangars 9 thru 14 and between stand 54 and hangar 15 MAX wingspan 118'/36m.

TWY A12 is closed for taxiing and towing of all ACFT types.

At RWY 06 extremity only RIGHT turn shall be carried out at reduced speed with flight crew's increased caution.

1.4. PARKING INFORMATION

Enter Vnukolo 1 apron stands 32, 34A, 38, 38C, 39, 40, 41, 51A, 54, 56, 56A, 57, 58, 59, 60, 61, 61B (temporary stand), 62B (temporary stand), 64, 65, 66, 66A and 66B by towing.

Exit Vnukovo 1 apron stands 1 thru 31, 23A, 24A, 26A, 32, 32A, 32B, 32C, 33, 34, 34A, 35, 38A, 38B, 38C, 38D, 38E, 39, 39A, 39B, 40, 41, 41A, 52, 53, 54, 55, 56, 56A, 57A, 58A, 59A, 60A, 61A, 65, 66, 66A, 66B, 67, 67A, 68, 68A, 69, 69A, 70, 70A, 71 and 71A by towing.

Enter Vnukovo 3 stands 1 thru 5, 8 thru 10 by towing.

Use Vnukovo 3 stands 73 thru 78 by towing.

Exit Vnukovo 3 stands 29 thru 31, 33, 34, 35 thru 72A and 90 thru 98 by towing.

Use Vnukovo 5 stands 501 thru 510 by towing.

Stands 62 thru 66 are available for loading and unloading of dangerous goods.

1.5. COMMUNICATION FAILURE PROCEDURES

In case of radio communication failure follow established procedures.

In all cases, crew can

- use mobile communication

  Flight Control Officer (Moscow TMA Control Center)
  Tel: +7 495 956 87 33
  +7 495 436 25 36
  +7 916 043 35 90
  Flight Control Officer (Moscow ACC)
  Tel: +7 495 956 87 34
  +7 495 436 26 62
  +7 916 043 36 16

- monitor LOM frequency for ATC instructions.

1.6. OTHER INFORMATION

Birds in vicinity of APT.

2. ARRIVAL

2.1. COMMUNICATION FAILURE PROCEDURES

2.1.1. COMMUNICATION FAILURE DURING ARRIVAL

Set transponder to 7600 and continue the flight maintaining the route and flight profile of the cleared (shor-test basic) RNAV STAR.

Execute approach according to the established procedure.

In case of missed approach carry out published missed approach procedure to the nearest holding, and follow paragraph 2.1.2.

Set transponder to 7700 if is necessary to deviate from the mentioned procedure.

2.1.2. COMMUNICATION FAILURE AFTER MISSED APPROACH

Continue maintaining the flight route and profile of missed approach procedure to the nearest holding area.

Enter the holding area at the upper published altitude at IAF, burn out fuel, if necessary.

After taking the decision to execute landing at the destination:

- Execute approach in accordance with the established procedure.

After taking the decision to proceed to the alternate AD in Moscow TMA:

- Proceed to WNK DVORDME climbing to transition altitude 10000′.
- Proceed to IAF of the alternate aerodrome in Moscow TMA via the following waypoints:

  Moscow/Sheremetyevo:

  LIKNI - GIGUN - NUZOR - PUFIK - ROLAZ - RIZNO - BEGEZ - LUNZA - EE048 - EE049 - EE050 - EE051 - TAFAZ - KEZVU (IAF).

  Moscow/Domodedovo:

  BITSA - IMZUP - KUPVE - NIDBE - IZVOK - IPKED - ZOVGO - ODZAG - GUFUZ - ALBOR (IAF).

  Ostafyevo:

  BUPOS - ORSIF - MEZER - NALFI - RAMZA - UKABE - FIDOT - RORUK (IAF).

  Ramenskoye:

  BITSA - IMZUP - GENKE - RT NDB - BW316 - W317 - BW318 - BW319 - ODLOR (IAF).
- At IAF enter the published, if available, or standard holding area.
- In the holding area descend from transition altitude 10000′ to the upper published approach procedure altitude at IAF.
- Execute approach according to the established procedure.

After taking the decision to proceed to the alternate AD outside Moscow TMA:

- Execute approach according to the established procedure to IF.
- Proceed from IF to the initiation point of the basic RNAV SID of the same RWY.
- Maintain flight route and profile of the basic RNAV SID to the maximum extent until leaving Moscow TMA.
- After leaving Moscow TMA reach the flight level specially established for flight without radio communication (FL 140, FL 150, FL 240, FL 250).

After taking the decision to proceed to the destination AD:

- Execute approach in accordance with the established procedure to IF.
- Proceed from IF to the initiation point of the basic RNAV SID of the same RWY.
- Maintain flight route and profile of the basic RNAV SID to the maximum extent until leaving Moscow TMA.
- After leaving Moscow TMA, reach flight level indicated in the flight plan.

2. ARRIVAL

2.2. CAT II OPERATIONS

RWYs 06, 24 and 19 approved for CAT II operations, special aircrew and ACFT certification required.

2.3. RWY OPERATION

2.3.1. MINIMUM RWY OCCUPANCY TIME

The pilot must vacate the RWY as fast as possible.

The time of RWY vacation after landing when friction coefficient is 0.45 or more:

On heading 012°:
- via TWY B5 - not more than 70 seconds.

On heading 057°:
- via TWY A5 - not more than 70 seconds.
- via TWY A7 - not more than 60 seconds.

On heading 192°:
- via TWY M1 - not more than 70 seconds.

On heading 237°:
- via RWY 01/19, TWY A10, A11, A12 - not more than 100 seconds.

Unless otherwise instructed by ATC, RWY vacation should be planned by flight crew considering LDA to the respective TWY indicated in the table.

RWY	TWY for RWY vacation	LDA
06	A9	4396' (1340m)
	A7	6378' (1944m)
	A5	8209' (2502m)
	A3	9708' (2959m)
24	A6	4154' (1266m)
	A10	8327' (2538m)
	A11	9843' (3000m)
01	B5	5879' (1792m)
	B4	7713' (2351m)
	B2	8409' (2563m)
19	M1	6850' (2088m)
	MAIN M2	7119' (2170m)
	B8	9429' (2874m)

2.3.2. AFTER LANDING

After RWY vacation the flight crew shall squawk ALT OFF or XPDR and keep this mode till ACFT parking onto a stand.

2.3.3. REDUCED RWY SEPARATION MINIMA

Reduced RWY separation minima are applied during flight operations to/from RWY 06/24 only in the day-time, within the time period that starts 30 minutes after sunrise and ends 30 minutes before sunset (LT), and are subject to the following meteorological conditions:

- visibility is at least 5km;
- the height of cloud base (ceiling) is 300m or above;
- tailwind component does not exceed 3m/s.

Normative friction coefficient on the RWY shall be 0.40 or above (estimated surface friction "good" or "medium to good").

Reduced RWY separation minima is applied based on the decision of the AD Flight Control Officer.

Information on application at the aerodrome must be included in ATIS broadcast.

2. ARRIVAL

ATS unit can issue landing clearance to flight crew of an arriving ACFT, provided there is reasonable assurance, that when this ACFT crosses the RWY THR:

- the preceding arriving ACFT has landed and has passed the point located at distance at least 2500m from the RWY THR, is in motion and will vacate the RWY without backtracking;
- the preceding departing ACFT is airborne and has passed the point located at distance at least 2500m from the RWY THR.

Landing clearance issued by ATS unit to flight crew of an arriving ACFT must contain information on type of the preceding departing ACFT or arriving ACFT that has already landed.

Flight crew must report sighting the preceding departing ACFT or arriving ACFT that has already landed, and acknowledge receipt of landing clearance, if requested by the ATS unit.

Pilot-in-command makes the final decision to execute landing. The obtained landing clearance does not entail an obligation to execute landing.

2.3.4. IFR PROCEDURES WITHIN MOSCOW (VNUKOVO) CTR

Clearance to execute visual approach for ACFT performing IFR flight shall be requested by the flight crew or shall be initiated by ATS unit. In the latter case flight crew's consent is required.

- the flight crew has reported of establishing a visual contact with the RWY and/or its references;
- the height of cloud base is 600m or above or the flight crew has reported that weather conditions allow to execute visual approach.

The flight crew of succeeding ACFT shall provide own acceptable separation interval between the preceding ACFT. If deemed necessary to increase separation interval, the flight crew shall inform ATS unit accordingly or take a decision to go around.

2.4. TAXIING PROCEDURES

Towing to stand 62 from the direction of stand 63 is permitted with operating radius of turn of not more than 85'/26m.

2.4.1. STANDARD TAXI ROUTES

FOR RWY 01:

Route 7: TWY M4, TR W.
Route 8: TWY M4, TWY B5, TR Z.
Route 9: TWY M4, TWY B5, TR Z, TR X, TR W.

FOR RWY 06:

Route 1: TWY M3, TWY A5, TR Z, TWY B5, TWY M4, TR W.
Route 2: TWY M3, TWY A5, TR Z (TR X, TR W).
Route 3: TWY M3, TWY A2, TR W.

FOR RWY 19:

Route 10: TWY M1, TR X, TR Z, TWY B5, TWY M4, TR W.
Route 11: TWY M1, TR X (TR Z, TR W).
Route 12: TWY M1, TWY M3, TWY A2, TR W.

FOR RWY 24:

Route 4: TWY M2, crossing of RWY 01/19 by ATC permission, TWY M1, TR X, TR Z, TWY B5, TWY M4, TR W.
Route 5: TWY M2, crossing of RWY 01/19 by ATC permission, TWY M1, TR X (TR Z, TR W).

2. ARRIVAL

2.4.2. CODED TAXI ROUTES (MONITOR PROCEDURES)

In all cases the flight crew shall inform Delivery controller about the estimated displaced position of take-off run on the RWY while obtaining ATC clearance. In case flight crew doesn't inform Delivery controller, GND controller gives instruction to take off from the RWY beginning taking into account limitations caused by airfield elements.

In case GND controller assigns a coded taxi route, monitor procedure is included in route description. In case the route is assigned indicating route elements, flight crew shall be informed about application of monitor procedure prior to start taxiing out of stand. Decision on application of monitor procedure shall be taken by the Flight Control Officer.

2.4.2.1. RWY 01

Route	Destination Point	Frequency	Routing via
Magenta 2	Vnukovo 3	120.450 GND 1	B5, M4, B5, Z, X, M1, hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller), M2

Taxiing to Vnukovo 3 Apron

Hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller).

ACFT shall stop at RWY 01/19 holding position maintaining listening watch on frequency of TWR controller without calling the controller. ACFT shall cross RWY01/19 by instruction of TWR controller and taxi to the stand on Vnukovo 3 apron, maintaining listening watch on frequency of GND controller.

2.4.2.2. RWY 06

Route	Destination Point	Frequency	Routing via
Blue 2	Vnukovo 1	120.450	A7/5/3, M3, Z, B5, M4, W

Taxiing to Vnukovo 1 Apron

Hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller). After crossing RWY 01/19 ACFT shall taxi to the stand under assistance of the Follow-me car maintaining listening watch on frequency of GND controller.

2.4.2.3. RWY 19

Route	Destination Point	Frequency	Routing via
Magenta 4	Vnukovo 1	120.450 GND 1	M1, X, W
Magenta 6			M1, X, Z

Taxiing to Vnukovo 1 Apron

ACFT shall vacate RWY via TWY M1 or via intersection of RWY 01/19 and RWY06/24 and taxi to the stand maintaining listening watch on frequency of GND controller.

ACFT shall vacate RWY via TWY B8.

Hold short of RWY 06/24, maintain listening watch on FREQ 118.300 MHz (TWR controller).

2. ARRIVAL

ACFT shall stop at RWY 06/24 holding position, maintaining listening watch on frequency of TWR controller without calling the controller. ACFT shall cross RWY06/24 by instruction of TWR controller and taxi to the stand, maintaining listening watch on frequency of GND controller.

Taxiing to Vnukovo 3 Apron

ACFT shall taxi to the stand under assistance of the Follow-me car, maintaining listening watch on frequency of GND controller.

2.4.2.4. RWY 24

Route	Destination Point	Frequency	Routing via
Blue 4	Vnukovo 1	120.450 GND 1	A10/11/13, M2, hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller), M1, X, W
Blue 6			A10/11/13, M2, hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller), M1, X, Z

Taxiing to Vnukovo 1 Apron

Hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller). After crossing RWY 01/19 ACFT shall taxi to the stand maintaining listening watch on frequency of GND controller.

Taxiing to Vnukovo 3 Apron

ACFT shall taxi to the stand under assistance of the Follow-me car maintaining listening watch on frequency of GND controller.

3. DEPARTURE

3.1. DE-ICING

For de-icing positions refer to 10-9 charts.

3.2. START-UP, TOWING AND TAXI PROCEDURES

3.2.1. START-UP

General

Parking onto start-up positions shall be executed in the direction indicated by the arrow.

Simultaneous use of two adjacent start-up positions of the same direction is prohibited.

During departure: the flight crew shall squawk ALT OFF or XPDR and keep this Mode till reaching the line-up position.

When towing of ACFT to engine start-up position is required, the flight crew shall advise stand number and request for towing clearance from GND controller.

Engine start-up shall be carried out at start-up points by the instruction of the technical specialist.

Flight crew can carry out engine start-up during towing of ACFT if this procedure is envisaged by Airplane Flight Manual and coordinated with the technical personnel of the tow-team.

3. DEPARTURE

Procedure

Flight crew of ACFT parking on Vnukovo 1 apron shall request the ATC clearance from Delivery 1, on other aprons from Delivery 3, not earlier than 15 minutes before time of departure indicated in the flight plan passing the following information to Delivery:

- flight number;
- the destination aerodrome;
- the ACFT stand number;
- ATIS code letter.

Note: Submission of a request for ATC clearance implies that all planned ground handling, passport control and customs clearance procedures were simultaneously completed by the prescribed TOBT.

Obtained ATC clearance is the permission to start up engines on the stand, to start up engines during towing, and to start up engines at start-up point.

After the confirmation of ATC clearance, the flight crew must immediately establish radio contact with GND controller and obtain instructions regarding sequence of ACFT taxxing from the stand, engine start-up operations, tow operations to start-up position, taxi operations to RWY holding position and movement of ACFT to de-icing treatment area.

If within 5 minutes after obtaining instructions regarding above operations, flight crew did not start to execute cleared operation, they must report to Delivery or GND controller, advising time of delay and reason.

In case of landing at Moscow/Vnukovo aerodrome as alternate and a change of the time earlier than indicated in the flight plan, a new time and slot must be coordinated with the APT services and flight plan message must be sent.

In case of ACFT departure from Vnukovo 1 or Vnukovo 3 apron 4, and de-icing treatment is required, 10 minutes shall be added to the time of departure for towing and engine start-up (departure considered as scheduled). Up to 15 minutes are given for taxiing and ensuring air traffic safety for all ACFT types.

Warning: ACFT taking off ahead of the departure time indicated in the flight plan shall be constituted as a breach of the Federal rules on the use of the airspace of the Russian Federation.

Flight crew is responsible for ensuring arrival of ACFT on de-icing treatment area.

3.2.2. TAXI PROCEDURES

Towing out of stands 64 and 65 is permitted in the direction of stands 42, 66 with operating radius of turn of not more than 85'/26m.

Taxiing out of stand 62A to taxi route Z under own engines power is permitted for ACFT with operating radius of turn of not more than 85'/26m.

3.2.3. STANDARD TAXI ROUTES

FOR RWY 01:

Route 19: TR W, TWY M4, TWY B5, TR Z, TR X, TWY M1, TWY A7, crossing of RWY 06/24 by ATC permission, TWY A8, TWY B8.

Route 20: TR W (TR Z), TR X, TWY M1, TWY A7, crossing of RWY 06/24 by ATC permission, TWY A8, TWY B8.

Route 21: TR W, TWY A2, TWY M3, TWY A7, crossing of RWY 06/24 by ATC permission, TWY A8, TWY B8.

FOR RWY 06:

Route 13: TR W, TWY M4, TWY B5, TR Z, TR X, TWY M1, crossing of RWY 01/19 by ATC permission, TWY M2.

Route 14: TR W (TR Z), TR X, TWY M1, crossing of RWY 01/19 by ATC permission, TWY M2.

3. DEPARTURE

FOR RWY 19:
Route 22: TR W, TWY M4.
Route 23: TR W, TR X, TR Z, TWY B5, TWY M4.

FOR RWY 24:
Route 16: TR W, TWY M4, TWY B5, TR Z, TWY A3.
Route 17: TR Z, TWY A3.
Route 18: TR W, TWY A2.

3.2.4. CODED TAXI ROUTES (MONITOR PROCEDURES)

In all cases the flight crew shall inform Vnukovo-Delivery controller about the estimated displaced position of take-off run on the RWY while obtaining ATC clearance. In case flight crew doesn't inform Delivery controller, GND controller gives instruction to take off from the RWY beginning taking into account limitations caused by airfield elements.

In case GND controller assigns a coded taxi route, monitor procedure is included in route description. In case the route is assigned indicating route elements, flight crew shall be informed about application of monitor procedure prior to start taxiing out of stand. Decision on application of monitor procedure shall be taken by the Flight Control Officer.

3.2.4.1. RWY 01

Route	Start Point	Frequency	Routing via
Orange 1	Vnukovo 1	120.450 GND 1	W/Z, X, M1, A7, hold short of RWY06/24, maintain listening watch on FREQ 118.300 MHz (TWR controller), A8, B8

Taxiing from Vnukovo 1 Apron

Hold short of RWY 06/24, maintain listening watch on FREQ 118.300 MHz (TWR controller).

ACFT shall stop at RWY 06/24 holding position maintaining listening watch on frequency of TWR controller without calling the controller. Control of an ACFT shall be transferred from GND controller to TWR controller.

ACFT shall cross RWY 06/24 by instruction of TWR controller and taxi to RWY 01 holding position.

Taxiing from Vnukovo 3 Apron

Hold on TWY C2/C3, maintain listening watch on FREQ 118.300 MHz (TWR controller).

ACFT shall stop at MAIN TWY M2 in order to cross RWY 01/19 (TWY C2, C3) maintaining listening watch on frequency of TWR controller without calling the controller. ACFT shall cross RWY 01/19 by instruction of TWR controller and taxi to RWY 06/24 holding position, maintaining listening watch on frequency of GND controller. Upon arrival to RWY06/24 holding position, control of an ACFT shall be transferred from GND controller to TWR controller.

3. DEPARTURE

3.2.4.2. RWY 06

Route	Start Point	Frequency	Routing via
Violet 1	Vnukovo 1	120.450 GND 1	W/Z, X, M1, hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller), M2, A13
Violet 3			W/Z, M3, M1, hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller), M2, A13

Taxiing from Vnukovo 1 Apron

Hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller).

ACFT shall stop at RWY 01/19 RWY holding position, maintaining listening watch on TWR controller's frequency without calling the controller. ACFT shall cross RWY 01/19 by instruction of TWR controller and taxi to RWY 06 holding position maintaining listening watch on frequency of GND controller. Upon arrival to RWY06 holding position, control of an ACFT shall be transferred from GND controller to TWR controller.

Taxiing from Vnukovo 3 Apron

Upon arrival to the holding position, control of an ACFT shall be transferred from GND controller to TWR controller.

3.2.4.3. RWY 19

Route	Start Point	Frequency	Routing via
Orange 3	Vnukovo 3	120.450 GND 1	M2, hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller), M1, X, Z, B5, M4, B1
Orange 5	Vnukovo 1		W, M3, Z, B5, M4, B1

Taxiing from Vnukovo 1 Apron

Upon arrival to RWY 19 holding position, control of an ACFT shall be transferred from GND controller to TWR controller.

Taxiing from Vnukovo 3 Apron

Hold on TWY C2/C3, maintain listening watch on FREQ 118.300 MHz (TWR controller).

ACFT shall stop at MAIN TWY M2 in order to cross RWY 01/19 (TWY C2, C3) maintaining listening watch on frequency of TWR controller without calling the controller. ACFT shall cross RWY 01/19 by instruction of TWR controller and taxi to RWY 19 holding position, maintaining listening watch on frequency of GND controller. Upon arrival to the holding position, control of an ACFT shall be transferred from GND controller to TWR controller.

3. DEPARTURE

3.2.4.4. RWY 24

Route	Start Point	Frequency	Routing via
Violet 5	Vnukovo 3	120.450 GND 1	M2, hold short of RWY 01/19, maintain listening watch on FREQ 118.300 MHz (TWR controller), M1, M3, A2
Violet 7	Vnukovo 1		W, M4, B5, Z, A3, C3, C2

Taxiing from Vnukovo 1 Apron

Upon arrival to the RWY-holding position, control of an ACFT shall be transferred from GND controller to TWR controller.

Taxiing from Vnukovo 3 Apron

Hold on TWY C2/C3, maintain listening watch on FREQ 118.300 MHz (TWR controller).

ACFT shall stop at MAIN TWY M2 in order to cross RWY 01/19 (C2, C3) maintaining listening watch on frequency of TWR controller without calling the controller. ACFT shall cross RWY 01/19 by instruction of TWR controller and taxi to RWY 24 holding position, maintaining listening watch on frequency of GND controller. Upon arrival to RWY 24 holding position, control of an ACFT shall be transferred from GND controller to TWR controller.

3.3. COMMUNICATION FAILURE PROCEDURES

3.3.1. COMMUNICATION FAILURE AFTER TAKE-OFF

Set transponder to 7600 and maintain flight route and profile of the cleared RNAV SID.

After taking decision to return to Moscow (Vnukovo):

- Proceed to SID termination point, and then to the significant point of the shortest basic STAR RNAV.
- Maintain flight route and profile of the basic RNAV STAR.
- Execute approach in accordance with the established procedure.

In case of missed approach follow paragraph 2.1.2.

After taking decision to proceed to destination AD:

- After leaving Moscow TMA, continue climbing to the flight level indicated in the flight plan.

Set transponder to 7700 if it is necessary to deviate from the mentioned procedure.

3.4. RWY OPERATION

3.4.1. MINIMUM RWY OCCUPANCY TIME

After obtaining line-up clearance the flight crew must execute taxiing to the assigned position on the RWY without delay.

Cockpit checks must be completed, if possible, before occupation of line-up position and ground checks which have to be done on the RWY shall be executed as soon as possible.

After ACFT occupation of the RWY the flight crew shall ensure commencement of ACFT movement for take-off within 10 seconds after obtaining the take-off clearance.

If unable to execute the above mentioned requirements, the flight crews must inform Tower about it.

3. DEPARTURE

During operation on two RWYs, the time from issuance of line-up clearance till start of take-off run:

On heading 012°:

- from TWY B8 and intersection with RWY 06/24 - not more than 30 seconds.
- from beginning of RWY 01 - not more than 120 seconds.

On heading 057°:

- from TWY A10, A11, A13 - not more than 60 seconds.

On heading 192°:

- from TWY B1, B2, B4 - not more than 60 seconds.

On heading 237°:

- from TWY A3 and A5 - not more than 60 seconds.
- from TWY A2 - not more than 90 seconds.

Using two RWYs, the time from issuance of controllers instruction to line up for take-off till the start of take-off run:

On heading 057° (take-off, landing) and on 012° (take-off):

- from TWY A10, A11, A13, M1, MAIN TWY M2 - not more than 60 seconds.

On heading 237° (take-off) and on 012° (landing):

- from TWY A3 and A5 - not more than 60 seconds.
- from TWY A2 - not more than 90 seconds.

On heading 192° (take-off) and 237° (landing):

- from TWY B1, B2, B4 - not more than 60 seconds.

On heading 057° (take-off) and 012° (landing):

- from TWY A10, A11, A13 - not more than 60 seconds.

3.4.2. INTERMEDIATE TAKE-OFF

If the pilot-in-command is ready to carry out take-off without stop on RWY, he has to report his decision to Tower at first contact at RWY holding position.

3.4.3. REDUCED RWY SEPARATION MINIMA

Reduced RWY separation minima are applied during flight operations to/from RWY 06/24 only in the day-time, within the time period that starts 30 minutes after sunrise and ends 30 minutes before sunset (LT), and are subject to the following meteorological conditions:

- visibility is at least 5km;
- the height of cloud base (ceiling) is 300m or above;
- tailwind component does not exceed 3m/s.

Normative friction coefficient on the RWY shall be 0.40 or above (estimated surface friction "good" or "medium to good").

Reduced RWY separation minima is applied based on the decision of the AD Flight Control Officer.

Information on application at the aerodrome must be included in ATIS broadcast.

ATS unit can issue take-off clearance to flight crew of a departing ACFT, provided there is reasonable assurance, that when this ACFT commences its take-off roll from the line-up position, the preceding departing ACFT is airborne and has passed the point located at a distance at least 2500m from the succeeding departing ACFT.

Take-off clearance issued by the ATS unit to flight crew of a succeeding departing ACFT must contain information on type of the preceding departing ACFT. ATS unit may instruct flight crew to comply with initial climb altitude restrictions.

Flight crew must report sighting the preceding departing ACFT and acknowledge receipt of take-off clearance, if requested by the ATS unit.

3. DEPARTURE

Pilot-in-command makes the final decision to execute take-off. The obtained take-off clearance does not entail an obligation for the flight crew to execute take-off.

In order to expedite ACFT line up after the approaching ACFT, VNUKOVO Tower controller may issue the conditional line-up clearance using the phrase:
"(Call sign), (ACFT type) on final, distance... km, cleared to line up (line up) RWY (designator) after landing ACFT".

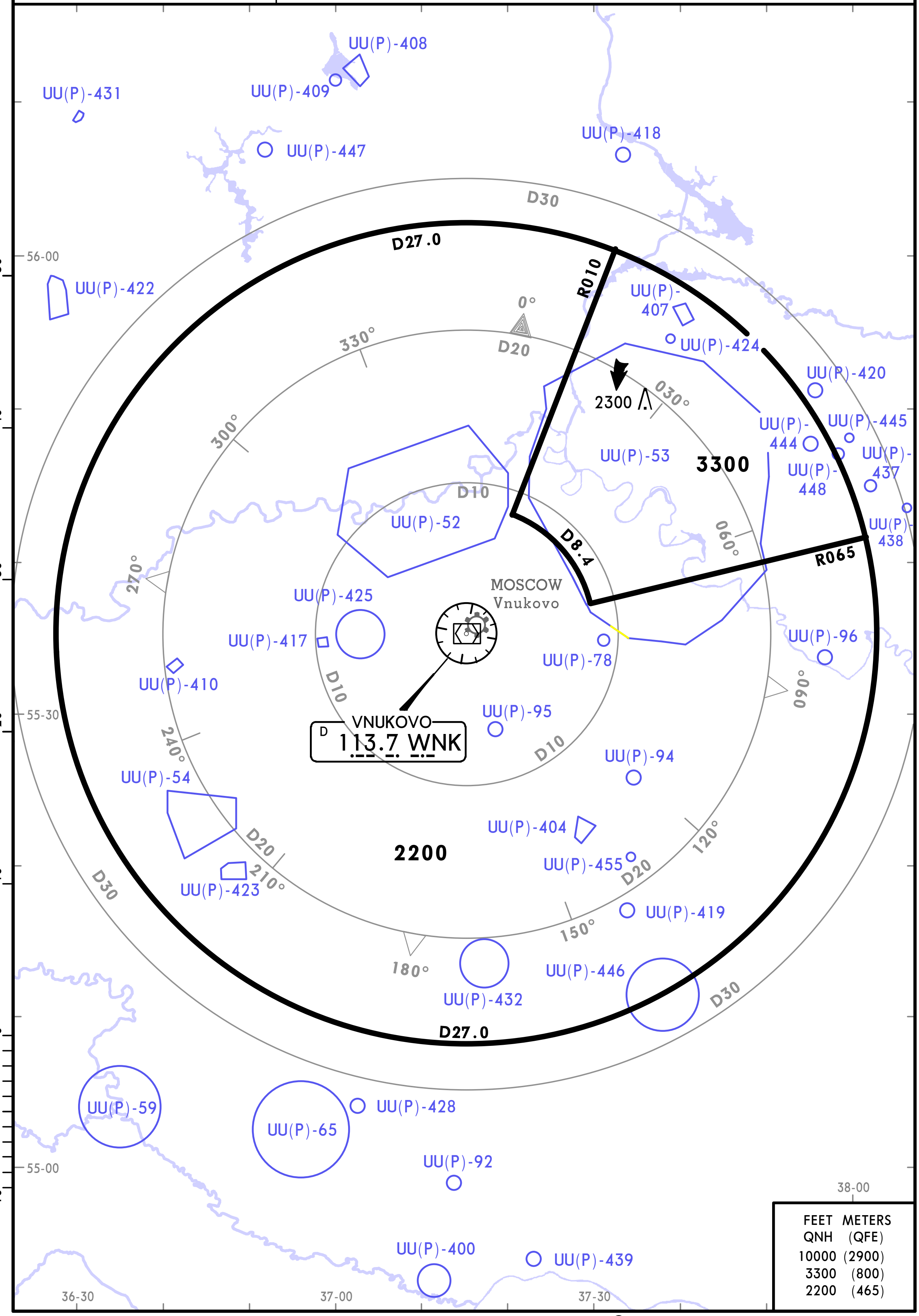
In order to expedite ACFT line up after the departing ACFT, VNUKOVO Tower controller may issue the conditional line-up clearance using the phrase:
"(Call sign), cleared to line up RWY (designator) in sequence".

VNUKOVO Radar (TWR)		
Sector A4	Sector WD	Sector D4
123.4	126.0	135.9

Apt Elev
685

Alt Set: hPa (MM on request) (QFE on request)
Trans level: FL110
FL120 if pressure is less than 1013 hPa (760mm)
FL130 if pressure is less than 977 hPa (733mm)
Trans alt: 10000

1. This chart may only be used for cross-checking of altitudes assigned while under RADAR control
2. When vectoring at 0° C or below, the minimum vectoring altitudes must be corrected by altimeter temperature correction.



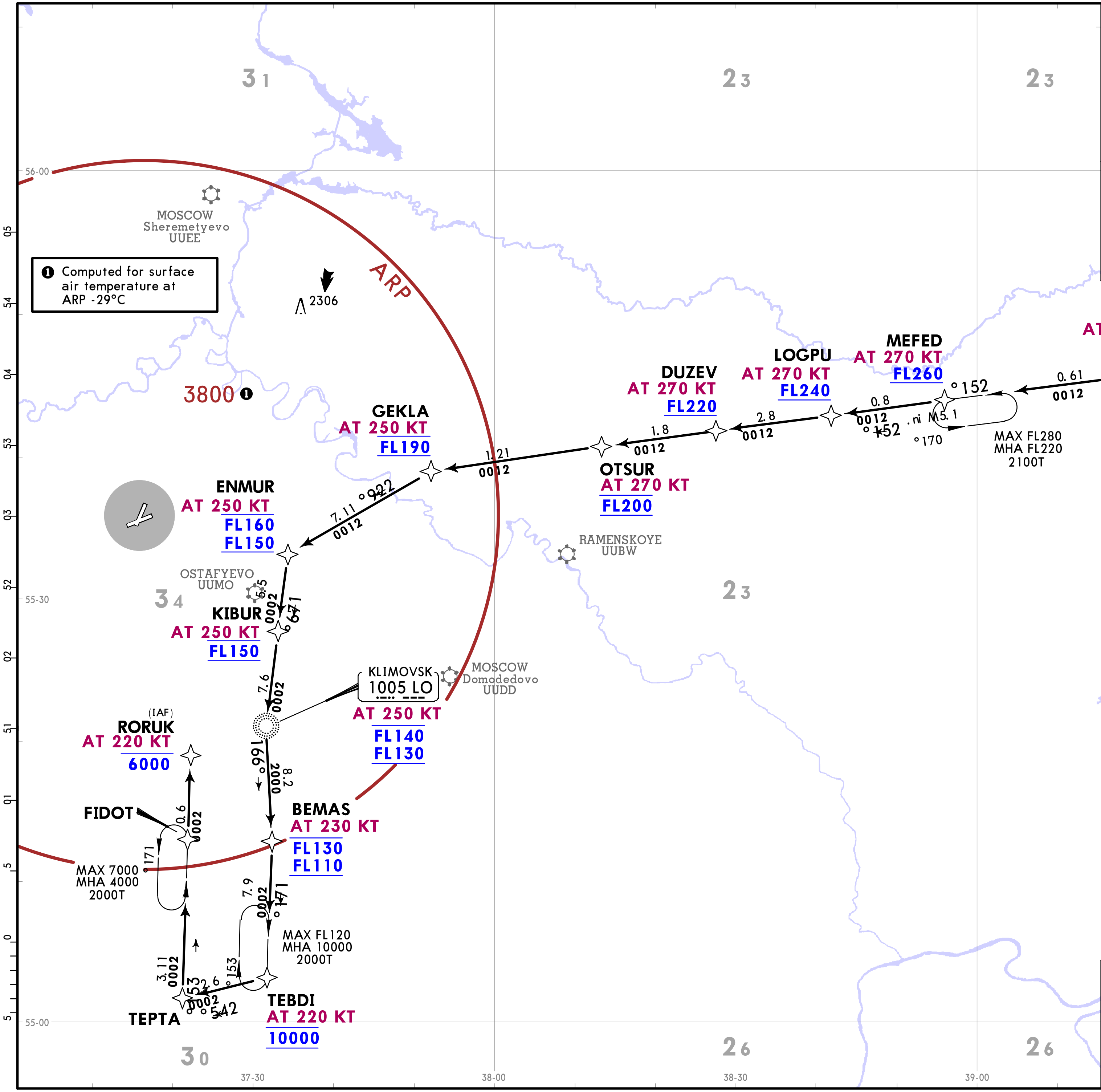
CHANGES: Prohibited areas.

ATIS	Apt Elev
131.850 (Russian 125.875)	685

Alt Set: hPa (MM on request)
Trans level: FL110
FL120 if pressure is less than 1013 hPa (760mm)
FL130 if pressure is less than 977 hPa (733mm)

RNAV 1 DME/DME or GNSS required

DIMGI 3A [DIMG3A]
DIMGI 3H ❷
RNAV ARRIVAL
(ALL RWYS)



❶ Computed for surface air temperature at ARP -29°C

❷ For fuel estimation:
DIMGI - VIVZO - MEFED - DUZEV - OTSUR - GEKLA - ENMUR - KIBUR - LO - BEMAS - FIDOT - RORUK.
Not to be used for navigation.

FEET QNH	METERS (QFE)
10000	(2860)
7000	(1945)
6000	(1640)
4000	(1030)

QFE values based on RWY 06 THR elevation

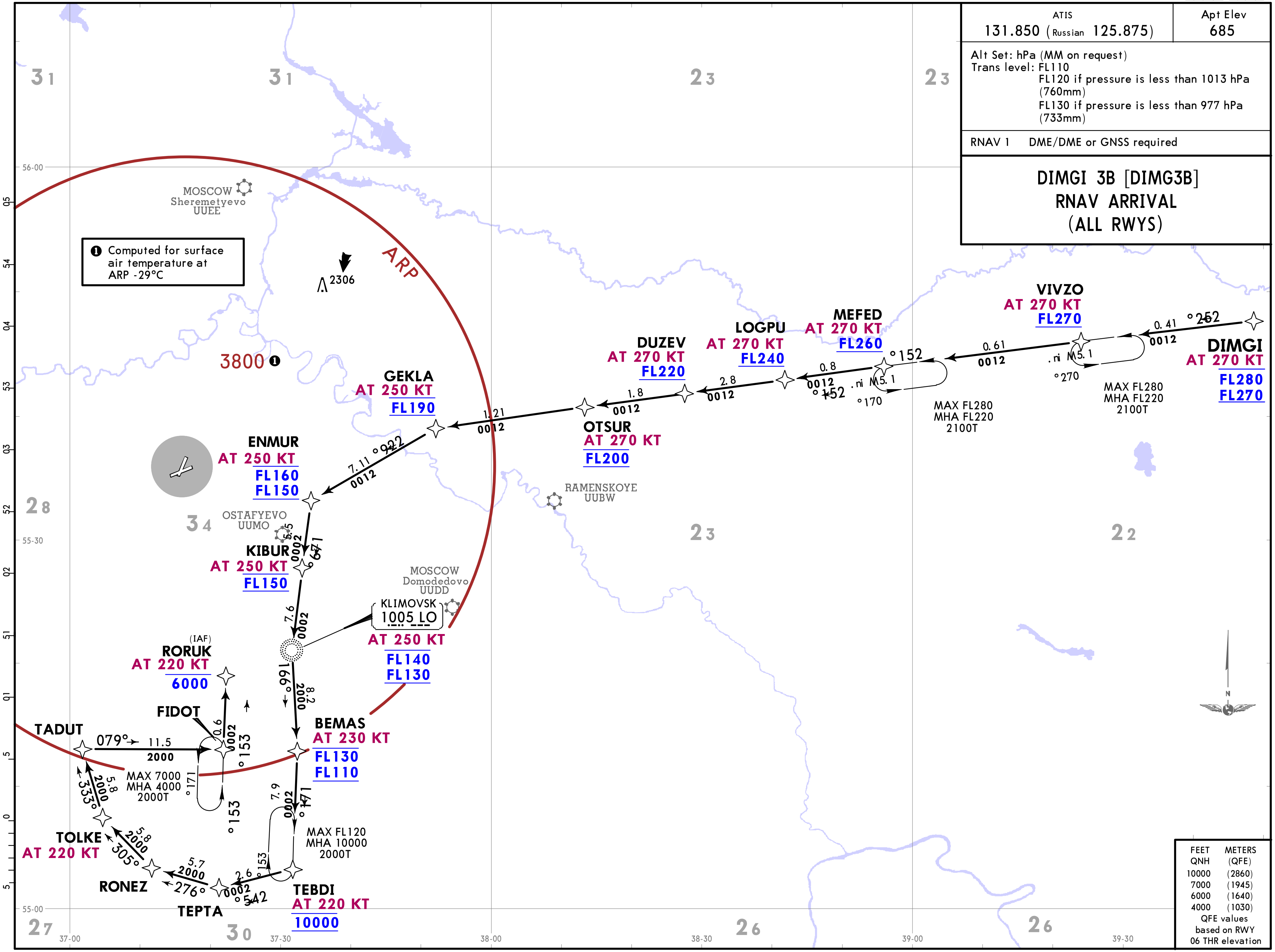
CHANGES: Nav-Data ident withdrawn.

ATIS	Apt Elev
131.850 (Russian 125.875)	685

Alt Set: hPa (MM on request)
Trans level: FL110
FL120 if pressure is less than 1013 hPa (760mm)
FL130 if pressure is less than 977 hPa (733mm)

RNAV 1 DME/DME or GNSS required

DIMGI 3B [DIMG3B]
RNAV ARRIVAL
(ALL RWYS)



FEET QNH	METERS (QFE)
10000	(2860)
7000	(1945)
6000	(1640)
4000	(1030)

QFE values based on RWY 06 THR elevation

CHANGES: None.

UUWW/VKO
VNUKOVO

JEPPESEN 1 NOV 24 (10-2B) MOSCOW, RUSSIA
RNAV STAR

LEKDO 3A [LEKD3A]
OLOPI 3A [OLOP3A]
OLOPI 3H ❷
RNAV ARRIVALS (ALL RWYS)

ATIS	Apt Elev
131.850 (Russian 125.875)	685

Alt Set: hPa (MM on request)
Trans level: FL110
FL120 if pressure is less than 1013 hPa (760mm)
FL130 if pressure is less than 977 hPa (733mm)

RNAV 1 DME/DME or GNSS required

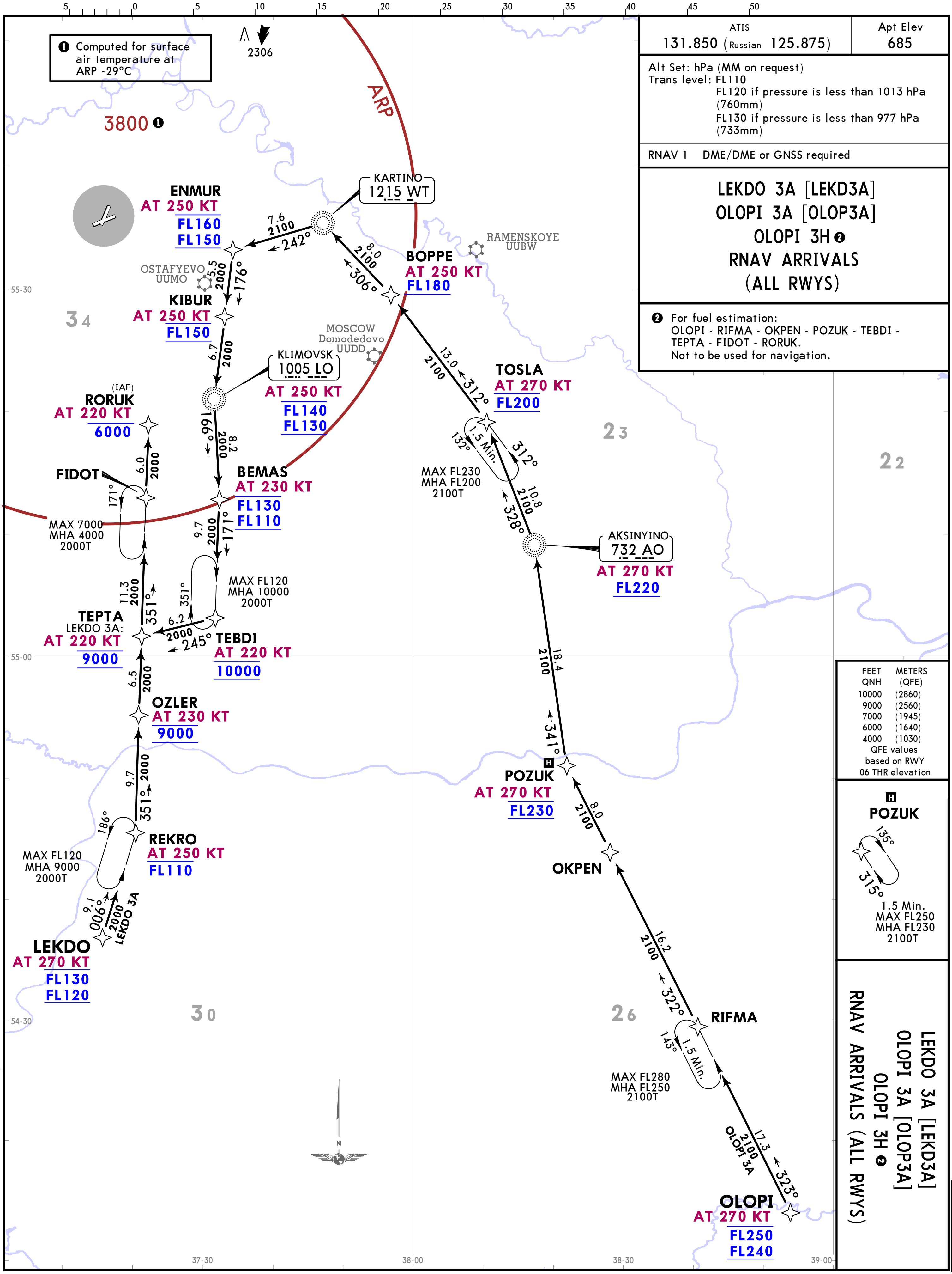
LEKDO 3A [LEKD3A]
OLOPI 3A [OLOP3A]
OLOPI 3H ❷
RNAV ARRIVALS (ALL RWYS)

❷ For fuel estimation:
OLOPI - RIFMA - OKPEN - POZUK - TEBDI - TEPTA - FIDOT - RORUK.
Not to be used for navigation.

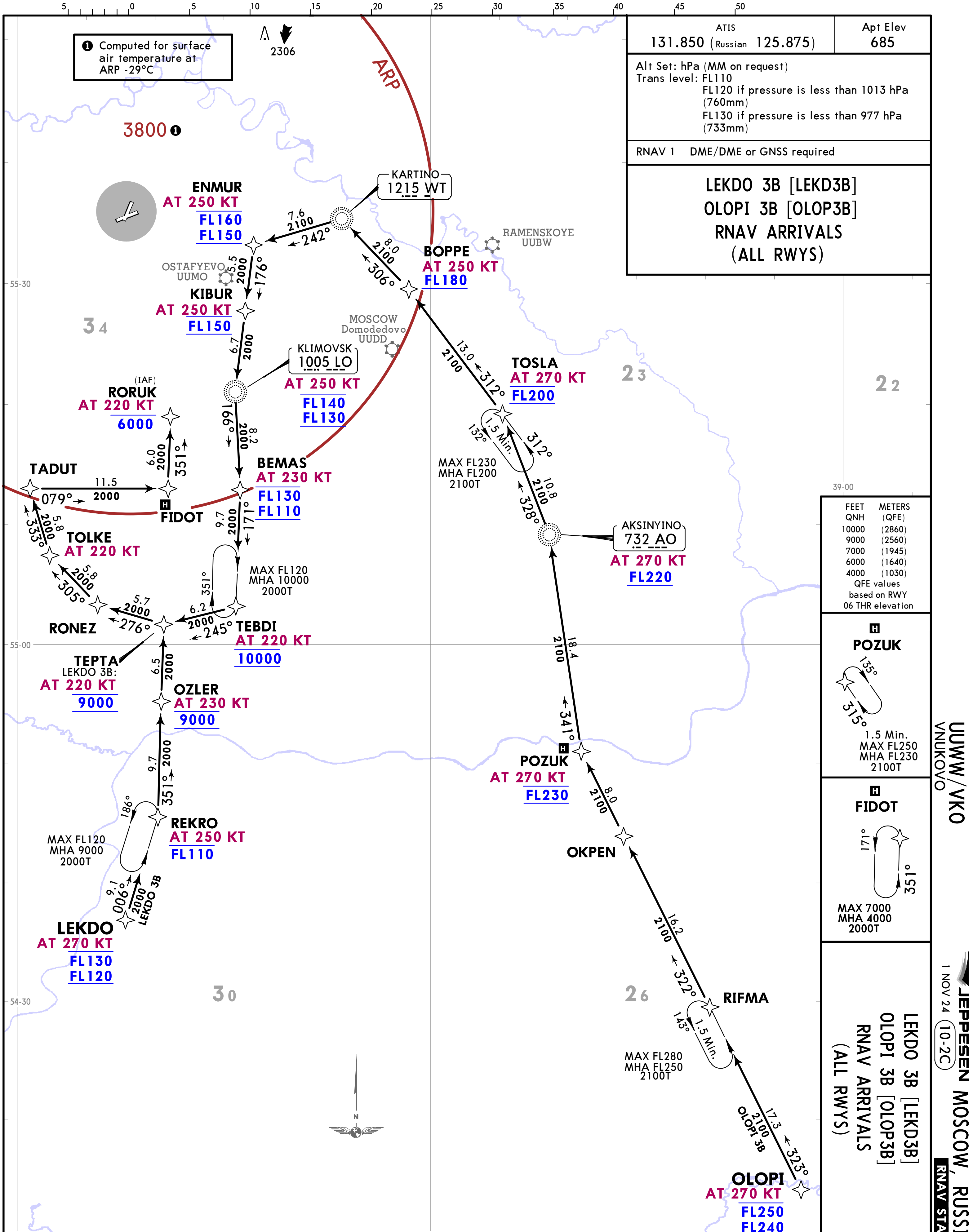
❶ Computed for surface air temperature at ARP -29°C

FEET QNH	METERS (QFE)
10000	(2860)
9000	(2560)
7000	(1945)
6000	(1640)
4000	(1030)

QFE values based on RWY 06 THR elevation



CHANGES: Nav-Data ident withdrawn.



UUWW/VKO
VNUKOVO
MOSCOW, RUSSIA
JEPPESEN
1 NOV 24
10-2C
RNAV STAR
ATIS
131.850 (Russian 125.875)
Apt Elev
685
Alt Set: hPa (MM on request)
Trans level: FL110
FL120 if pressure is less than 1013 hPa (760mm)
FL130 if pressure is less than 977 hPa (733mm)
RNAV 1 DME/DME or GNSS required
LEKDO 3B [LEKD3B]
OLOPI 3B [OLOP3B]
RNAV ARRIVALS
(ALL RWYS)
CHANGES: None.
❶ Computed for surface air temperature at ARP -29°C
3800 ❶
2306
ARP
KARTINO
1215 WT
RAMENSKOYE
UUBW
OSTAFYEVO
UUMO
MOSCOW
Domodedovo
UUDD
KLIMOVSK
1005 LO
AKSINYINO
732 AO
ENMUR
AT 250 KT
FL160
FL150
KIBUR
AT 250 KT
FL150
BOPPE
AT 250 KT
FL180
TOSLA
AT 270 KT
FL200
RORUK (IAF)
AT 220 KT
6000
TADUT
FIDOT
BEMAS
AT 230 KT
FL130
FL110
TOLKE
AT 220 KT
RONEZ
TEBDI
AT 220 KT
10000
TEPTA
LEKDO 3B:
AT 220 KT
9000
OZLER
AT 230 KT
9000
REKRO
AT 250 KT
FL110
LEKDO
AT 270 KT
FL130
FL120
LEKDO 3B
POZUK
AT 270 KT
FL230
OKPEN
RIFMA
OLOPI
AT 270 KT
FL250
FL240
OLOPI 3B
AT 270 KT
FL220
MAX FL230
MHA FL200
2100T
MAX FL120
MHA 10000
2000T
MAX FL120
MHA 9000
2000T
MAX FL280
MHA FL250
2100T
1.5 Min.
FEET QNH | METERS (QFE)
10000 (2860)
9000 (2560)
7000 (1945)
6000 (1640)
4000 (1030)
QFE values based on RWY 06 THR elevation
POZUK
135°
315°
1.5 Min.
MAX FL250
MHA FL230
2100T
FIDOT
171°
351°
MAX 7000
MHA 4000
2000T
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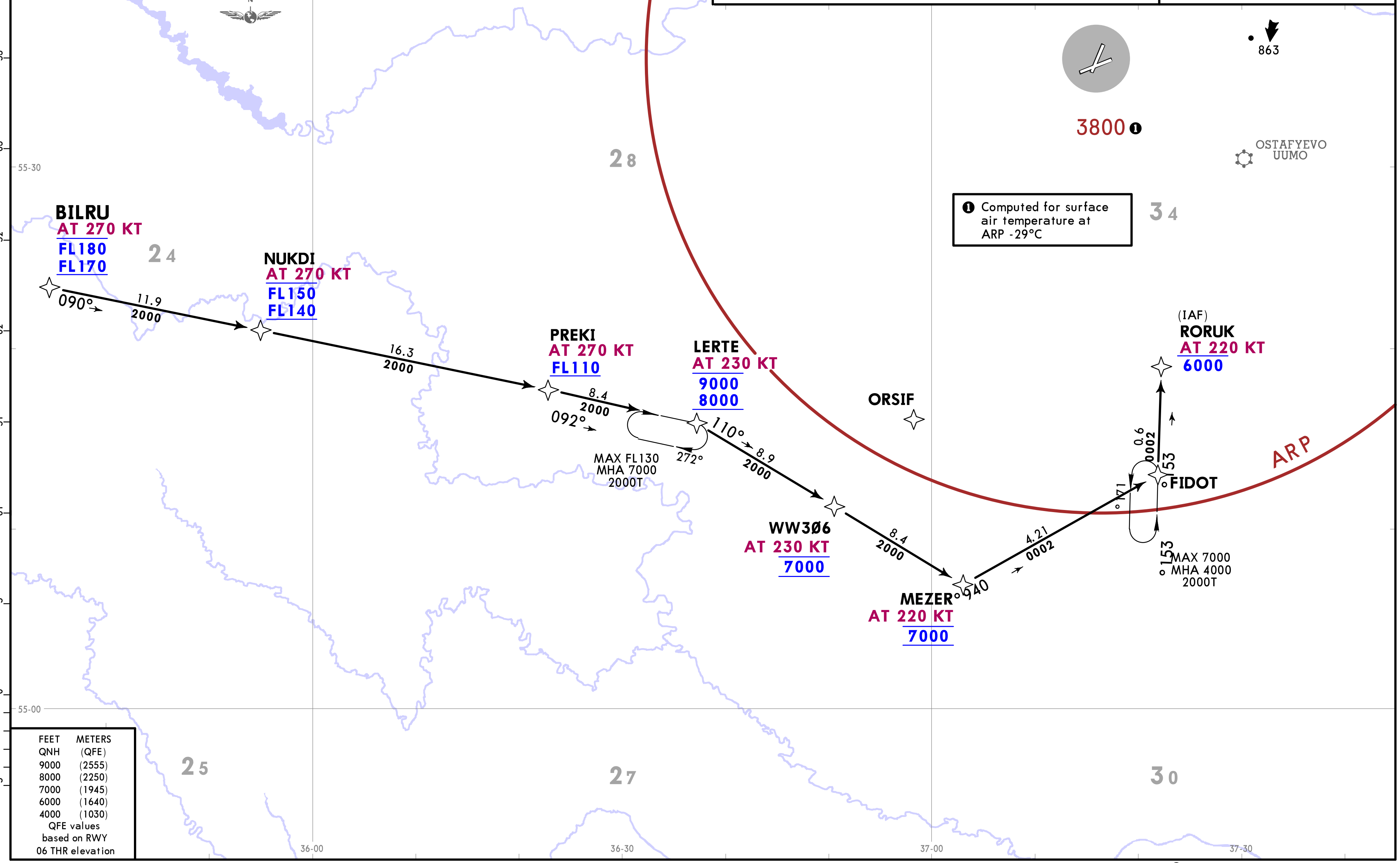
ATIS	Alt Set: hPa (MM on request)
131.850 (Russian 125.875)	Trans level: FL110 FL120 if pressure is less than 1013 hPa (760mm) FL130 if pressure is less than 977 hPa (733mm)
Apt Elev 685	RNAV 1 DME/DME or GNSS required

BILRU 3A [BILR3A], BILRU 3H❷
RNAV ARRIVAL
(ALL RWYS)

❷ For fuel estimation:
BILRU - NUKDI - PREKI - ORSIF - FIDOT - RORUK.
Not to be used for navigation.



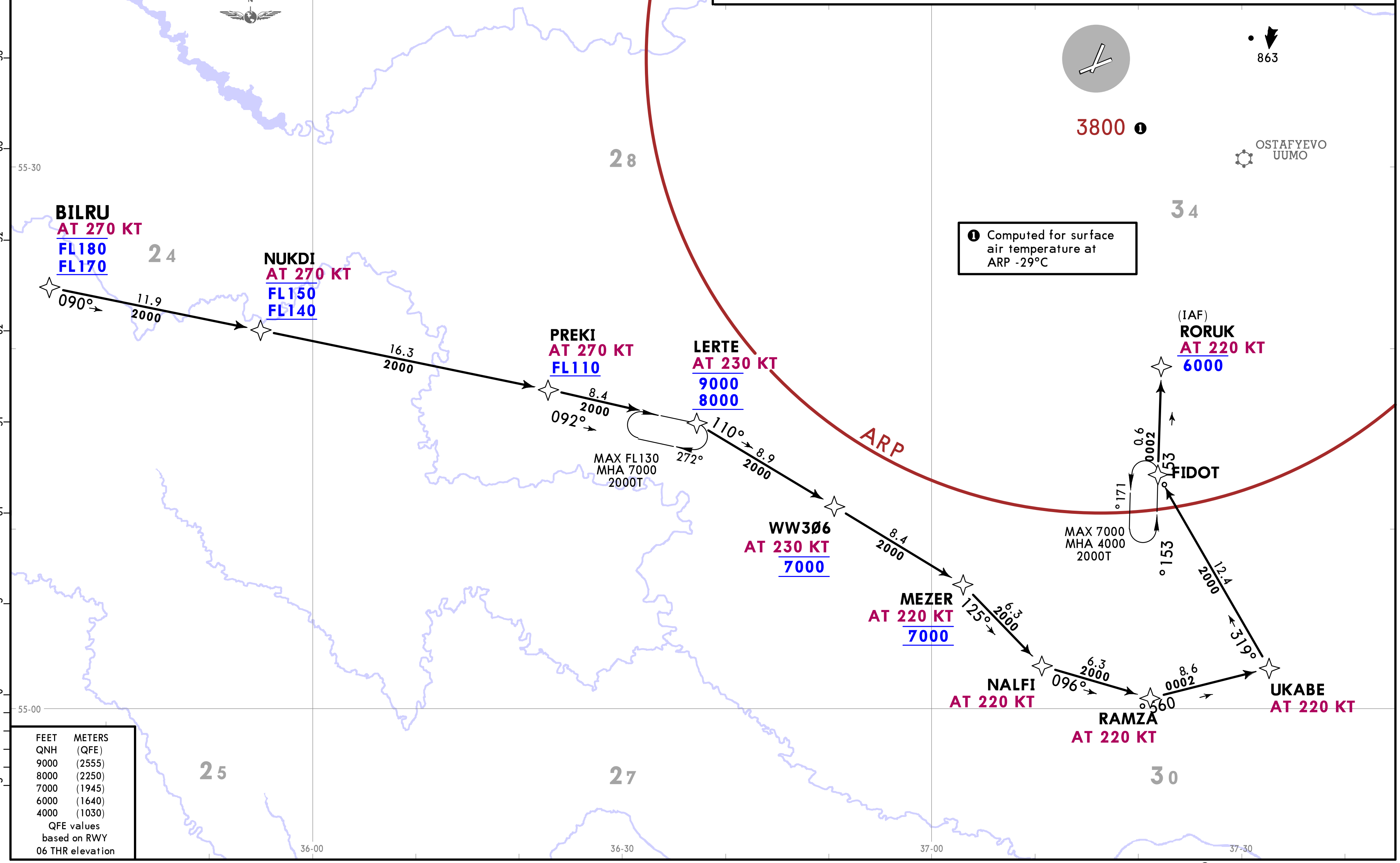
FEET QNH	METERS (QFE)
9000	(2555)
8000	(2250)
7000	(1945)
6000	(1640)
4000	(1030)

QFE values based on RWY 06 THR elevation

CHANGES: Nav-Data ident withdrawn.

ATIS	Alt Set: hPa (MM on request)
131.850 (Russian 125.875)	Trans level: FL110 FL120 if pressure is less than 1013 hPa (760mm) FL130 if pressure is less than 977 hPa (733mm)
Apt Elev 685	RNAV 1 DME/DME or GNSS required

BILRU 3B [BILR3B]
RNAV ARRIVAL
(ALL RWYS)



FEET QNH	METERS (QFE)
9000	(2555)
8000	(2250)
7000	(1945)
6000	(1640)
4000	(1030)

QFE values based on RWY 06 THR elevation

CHANGES: None.

ATIS	Apt Elev
131.850 (Russian 125.875)	685

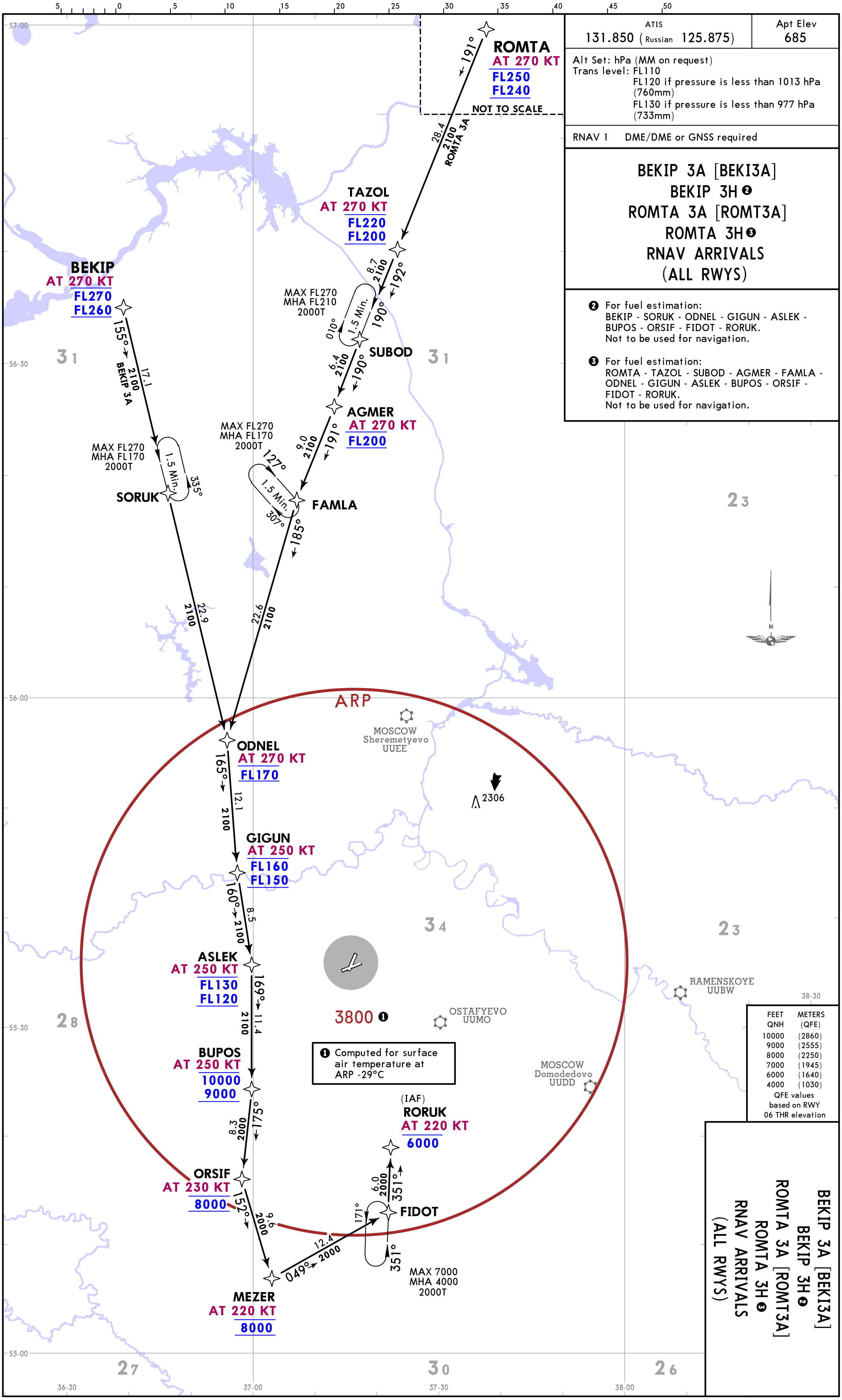
Alt Set: hPa (MM on request)
Trans level: FL110
FL120 if pressure is less than 1013 hPa (760mm)
FL130 if pressure is less than 977 hPa (733mm)

RNAV 1 DME/DME or GNSS required

BEKIP 3A [BEKI3A]
BEKIP 3H ❷
ROMTA 3A [ROMT3A]
ROMTA 3H ❸
RNAV ARRIVALS
(ALL RWYS)

❷ For fuel estimation:
BEKIP - SORUK - ODNEL - GIGUN - ASLEK - BUPOS - ORSIF - FIDOT - RORUK.
Not to be used for navigation.

❸ For fuel estimation:
ROMTA - TAZOL - SUBOD - AGMER - FAMLA - ODNEL - GIGUN - ASLEK - BUPOS - ORSIF - FIDOT - RORUK.
Not to be used for navigation.



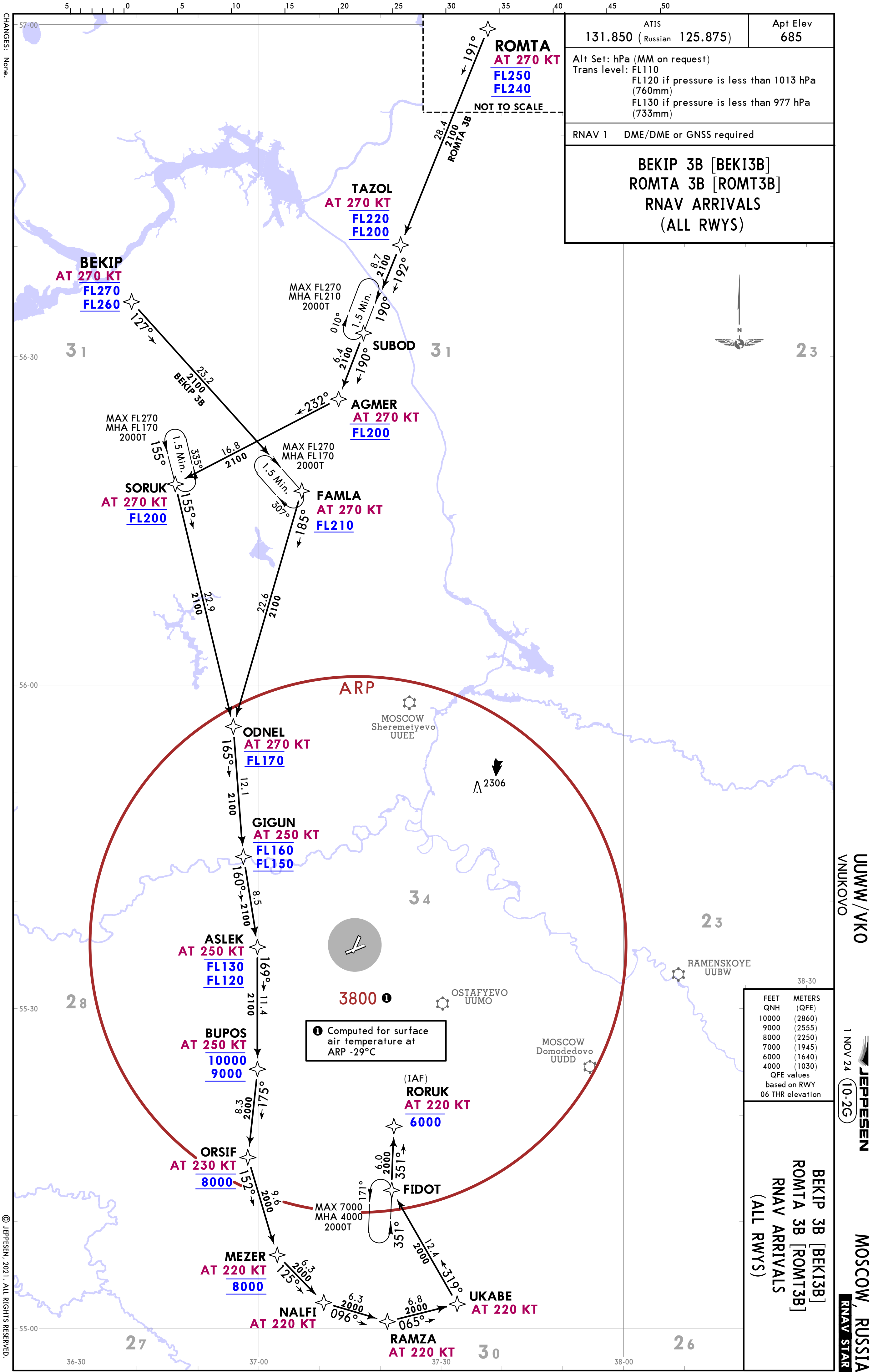
FEET QNH	METERS (QFE)
10000	(2860)
9000	(2555)
8000	(2250)
7000	(1945)
6000	(1640)
4000	(1030)

QFE values based on RWY 06 THR elevation

UUWW/VKO
VNUKOVO

JEPPESEN 1 NOV 24 (10-2F)
MOSCOW, RUSSIA
RNAV STAR

BEKIP 3A [BEKI3A]
BEKIP 3H ❷
ROMTA 3A [ROMT3A]
ROMTA 3H ❸
RNAV ARRIVALS
(ALL RWYS)


UUWW/VKO
VNUKOVO
MOSCOW, RUSSIA
RNAV STAR
JEPPESEN
1 NOV 24 10-2G
BEKIP 3B [BEKI3B]
ROMTA 3B [ROMT3B]
RNAV ARRIVALS
(ALL RWYS)
ATIS
131.850 (Russian 125.875)
Apt Elev
685
Alt Set: hPa (MM on request)
Trans level: FL110
FL120 if pressure is less than 1013 hPa (760mm)
FL130 if pressure is less than 977 hPa (733mm)
RNAV 1 DME/DME or GNSS required
CHANGES: None.
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NOT TO SCALE
ROMTA
AT 270 KT
FL250
FL240
191°
28.4
2100
ROMTA 3B
TAZOL
AT 270 KT
FL220
FL200
8.7
2100
192°
MAX FL270
MHA FL210
2000T
1.5 Min.
010°
190°
SUBOD
6.4
2100
190°
AGMER
AT 270 KT
FL200
232°
BEKIP
AT 270 KT
FL270
FL260
127°
23.2
2100
BEKIP 3B
MAX FL270
MHA FL170
2000T
1.5 Min.
155°
335°
16.8
2100
SORUK
AT 270 KT
FL200
155°
MAX FL270
MHA FL170
2000T
1.5 Min.
307°
FAMLA
AT 270 KT
FL210
185°
22.9
2100
22.6
2100
ARP
MOSCOW
Sheremetyevo
UUEE
2306
ODNEL
AT 270 KT
FL170
165°
12.1
2100
GIGUN
AT 250 KT
FL160
FL150
160°
8.5
2100
ASLEK
AT 250 KT
FL130
FL120
169°
11.4
2100
3800
Computed for surface air temperature at ARP -29°C
OSTAFYEVO
UUMO
MOSCOW
Domodedovo
UUDD
RAMENSKOYE
UUBW
BUPOS
AT 250 KT
10000
9000
175°
8.3
2000
(IAF)
RORUK
AT 220 KT
6000
6.0
2000
351°
ORSIF
AT 230 KT
8000
152°
9.6
2000
FIDOT
171°
MAX 7000
MHA 4000
2000T
351°
12.4
2000
319°
MEZER
AT 220 KT
8000
6.3
2000
125°
NALFI
AT 220 KT
6.3
2000
096°
RAMZA
AT 220 KT
6.8
2000
065°
UKABE
AT 220 KT
FEET QNH | METERS (QFE)
10000 (2860)
9000 (2555)
8000 (2250)
7000 (1945)
6000 (1640)
4000 (1030)
QFE values based on RWY 06 THR elevation

CHANGES: Initial climb clearance withdrawn.

UUWW/VKO VNUKOVO

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

RILPO 3A [RILP3A]

CAT A, B & C

(RWY 01)

RILPO 3B [RILP3B]

(RWY 06)

RNAV DEPARTURES

Close-in Obstacles
RWY 01: MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 06: MAX elevation 746 - group of obstacles at 0.4 NM from DER to the RIGHT of take-off heading.
These SIDs require minimum climb gradients of
RILPO 3A: 5.1% up to 1200 due to obstacles.
8.8% up to FL120 due to airspace structure.
RILPO 3B: 5.7% up to 6000 due to airspace structure.

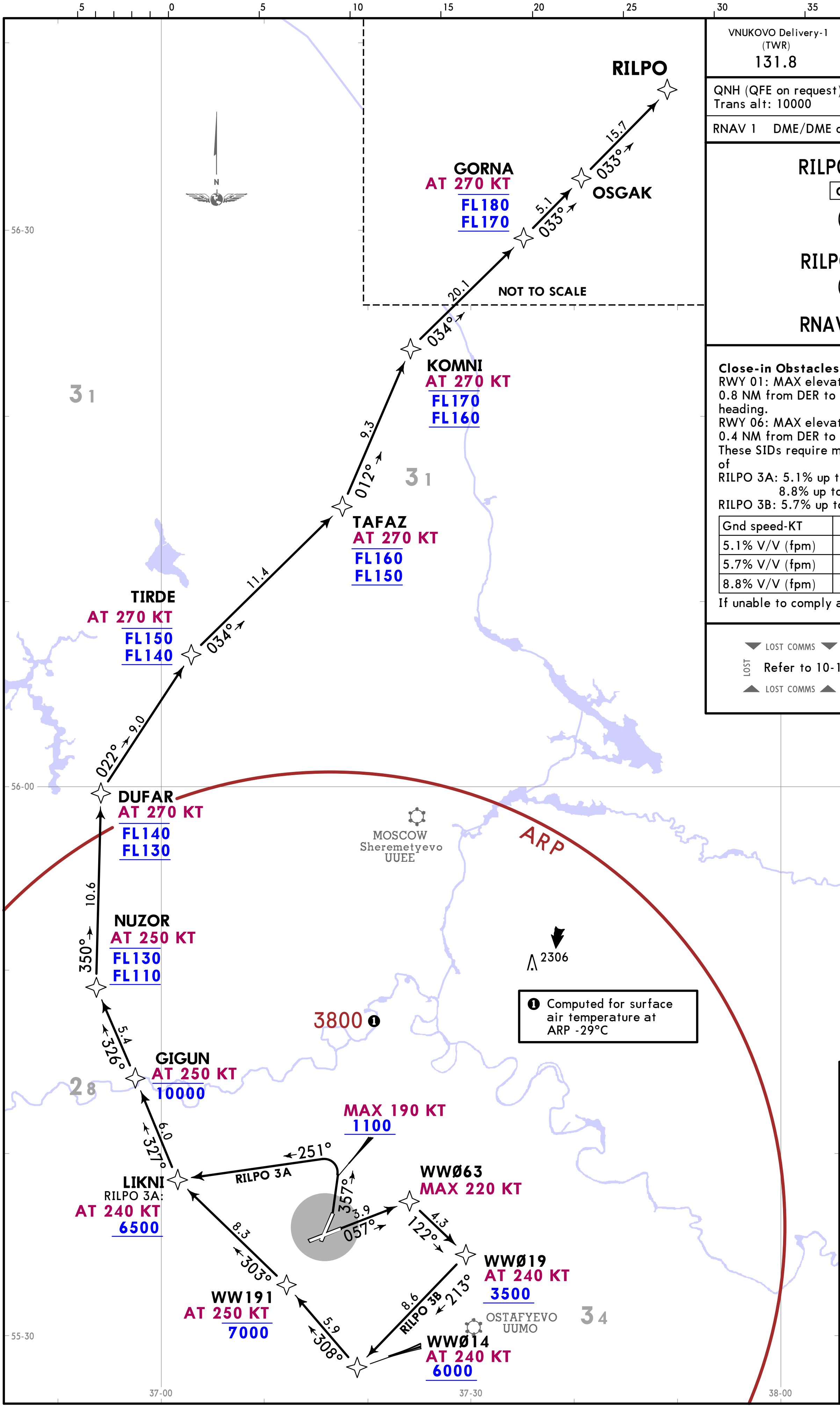
Gnd speed-KT	75	100	150	200	250	300
5.1% V/V (fpm)	387	516	775	1033	1291	1549
5.7% V/V (fpm)	433	577	866	1154	1443	1732
8.8% V/V (fpm)	668	891	1337	1782	2228	2673

If unable to comply advise VNUKOVO - Delivery

LOST COMMS LOST COMMS
LOST Refer to 10-1P pages. COMMS
LOST COMMS LOST COMMS

FEET	METERS
QNH	(QFE)
1100	(145)
1200	(175)
3500	(875)
6000	(1640)
6500	(1795)
7000	(1945)
10000	(2860)

QFE values based on RWY 06 THR elevation



RILPO 3A [RILP3A]
CAT A, B & C
(RWY 01)
RILPO 3B [RILP3B]
(RWY 06)
RNAV DEPARTURES

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

RILPO 3C [RILP3C] (RWY 19)
RILPO 3D [RILP3D] (RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.
These SIDs require minimum climb gradients of
RILPO 3C: 6.6% up to FL110 due to airspace structure.
RILPO 3D: 6.9% up to FL110 due to airspace structure.

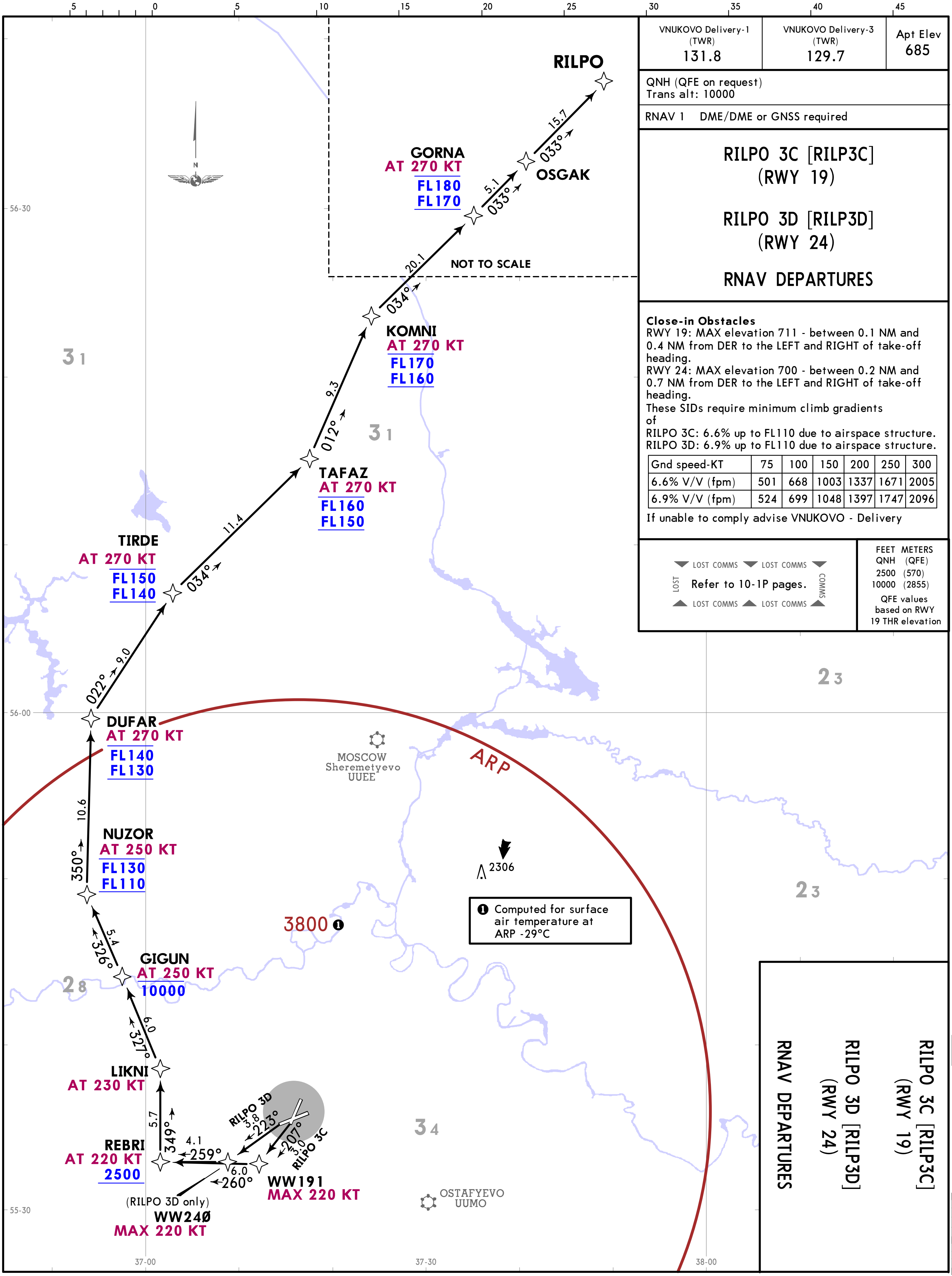
Gnd speed-KT	75	100	150	200	250	300
6.6% V/V (fpm)	501	668	1003	1337	1671	2005
6.9% V/V (fpm)	524	699	1048	1397	1747	2096

If unable to comply advise VNUKOVO - Delivery

LOST COMMS
Refer to 10-1P pages.

FEET QNH	METERS (QFE)
2500	(570)
10000	(2855)

QFE values based on RWY 19 THR elevation



UUWW/VKO
VNUKOVO

JEPPESEN MOSCOW, RUSSIA

5 JUL 24 10-3B Eff 11 Jul

RNAV SID

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

RILPO 3W [RILP3W]
RNAV DEPARTURE
(RWY 01)

Close-in Obstacles
MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.

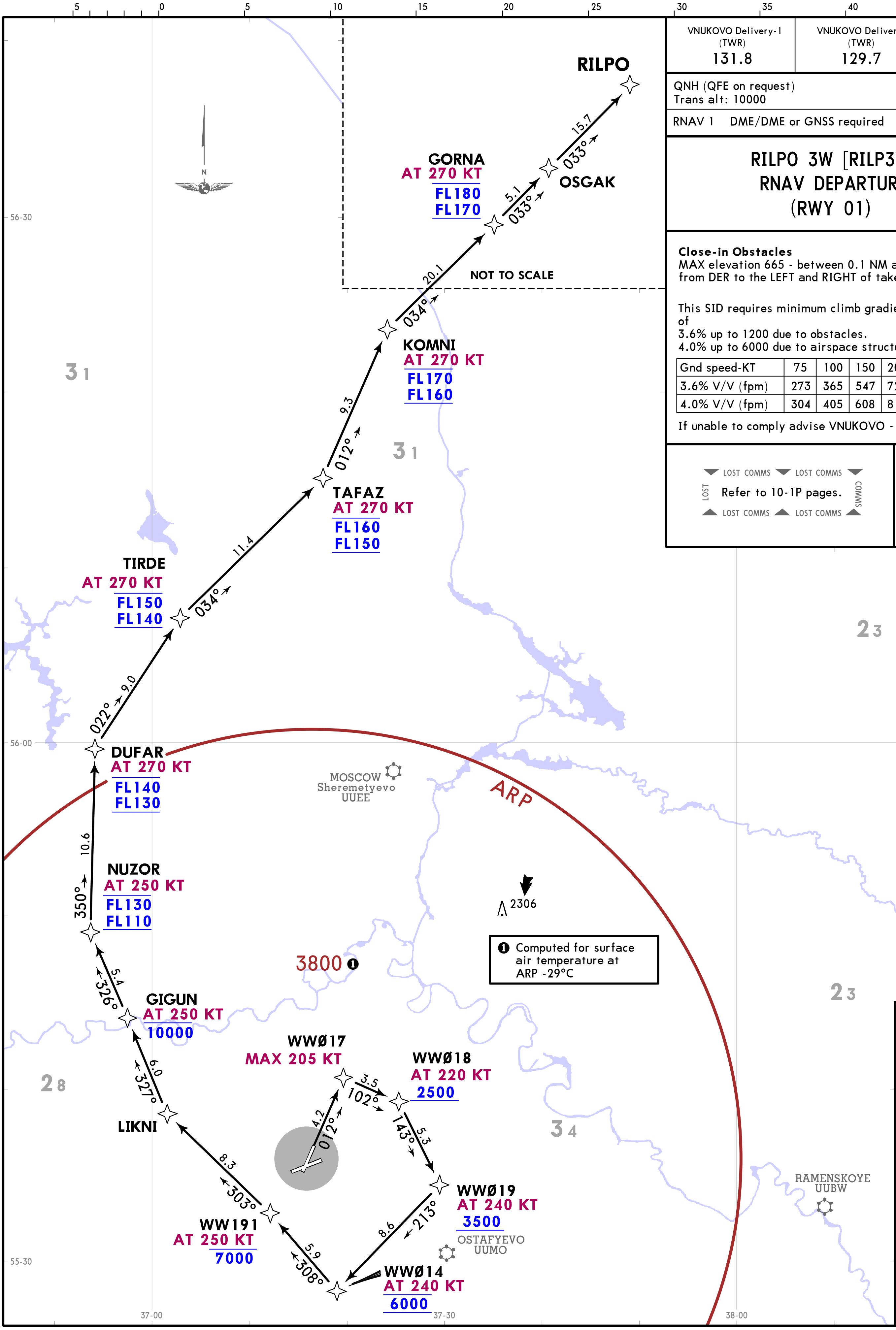
This SID requires minimum climb gradient of
3.6% up to 1200 due to obstacles.
4.0% up to 6000 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
3.6% V/V (fpm)	273	365	547	729	911	1094
4.0% V/V (fpm)	304	405	608	810	1013	1215

If unable to comply advise VNUKOVO - Delivery

LOST COMMS LOST COMMS
LOST Refer to 10-1P pages. COMMS
LOST COMMS LOST COMMS

FEET QNH	METERS (QFE)
1200	(175)
2500	(570)
3500	(875)
6000	(1640)
7000	(1945)
10000	(2860)



RILPO 3W [RILP3W]
(RWY 01)
RNAV DEPARTURE

CHANGES: Initial climb clearance withdrawn.

CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)
131.8	129.7

Apt Elev 685	QNH (QFE on request) Trans alt: 10000
	RNAV 1 DME/DME or GNSS required

RILPO 3Y [RILP3Y] (RWY 19)
RILPO 3Z [RILP3Z] (RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.

These SIDs require minimum climb gradients of
RILPO 3Y: 4.1% up to 7000 due to airspace structure.
RILPO 3Z: 4.2% up to 7000 due to airspace structure.

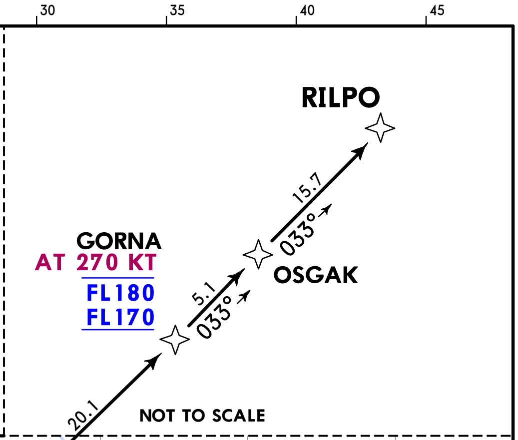
Gnd speed-KT	75	100	150	200	250	300
4.1% V/V (fpm)	311	415	623	830	1038	1246
4.2% V/V (fpm)	319	425	638	851	1063	1276

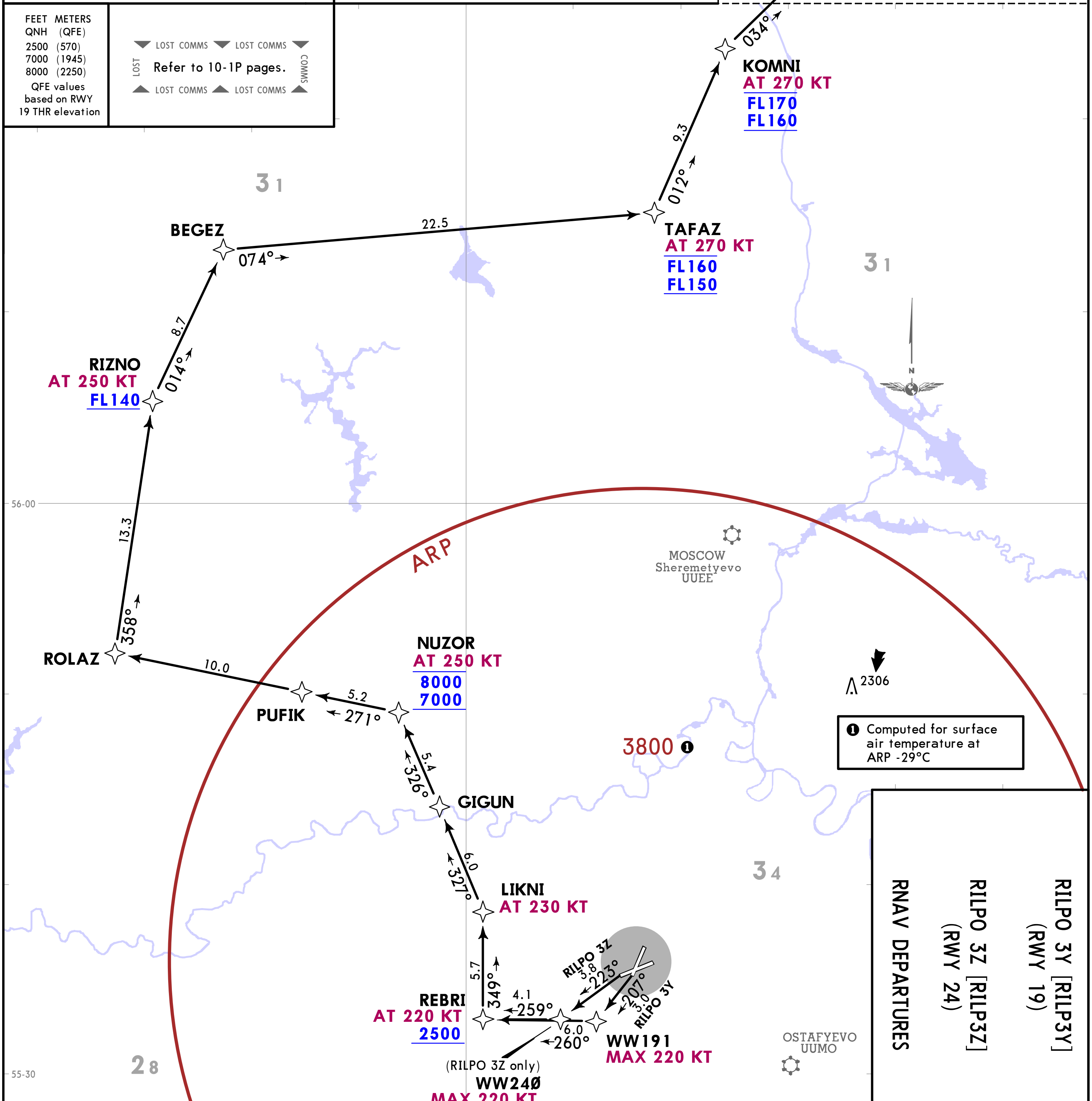
If unable to comply advise VNUKOVO - Delivery

FEET	METERS
QNH	(QFE)
2500	(570)
7000	(1945)
8000	(2250)

QFE values based on RWY 19 THR elevation

LOST COMMS LOST COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS





RILPO 3Y [RILP3Y] (RWY 19)
RILPO 3Z [RILP3Z] (RWY 24)
RNAV DEPARTURES

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

KOGOM 3A [KOGO3A]
OLMUN 3A [OLMU3A]
(RWY 01)

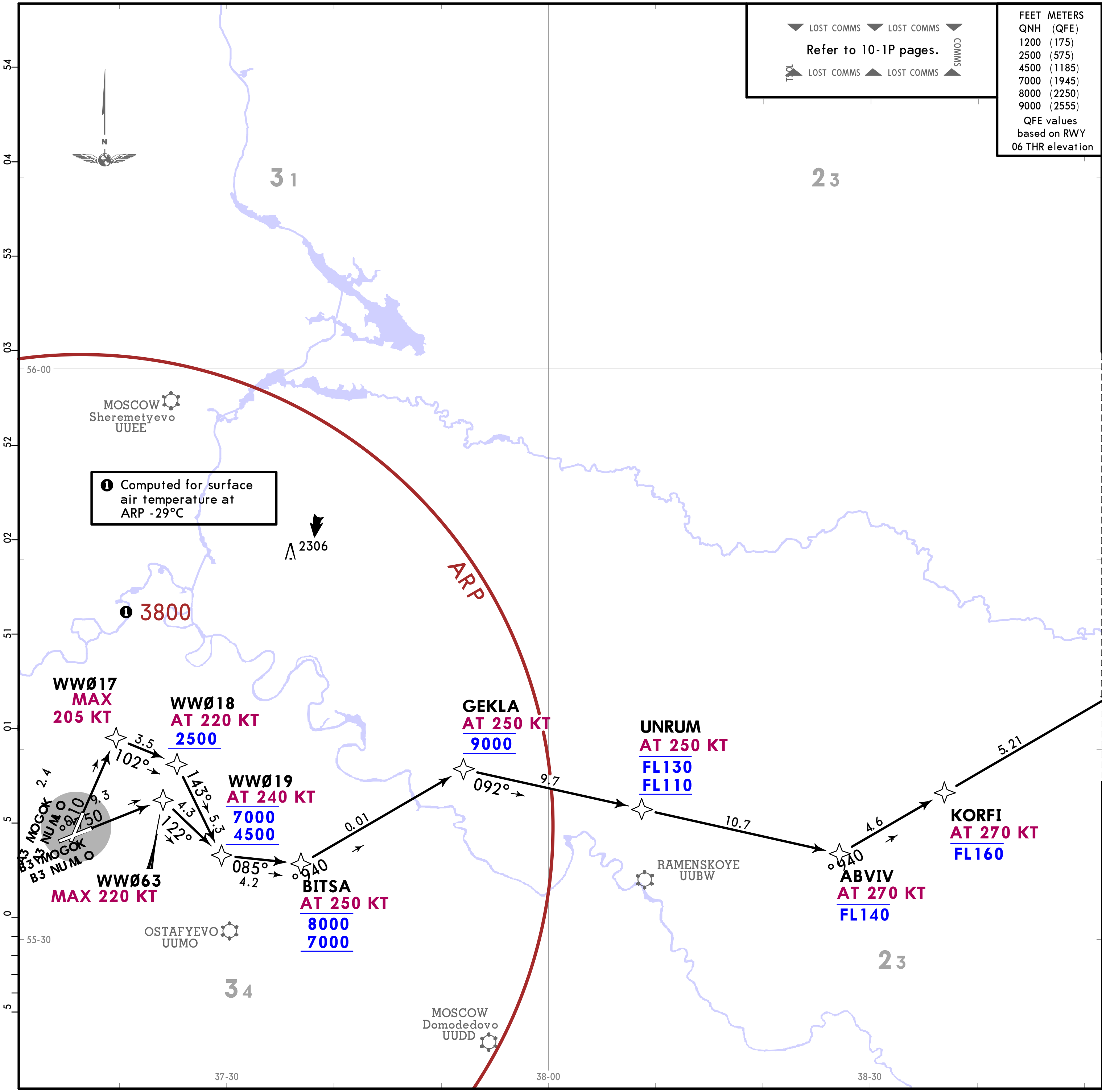
KOGOM 3B [KOGO3B]
OLMUN 3B [OLMU3B]
(RWY 06)

RNAV DEPARTURES

FEET QNH	METERS (QFE)
1200	(175)
2500	(575)
4500	(1185)
7000	(1945)
8000	(2250)
9000	(2555)

QFE values based on RWY 06 THR elevation

LOST COMMS — Refer to 10-1P pages.



Close-in Obstacles
RWY 01: MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 06: MAX elevation 746 - group of obstacles at 0.4 NM from DER to the RIGHT of take-off heading.

These SIDs require minimum climb gradients of
KOGOM 3A, OLMUN 3A:
3.6% up to 1200 due to obstacles.
6.2% up to 7000 due to airspace structure.
KOGOM 3B, OLMUN 3B:
8.5% up to 7000 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
3.6% V/V (fpm)	273	365	547	729	911	1094
6.2% V/V (fpm)	471	628	942	1256	1570	1884
8.5% V/V (fpm)	646	861	1291	1722	2152	2582

If unable to comply advise VNUKOVO - Delivery

CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

FEET QNH	METERS (QFE)
2500	(570)
5000	(1335)
7000	(1945)
8000	(2250)
9000	(2555)

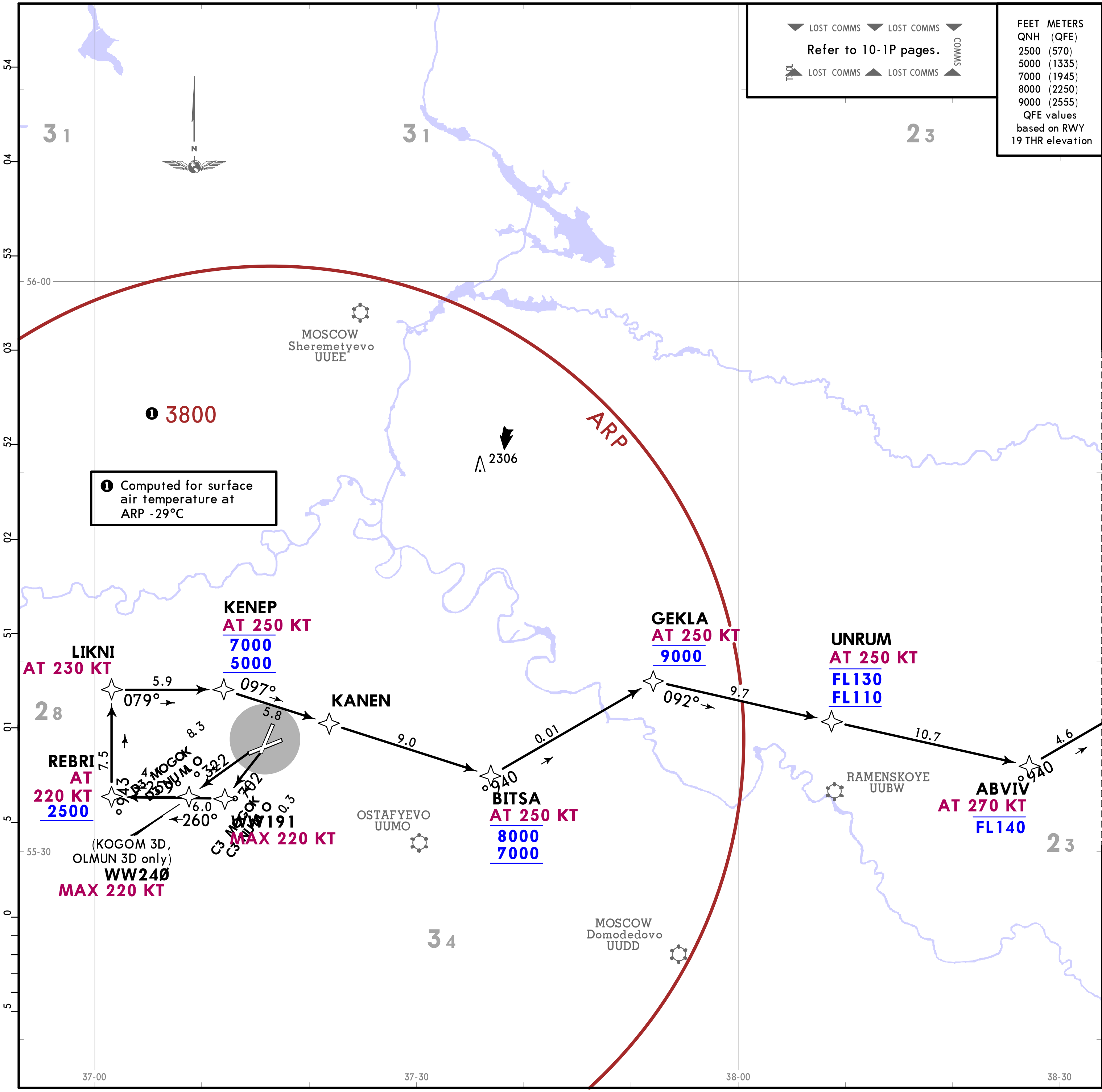
QFE values based on RWY 19 THR elevation

LOST COMMS LOST COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS

KOGOM 3C [KOGO3C]
OLMUN 3C [OLMU3C]
(RWY 19)

KOGOM 3D [KOGO3D]
OLMUN 3D [OLMU3D]
(RWY 24)

RNAV DEPARTURES



Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.

These SIDs require a minimum climb gradient of
4.0% up to 5000 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
4.0% V/V (fpm)	304	405	608	810	1013	1215

If unable to comply advise VNUKOVO - Delivery

CHANGES: Initial climb clearance withdrawn.

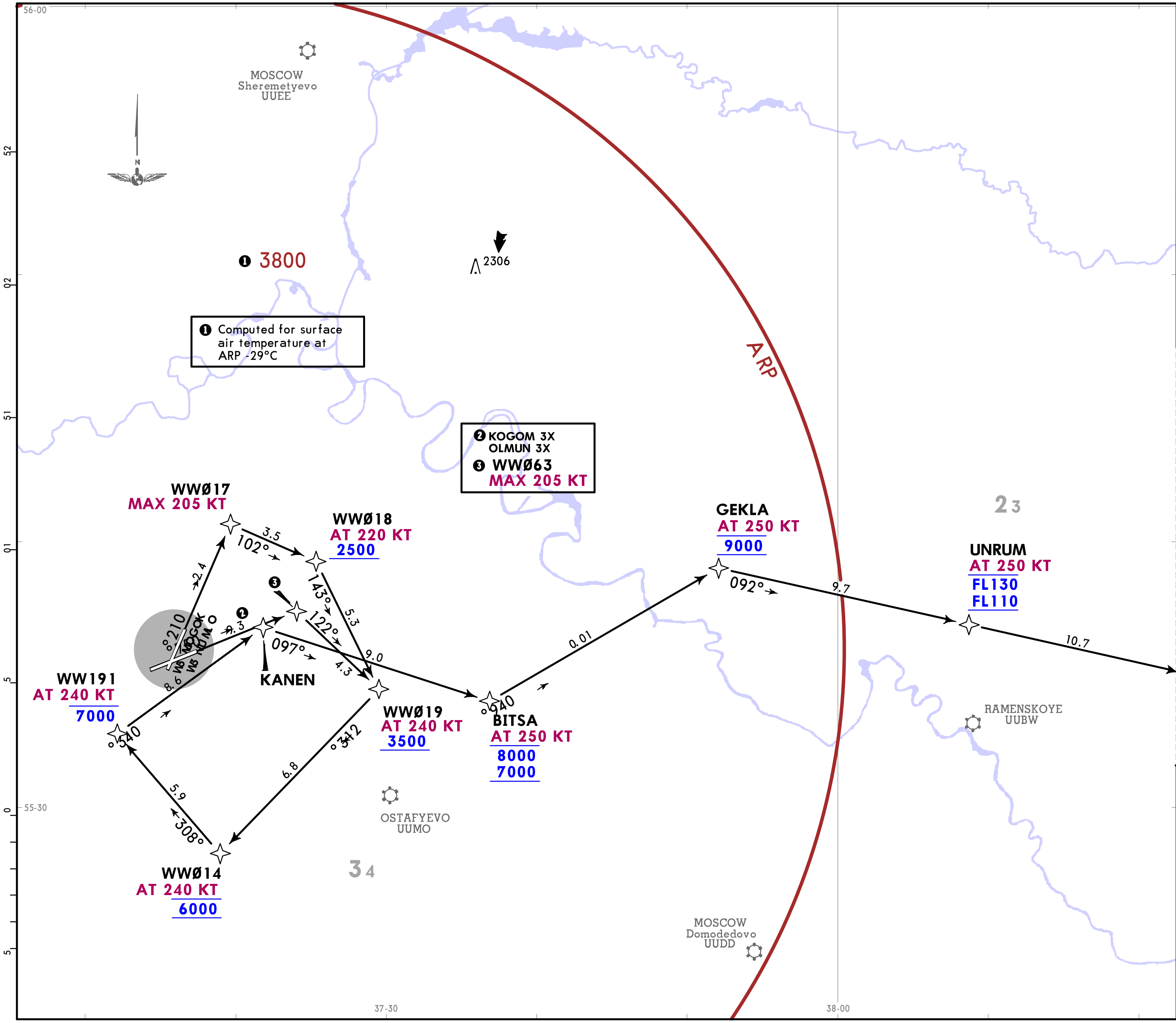
VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev	QNH (QFE on request) Trans alt: 10000
131.8	129.7	685	RNAV 1 DME/DME or GNSS required

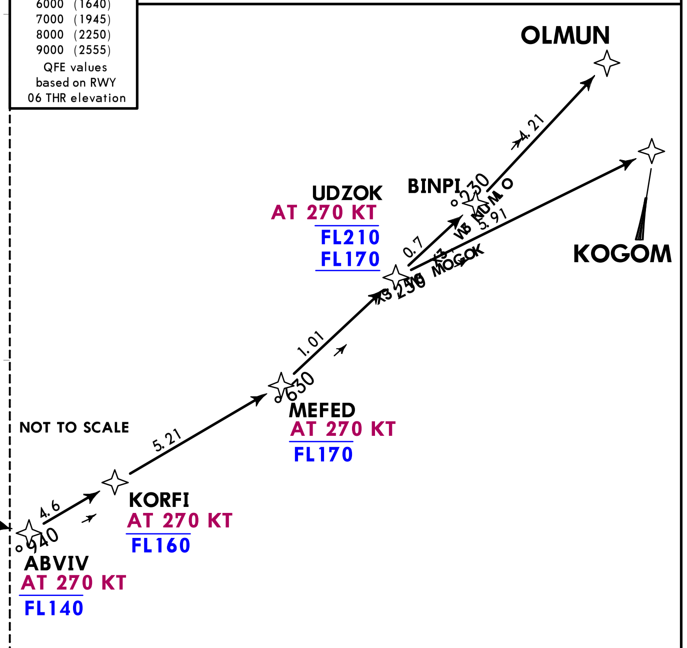
KOGOM 3W [KOGO3W] OLMUN 3W [OLMU3W] (RWY 01)
KOGOM 3X [KOGO3X] OLMUN 3X [OLMU3X] (RWY 06)
RNAV DEPARTURES

FEET QNH	METERS (QFE)
1200	(175)
2500	(575)
3500	(880)
6000	(1640)
7000	(1945)
8000	(2250)
9000	(2555)

QFE values based on RWY 06 THR elevation

LOST COMMS — Refer to 10-1P pages.





Close-in Obstacles
RWY 01: MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 06: MAX elevation 746 - group of obstacles at 0.4 NM from DER to the RIGHT of take-off heading.

These SIDs require minimum climb gradients of
KOGOM 3W, KOGOM 3X, OLMUN 3W, OLMUN 3X:
5.3% up to 6000 due to airspace structure.
KOGOM 3W, OLMUN 3W:
3.6% up to 1200 due to obstacles.

Gnd speed-KT	75	100	150	200	250	300
3.6% V/V (fpm)	273	365	547	729	911	1094
5.3% V/V (fpm)	403	537	805	1073	1342	1610

If unable to comply advise VNUKOVO - Delivery

CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

EMGAS 3A [EMGA3A]
TOGMO 3A [TOGM3A]
(RWY 01)

EMGAS 3B [EMGA3B]
TOGMO 3B [TOGM3B]
(RWY 06)

RNAV DEPARTURES

Close-in Obstacles
RWY 01: MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 06: MAX elevation 746 - group of obstacles at 0.4 NM from DER to the RIGHT of take-off heading.

These SIDs require minimum climb gradients of
EMGAS 3A, EMGAS 3B, TOGMO 3A, TOGMO 3B:
5.4% up to 6000 due to airspace structure.
EMGAS 3A & TOGMO 3A:
3.6% up to 1200 due to obstacles.

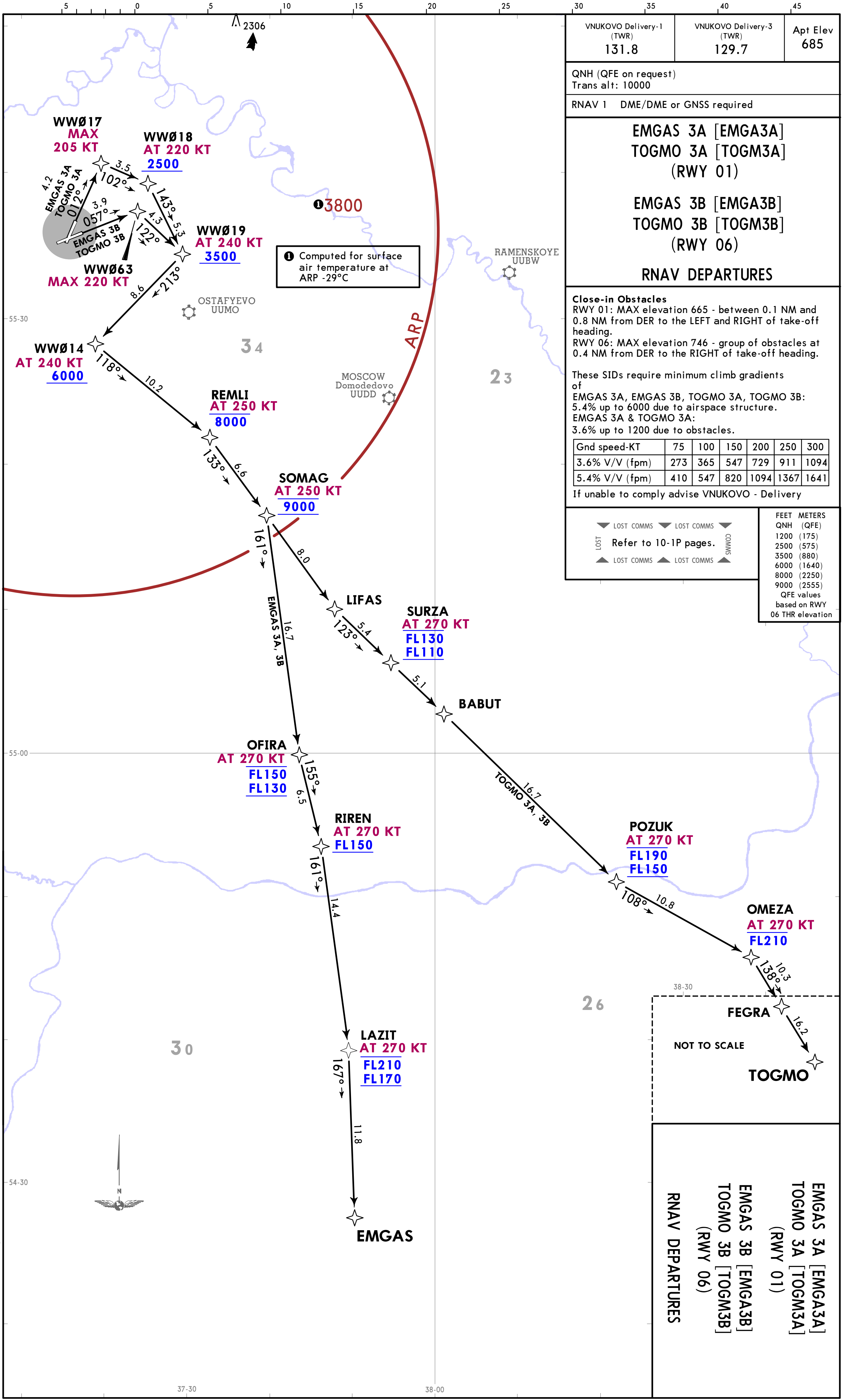
Gnd speed-KT	75	100	150	200	250	300
3.6% V/V (fpm)	273	365	547	729	911	1094
5.4% V/V (fpm)	410	547	820	1094	1367	1641

If unable to comply advise VNUKOVO - Delivery

LOST COMMS LOST COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS

FEET QNH	METERS (QFE)
1200	(175)
2500	(575)
3500	(880)
6000	(1640)
8000	(2250)
9000	(2555)

QFE values based on RWY 06 THR elevation



CHANGES: Initial climb clearance withdrawn, chart redrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

EMGAS 3C [EMGA3C]
TOGMO 3C [TOGM3C]
(RWY 19)
EMGAS 3D [EMGA3D]
TOGMO 3D [TOGM3D]
(RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.

These SIDs require minimum climb gradients of
8.2% up to 8000 due to airspace structure.

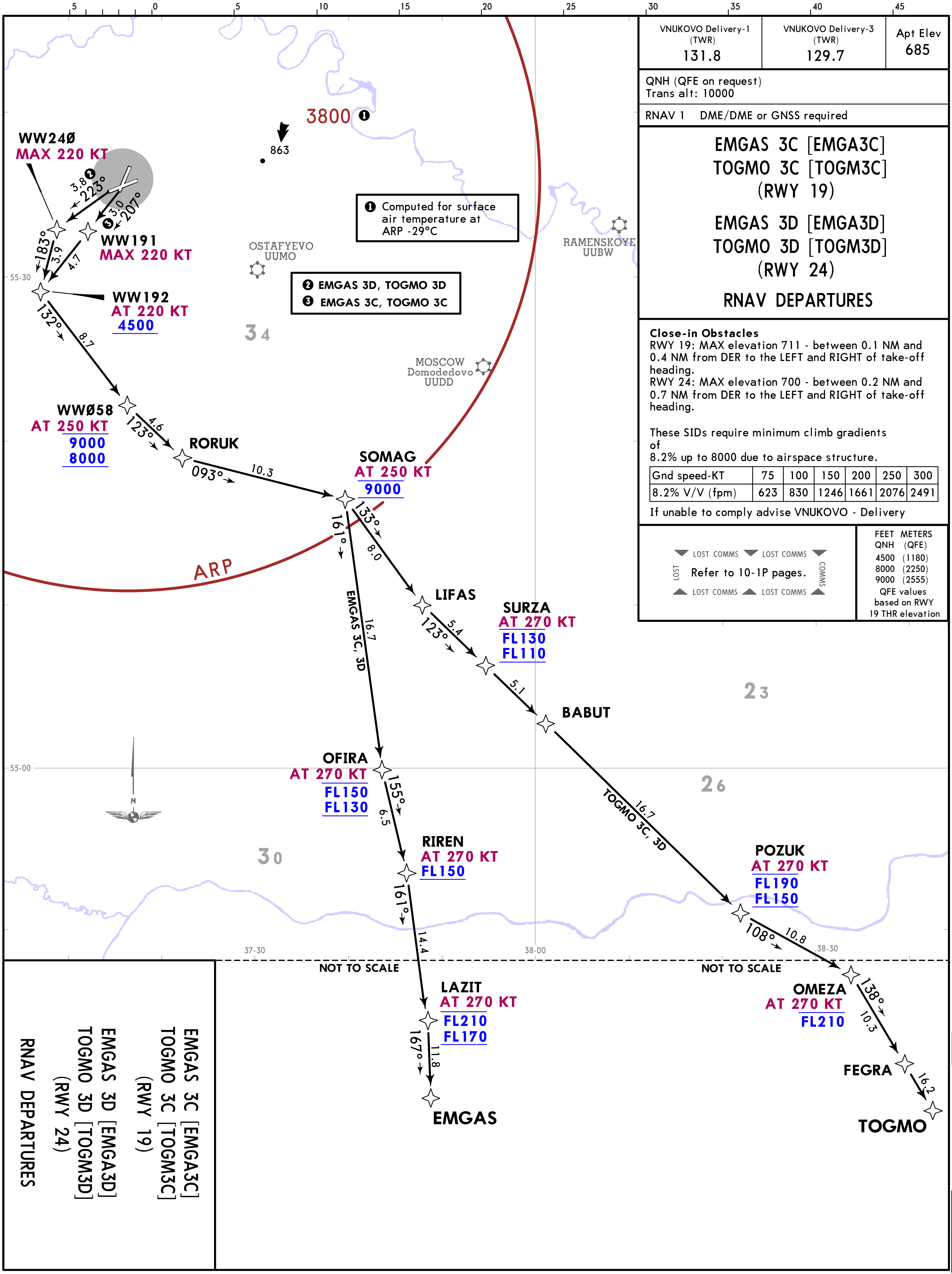
Gnd speed-KT	75	100	150	200	250	300
8.2% V/V (fpm)	623	830	1246	1661	2076	2491

If unable to comply advise VNUKOVO - Delivery

LOST COMMS LOST COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS

FEET	METERS
QNH	(QFE)
4500	(1180)
8000	(2250)
9000	(2555)

QFE values based on RWY 19 THR elevation



EMGAS 3C [EMGA3C]
TOGMO 3C [TOGM3C]
(RWY 19)
EMGAS 3D [EMGA3D]
TOGMO 3D [TOGM3D]
(RWY 24)
RNAV DEPARTURES

CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

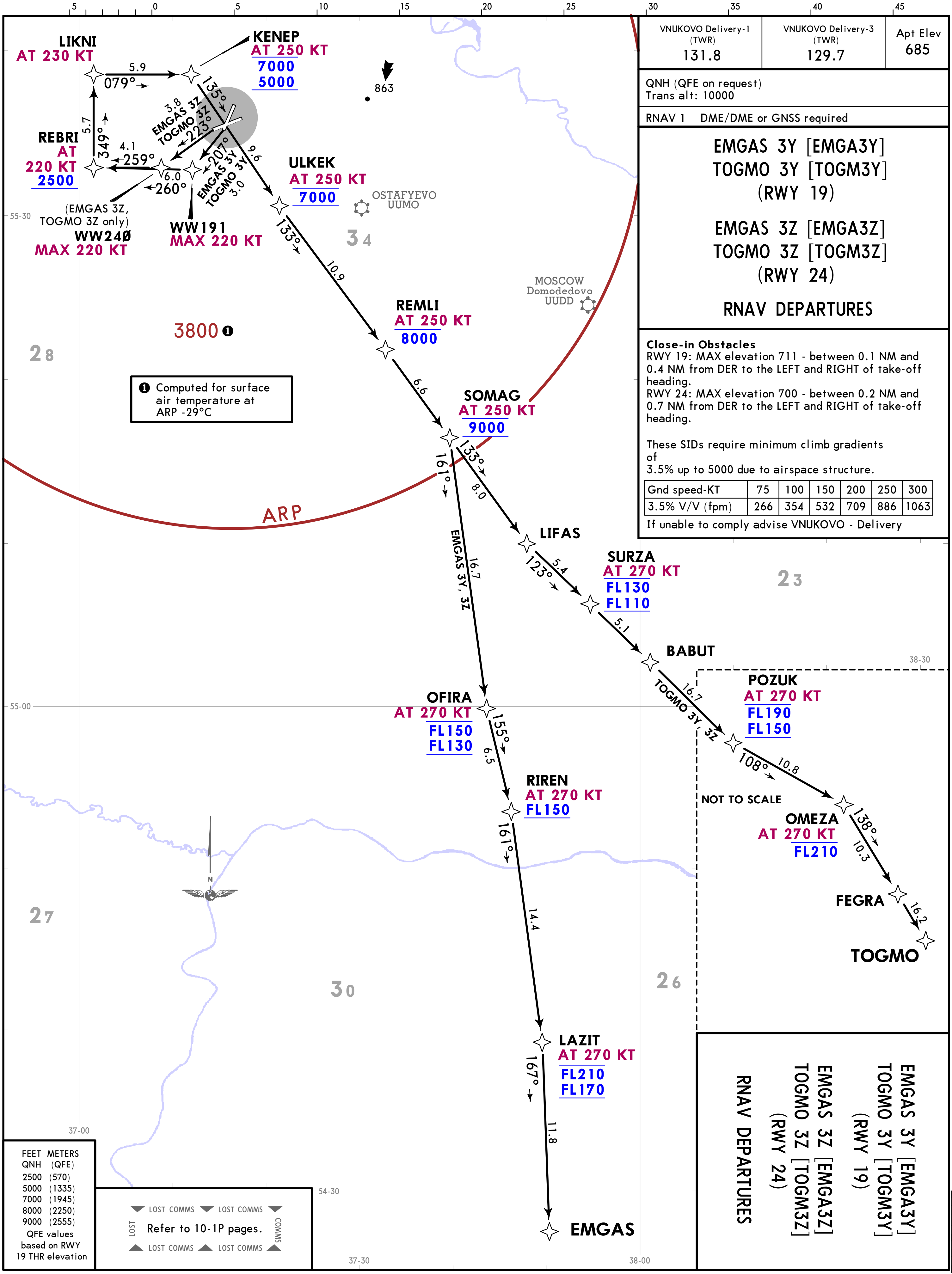
EMGAS 3Y [EMGA3Y]
TOGMO 3Y [TOGM3Y]
(RWY 19)
EMGAS 3Z [EMGA3Z]
TOGMO 3Z [TOGM3Z]
(RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.

These SIDs require minimum climb gradients of
3.5% up to 5000 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
3.5% V/V (fpm)	266	354	532	709	886	1063

If unable to comply advise VNUKOVO - Delivery



VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

SOTOG 3A [SOTO3A] (RWY 01)

SOTOG 3B [SOTO3B] (RWY 06)

RNAV DEPARTURES

Close-in Obstacles
RWY 01: MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 06: MAX elevation 746 - group of obstacles at 0.4 NM from DER to the RIGHT of take-off heading.

These SIDs require minimum climb gradients of
SOTOG 3A, SOTOG 3B:
6.5% up to FL120 due to airspace structure.
SOTOG 3A: 3.6% up to 1200 due to obstacles.

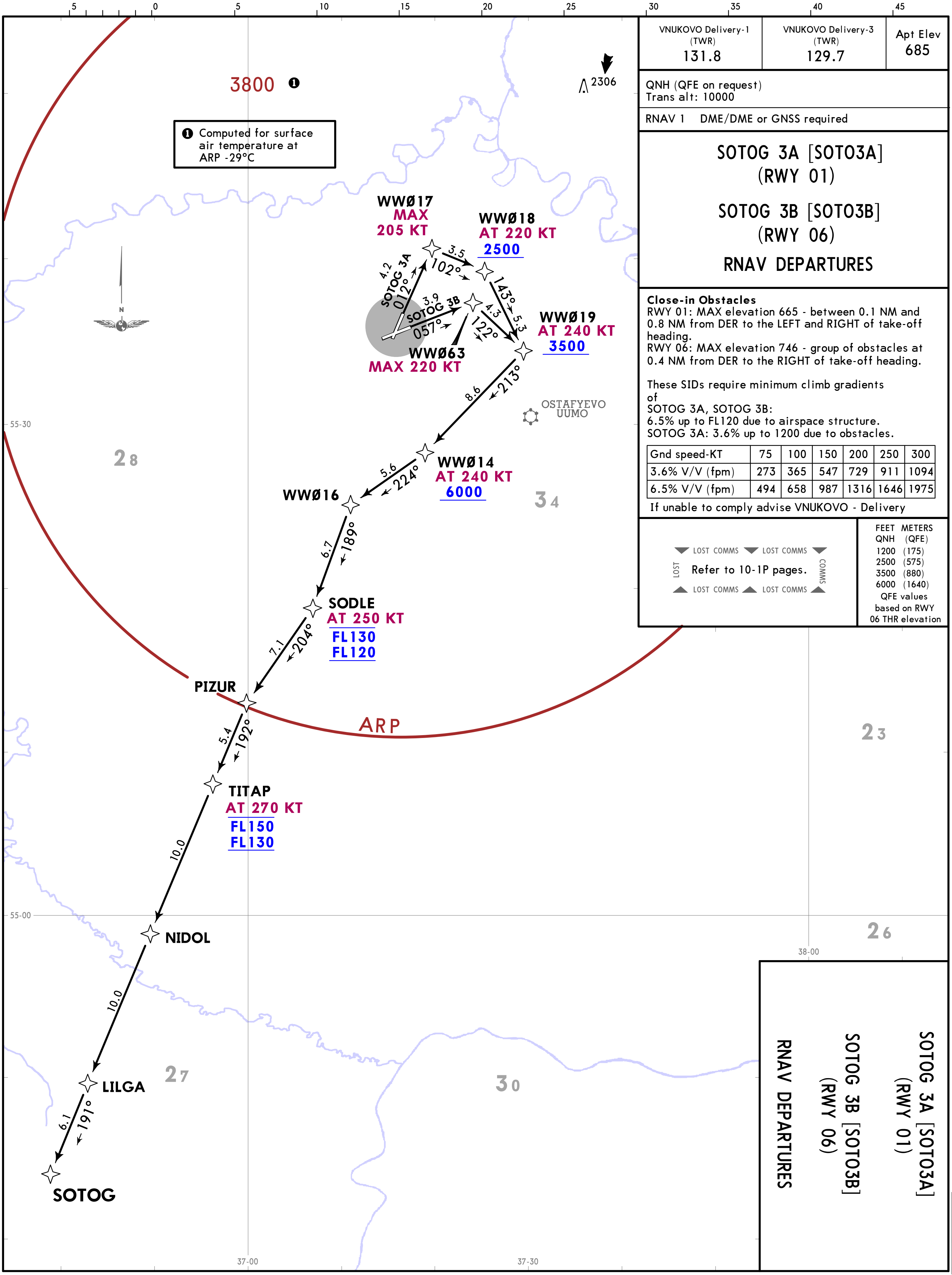
Gnd speed-KT	75	100	150	200	250	300
3.6% V/V (fpm)	273	365	547	729	911	1094
6.5% V/V (fpm)	494	658	987	1316	1646	1975

If unable to comply advise VNUKOVO - Delivery

LOST COMMS LOST COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS

FEET QNH	METERS (QFE)
1200	(175)
2500	(575)
3500	(880)
6000	(1640)

QFE values based on RWY 06 THR elevation



CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

SOTOG 3C [SOTO3C] (RWY 19)
SOTOG 3D [SOTO3D] (RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.

These SIDs require a minimum climb gradient of
5.0% up to FL120 due to airspace structure.

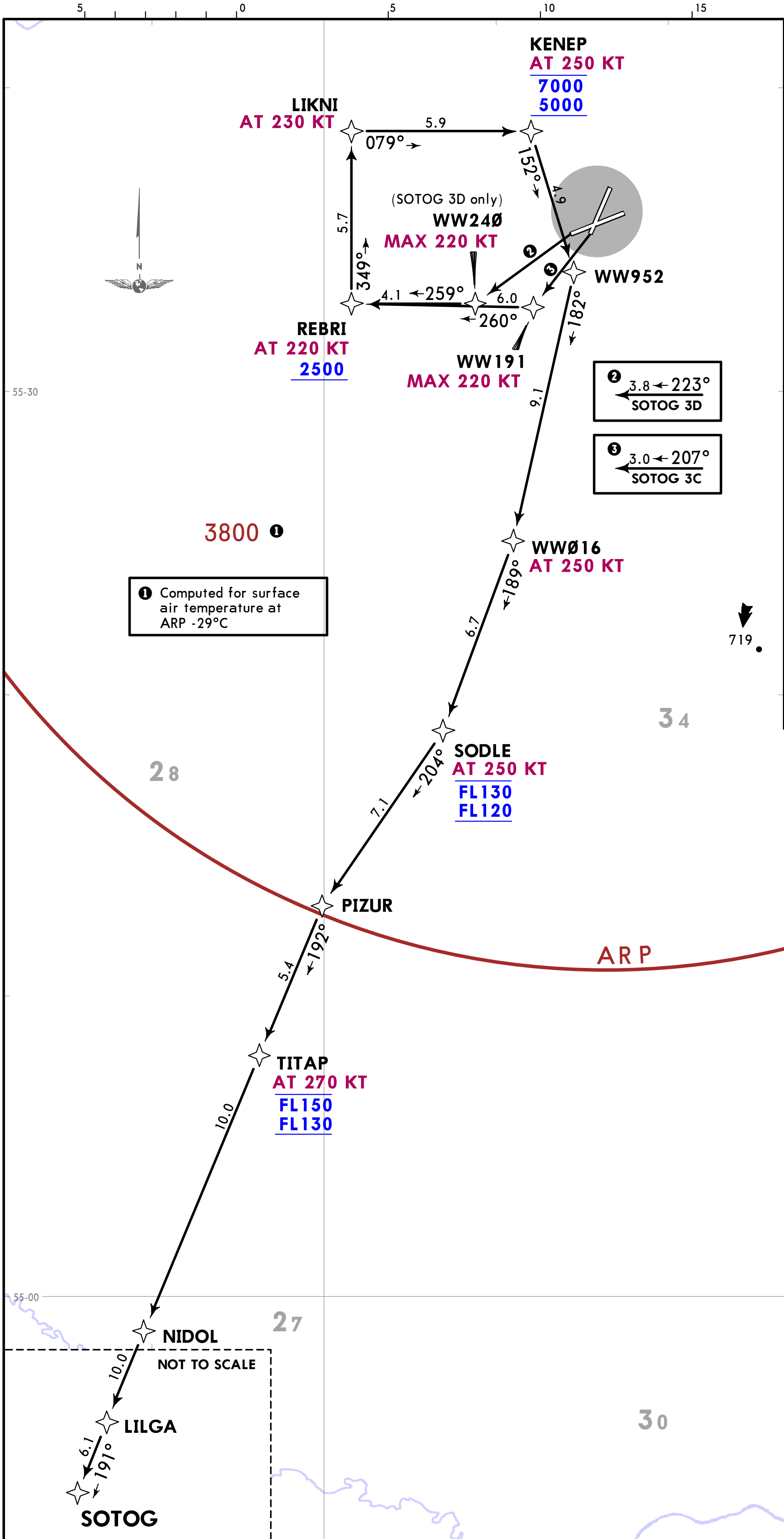
Gnd speed-KT	75	100	150	200	250	300
5.0% V/V (fpm)	380	506	760	1013	1266	1519

If unable to comply advise VNUKOVO - Delivery

LOST COMMS Refer to 10-1P pages.

FEET	METERS
QNH	(QFE)
2500	(570)
5000	(1335)
7000	(1945)

QFE values based on RWY 19 THR elevation



UUWW/VKO
VNUKOVO

RNAV DEPARTURES
SOTOG 3D [SOTO3D] (RWY 24)
SOTOG 3C [SOTO3C] (RWY 19)

CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

SOTOG 3W [SOTO3W] (RWY 01)
SOTOG 3X [SOTO3X] (RWY 06)
RNAV DEPARTURES

Close-in Obstacles
RWY 01: MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 06: MAX elevation 746 - group of obstacles at 0.4 NM from DER to the RIGHT of take-off heading.

SOTOG 3W:
This SID requires minimum climb gradients of
3.6% up to 1200 due to obstacles.
4.1% up to 6000 due to airspace structure.

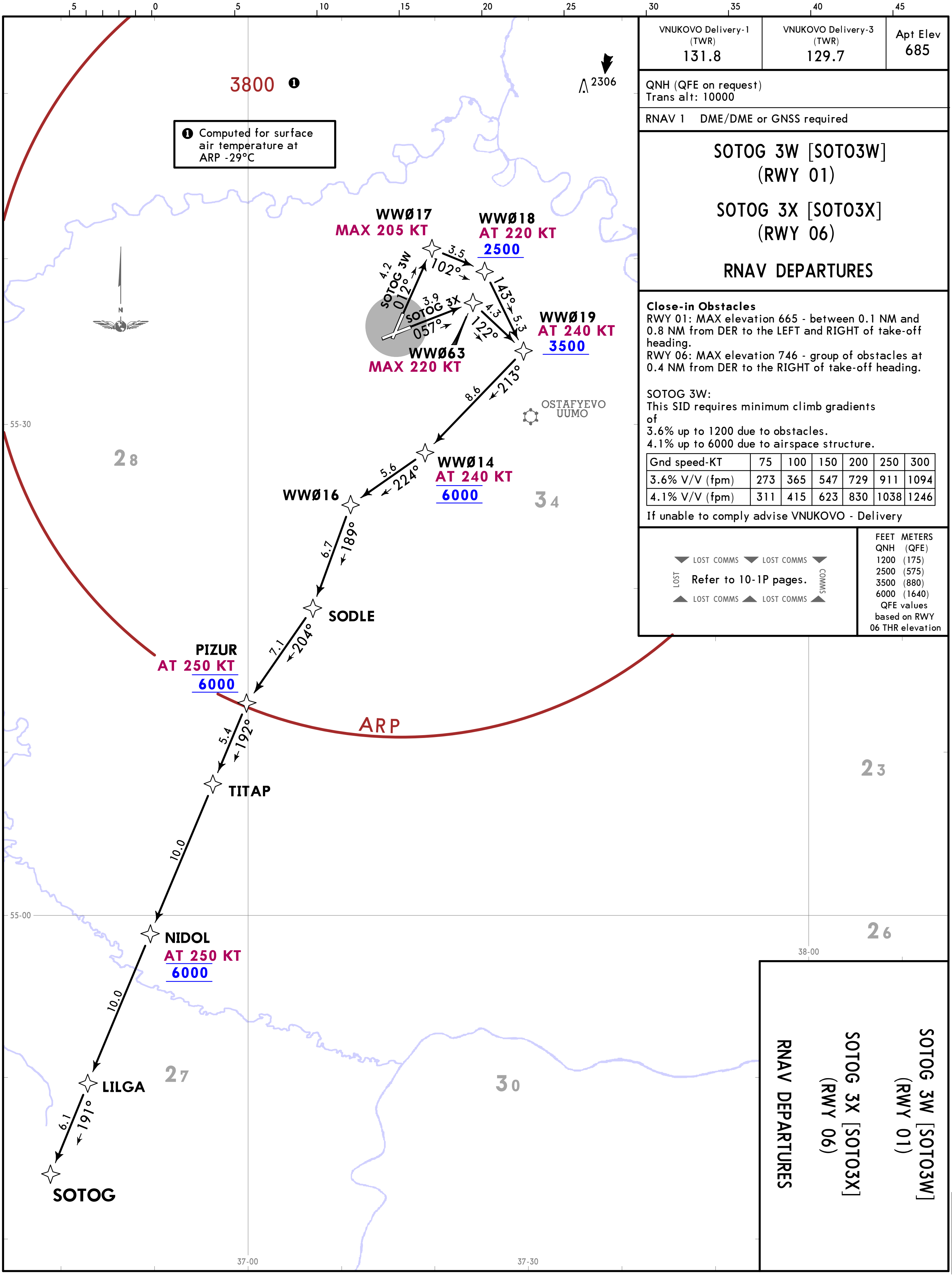
Gnd speed-KT	75	100	150	200	250	300
3.6% V/V (fpm)	273	365	547	729	911	1094
4.1% V/V (fpm)	311	415	623	830	1038	1246

If unable to comply advise VNUKOVO - Delivery

LOST COMMS LOST COMMS
Refer to 10-1P pages.

FEET QNH	METERS (QFE)
1200	(175)
2500	(575)
3500	(880)
6000	(1640)

QFE values based on RWY 06 THR elevation



SOTOG 3W [SOTO3W] (RWY 01)
SOTOG 3X [SOTO3X] (RWY 06)
RNAV DEPARTURES

CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

SOTOG 3Y [SOTO3Y] (RWY 19)
SOTOG 3Z [SOTO3Z] (RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.

These SIDs require a minimum climb gradient of 4.2% up to 2500 due to airspace structure.

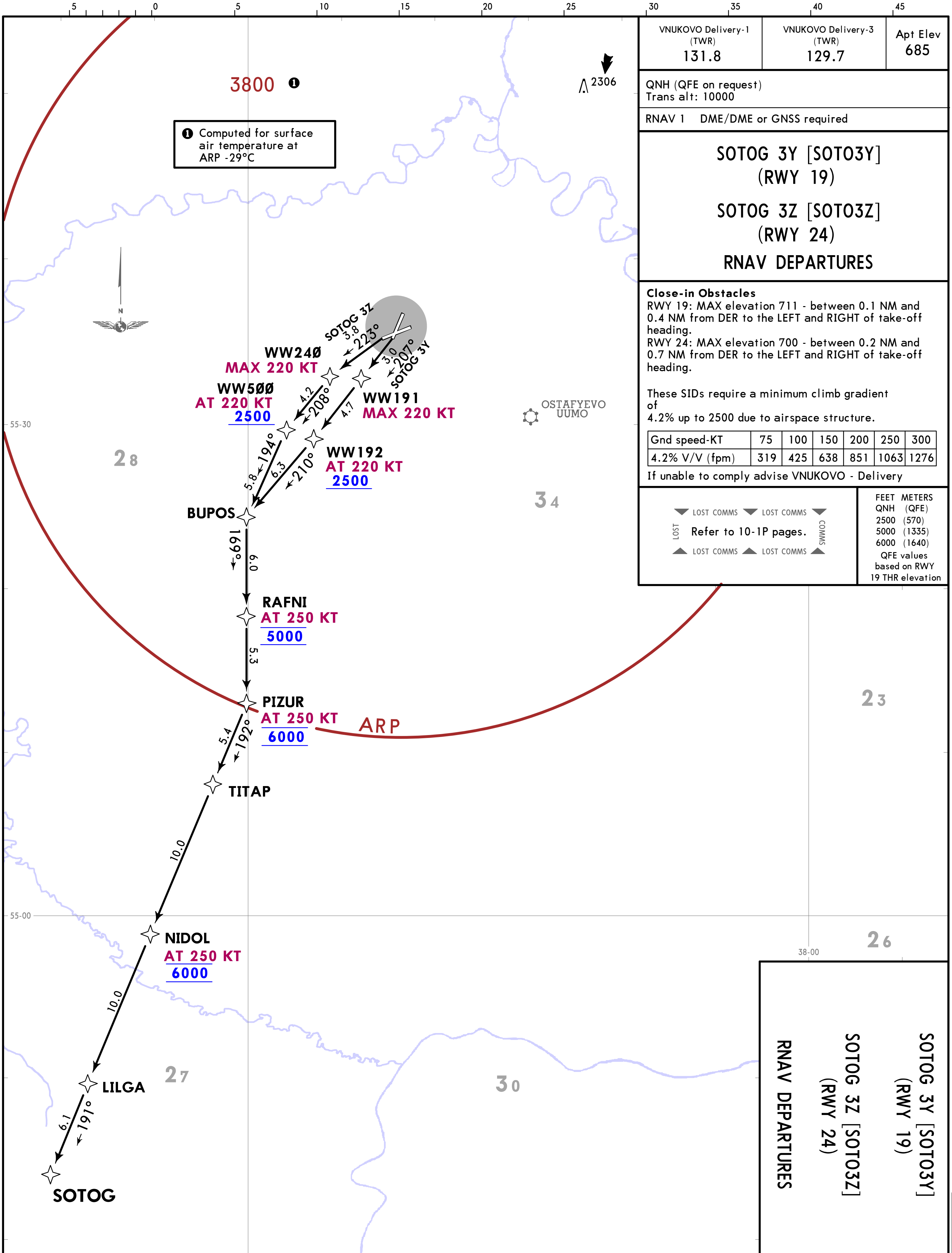
Gnd speed-KT	75	100	150	200	250	300
4.2% V/V (fpm)	319	425	638	851	1063	1276

If unable to comply advise VNUKOVO - Delivery

LOST COMMS LOST COMMS
LOST COMMS Refer to 10-1P pages. COMMS
LOST COMMS LOST COMMS

FEET QNH	METERS (QFE)
2500	(570)
5000	(1335)
6000	(1640)

QFE values based on RWY 19 THR elevation



SOTOG 3Y [SOTO3Y] (RWY 19)
SOTOG 3Z [SOTO3Z] (RWY 24)
RNAV DEPARTURES

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

POKAG 3A [POKA3A]
SUNUN 3A [SUNU3A]
(RWY 01)

CAT A, B & C

POKAG 3B [POKA3B]
SUNUN 3B [SUNU3B]
(RWY 06)

RNAV DEPARTURES

Close-in Obstacles
RWY 01: MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 06: MAX elevation 746 - group of obstacles at 0.4 NM from DER to the RIGHT of take-off heading.
These SIDs require minimum climb gradients of
POKAG 3A, SUNUN 3A:
5.1% up to 1200 due to obstacles.
8.8% up to FL120 due to airspace structure.
POKAG 3B, SUNUN 3B:
5.7% up to 6000 due to airspace structure.

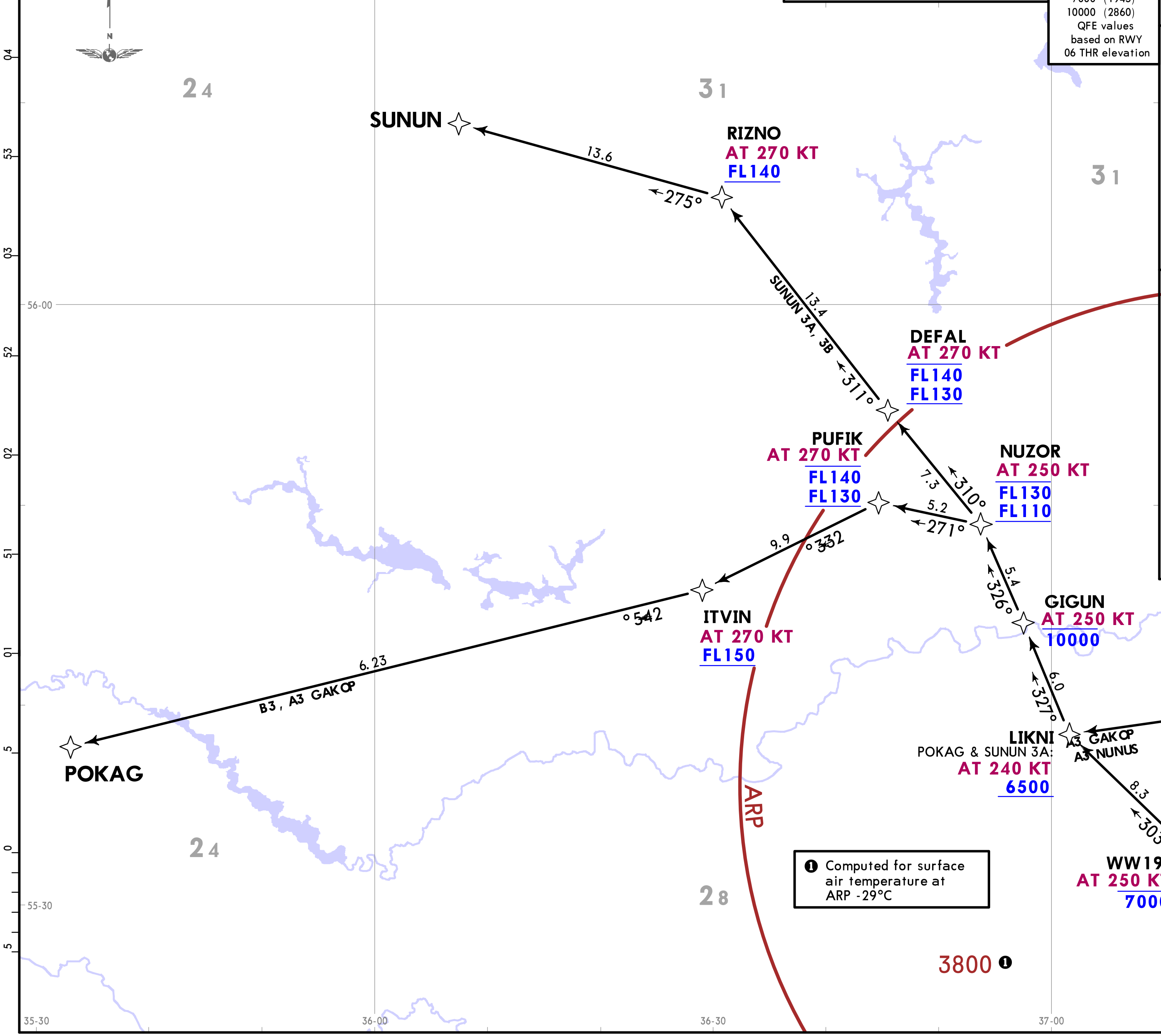
Gnd speed-KT	75	100	150	200	250	300
5.1% V/V (fpm)	387	516	775	1033	1291	1549
5.7% V/V (fpm)	433	577	866	1154	1443	1732
8.8% V/V (fpm)	668	891	1337	1782	2228	2673

If unable to comply advise VNUKOVO - Delivery

FEET QNH	METERS (QFE)
1100	(145)
1200	(175)
3500	(875)
6000	(1640)
6500	(1795)
7000	(1945)
10000	(2860)

QFE values based on RWY 06 THR elevation

LOST COMMS LOST COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS



CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

FEET QNH	METERS (QFE)
2500	(570)
10000	(2855)

QFE values based on RWY 19 THR elevation

LOST COMMS LOST COMMS COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS

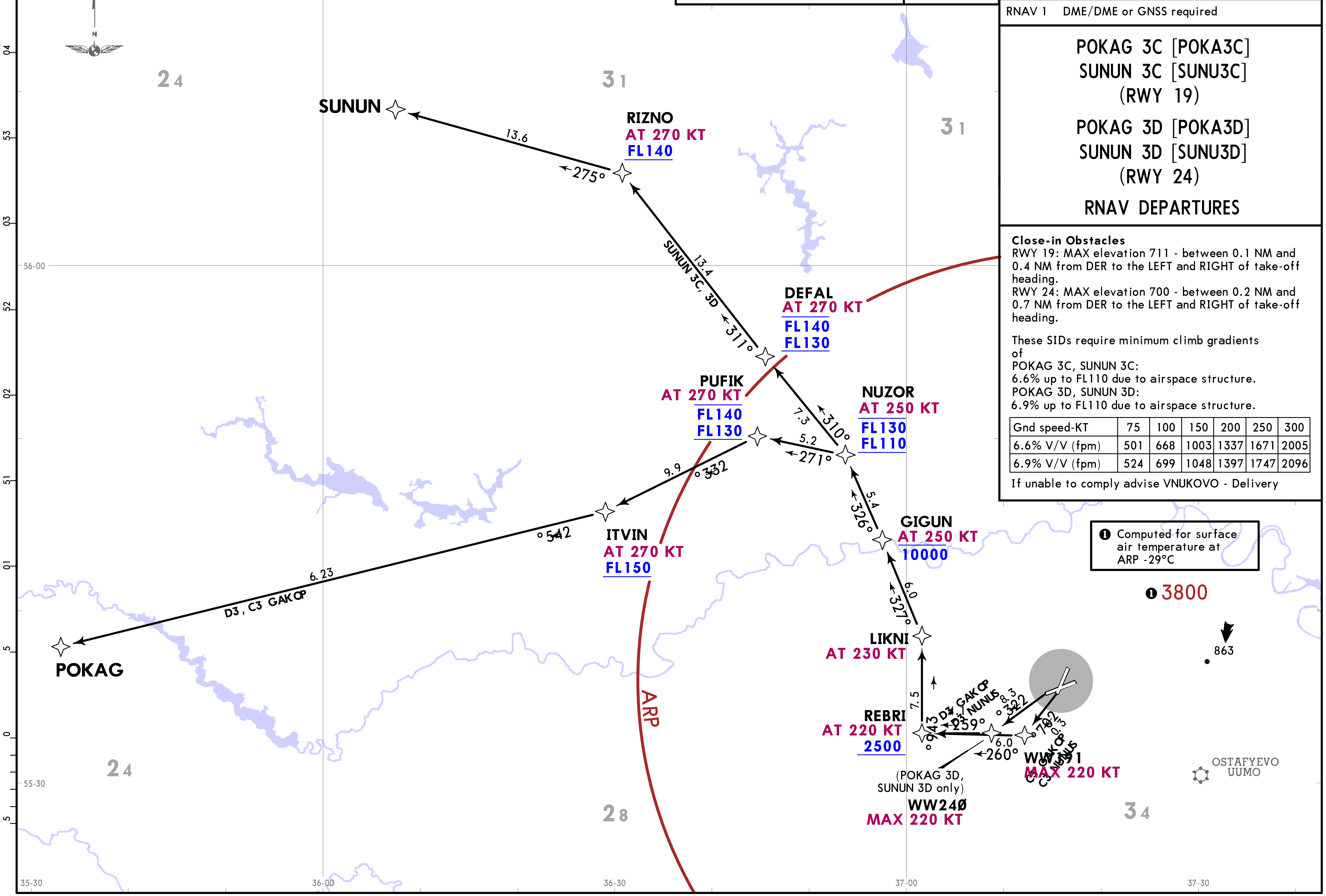
POKAG 3C [POKA3C]
SUNUN 3C [SUNU3C]
(RWY 19)
POKAG 3D [POKA3D]
SUNUN 3D [SUNU3D]
(RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.

These SIDs require minimum climb gradients of
POKAG 3C, SUNUN 3C:
6.6% up to FL110 due to airspace structure.
POKAG 3D, SUNUN 3D:
6.9% up to FL110 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
6.6% V/V (fpm)	501	668	1003	1337	1671	2005
6.9% V/V (fpm)	524	699	1048	1397	1747	2096

If unable to comply advise VNUKOVO - Delivery



CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

POKAG 3W [POKA3W]
SUNUN 3W [SUNU3W]
RNAV DEPARTURES
(RWY 01)

Close-in Obstacles
MAX elevation 665 - between 0.1 NM and 0.8 NM from DER to the LEFT and RIGHT of take-off heading.

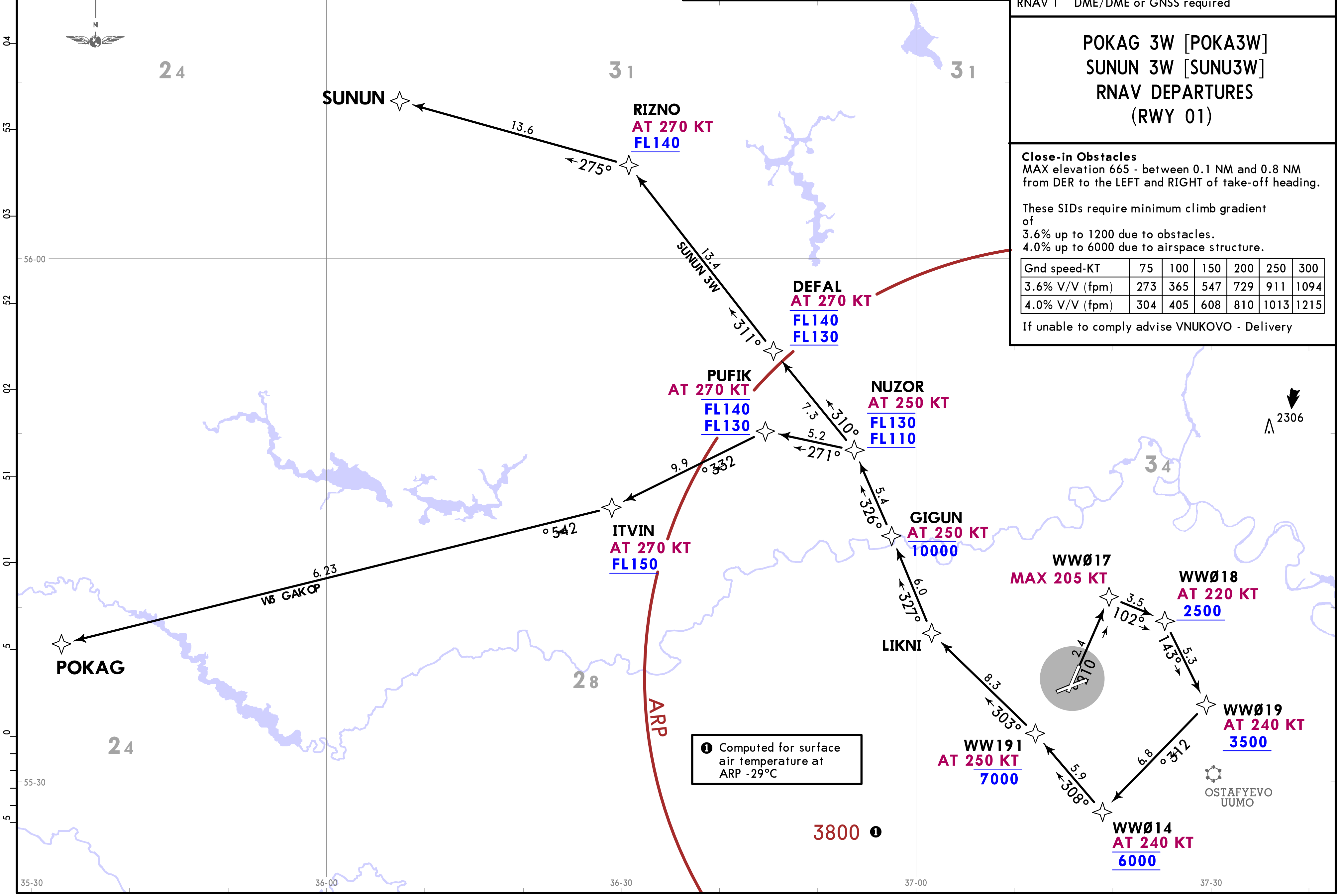
These SIDs require minimum climb gradient of
3.6% up to 1200 due to obstacles.
4.0% up to 6000 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
3.6% V/V (fpm)	273	365	547	729	911	1094
4.0% V/V (fpm)	304	405	608	810	1013	1215

If unable to comply advise VNUKOVO - Delivery

FEET QNH	METERS (QFE)
1200	(175)
2500	(570)
3500	(875)
6000	(1640)
7000	(1945)
10000	(2860)

LOST COMMS LOST COMMS COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS



❶ Computed for surface air temperature at ARP -29°C

CHANGES: Initial climb clearance withdrawn.

VNUKOVO Delivery-1 (TWR)	VNUKOVO Delivery-3 (TWR)	Apt Elev
131.8	129.7	685

QNH (QFE on request)
Trans alt: 10000

RNAV 1 DME/DME or GNSS required

FEET QNH	METERS (QFE)
2500	(570)
7000	(1945)
8000	(2250)

QFE values based on RWY 19 THR elevation

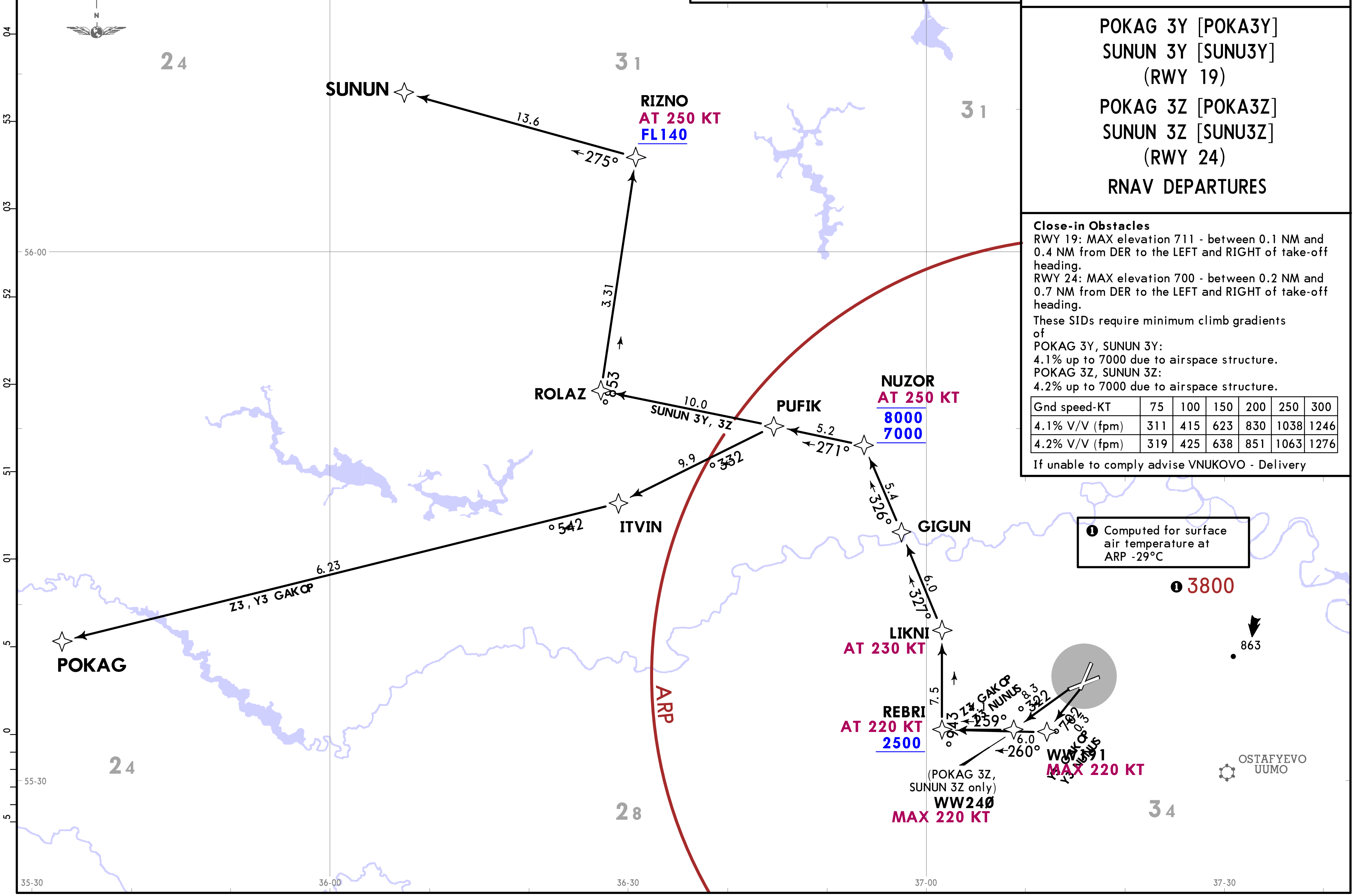
LOST COMMS LOST COMMS COMMS
Refer to 10-1P pages.
LOST COMMS LOST COMMS

POKAG 3Y [POKA3Y]
SUNUN 3Y [SUNU3Y]
(RWY 19)
POKAG 3Z [POKA3Z]
SUNUN 3Z [SUNU3Z]
(RWY 24)
RNAV DEPARTURES

Close-in Obstacles
RWY 19: MAX elevation 711 - between 0.1 NM and 0.4 NM from DER to the LEFT and RIGHT of take-off heading.
RWY 24: MAX elevation 700 - between 0.2 NM and 0.7 NM from DER to the LEFT and RIGHT of take-off heading.
These SIDs require minimum climb gradients of
POKAG 3Y, SUNUN 3Y:
4.1% up to 7000 due to airspace structure.
POKAG 3Z, SUNUN 3Z:
4.2% up to 7000 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
4.1% V/V (fpm)	311	415	623	830	1038	1246
4.2% V/V (fpm)	319	425	638	851	1063	1276

If unable to comply advise VNUKOVO - Delivery



Apt Elev 685	Trans alt: 10000 QNH (QFE on request)

FLIGHT ROUTES BETWEEN AERODROMES WITHIN MOSCOW TMA

1. Departure instructions provide ACFT vectoring to the significant point on the route (the first waypoint in the flight plan);
2. Flights within CTRs shall be carried out via waypoints, separated by letters DCT in the flight plan, to IAF of the destination aerodrome (in accordance with the information published for appropriate departure aerodrome);
3. Approach shall be executed from IAF of the destination aerodrome:
   - Moscow/Sheremetyevo - KEZVU (IAF)
   - Moscow/Domodedovo - ALBOR (IAF)
   - Moscow/Vnukovo - RORUK (IAF)
   - Ostafyevo - RORUK (IAF)
   - Ramenskoye - ODLOR (IAF).

Departure To	ROUTING
Moscow/ Sheremetyevo	LIKNI - GIGUN - NUZOR - PUFIK - ROLAZ - RIZNO - BEGEZ - LUNZA - EEØ48 - EEØ49 - EEØ5Ø - EEØ51 - TAFAZ - KEZVU (IAF).
Moscow/ Domodedovo	BITSA - IMZUP - KUPVE - NIDBE - IZVOK - IPKED - ZOVGO - ODZAG - GUFUZ - ALBOR (IAF).
Ostafyevo	BUPOS - ORSIF - MEZER - NALFI - RAMZA - UKABE - FIDOT - RORUK (IAF).
Ramenskoye	BITSA - IMZUP - GENKE - RT NDB - BW316 - BW317 - BW318 - BW319 - ODLOR (IAF).

NOISE ABATEMENT

LT minus 3 HOURS = UTC (Z)

GENERAL

Noise abatement procedures during take-off, climb and approach shall be employed by all ACFT.

Noise abatement procedures shall not be employed at the expense of compromising safety of flight operations.

Maintain the assigned SID and STAR routes, and in case of deviation from them, immediately join the assigned flight track.

ARRIVALS

Great rates of descent should be avoided (if possible) immediately prior to the final approach segment.

Change of ACFT configuration and flight speed within noise abatement procedures is permitted in accordance with the requirements of the Aeroplane Flight Manual.

Flying below ILS glide path is PROHIBITED, during instrument and visual approach.

Noise abatement procedures should not involve employment of speed greater than the indicated airspeed of descent prescribed in the Aeroplane Flight Manual.

USE OF REVERSE TRUST

Engines reverse thrust at idle power is recommended for ACFT landing at the AD from 2300-0700 LT except for safety related cases.

DEPARTURES

Runway threshold displacement must not be used as a noise abatement measure.

NADP 1 is employed for ACFT departing from RWY 01 and RWY 06, NADP 2 is employed for ACFT departing from RWY 19 and RWY 24.

ACFT with partial load shall take-off at rated engines power, in accordance with the Aeroplane Flight Manual.

Aircraft turn from 500' (150 m)to 1000' (300 m) AAL shall be carried out with a 15° bank, from 1000' (300 m) to 2960' (900 m) with a 20° bank and from 2960'(900 m) with a 25° bank or at angular speed of 3°/sec.

Change of flight direction (course) after take-off is permitted only after reaching 500' (150 m) AAL.

APU USAGE

ACFT APU run-up on designated stands not equipped with jet blast deflectors and sound barriers is PROHIBITED.

Also, use of ACFT APU between 2300-0700 LT is PROHIBITED.

Use of APU should be avoided and/or restricted on stands equipped with ACFT ground power units and preconditioned air systems from 2300-0700 LT after ACFT is parked on stand or before ACFT leaves the stand .

PREFERENTIAL RWY SYSTEM DURING NIGHT PERIOD

Within noise abatement procedures the term NIGHT indicates the period from 2300-0700 LT.

RWY 19 and RWY 06 should not be used as preferential RWYs for landing operations, RWY 01 should not be used as preferential RWY for take-off operations.

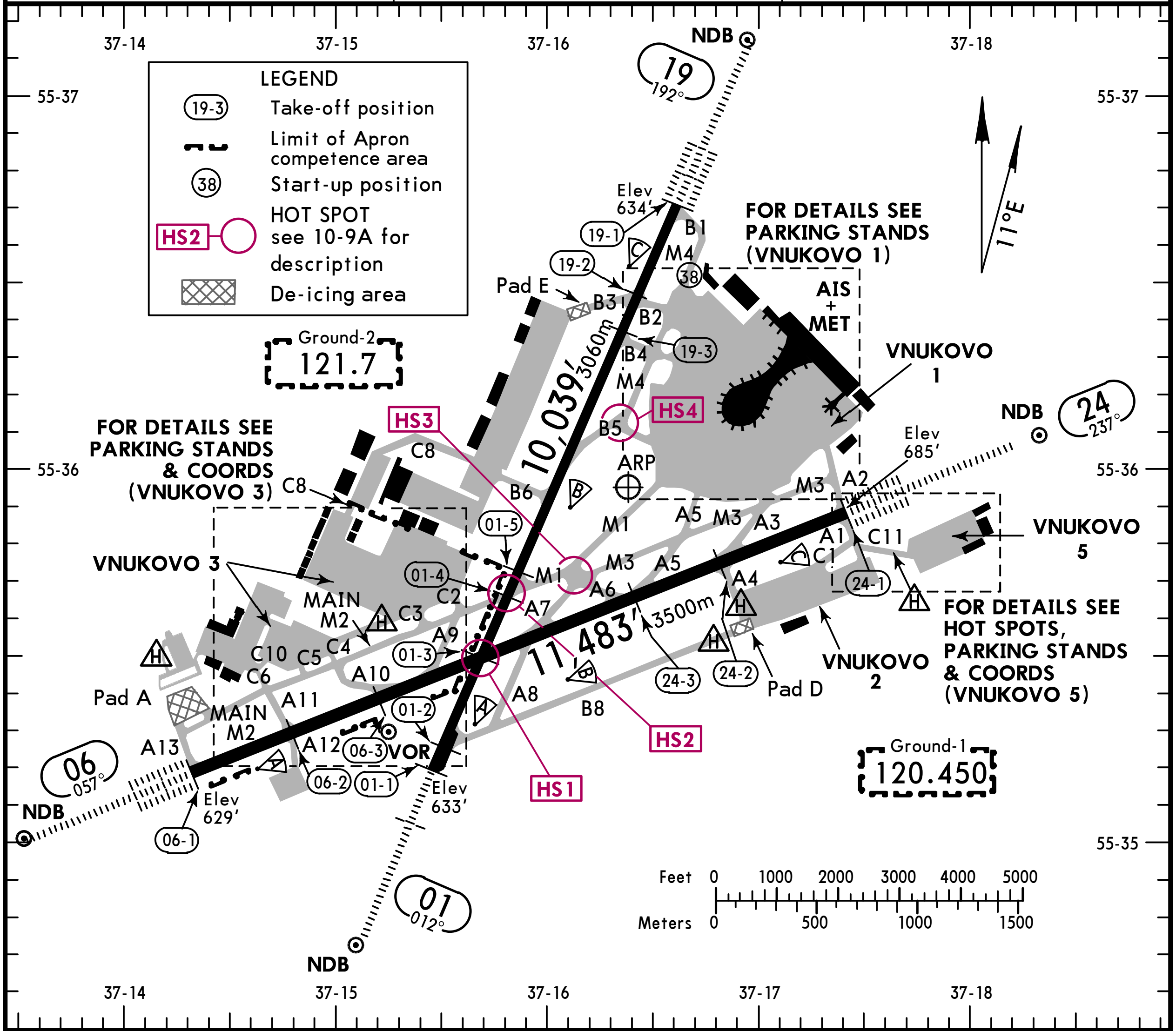
Deviations from the given restrictions are possible in view of RWY 06/24 closure, adverse weather conditions or operating limitations.

Arrivals and departures of ACFT that comply with noise certification requirements specified in ICAO Annex 16, Chapter 3 are permitted during that time.

Restrictions

Flight operations of Tu-134, Tu-154B and Il-86 ACFT are prohibited, except for flights operated for the purpose of transport of Heads of State, provision of medical and safety and rescue assistance.

ATIS Departure	VNUKOVO Delivery-1 (TWR) VNUKOVO 1	VNUKOVO Delivery-3 (TWR) VNUKOVO 2, 3 and 5
124.45 (Russian 127.8)	131.8	129.7
VNUKOVO Ground-1 (TWR)	VNUKOVO Ground-2 (TWR)	Tower
120.450	121.7	118.3



ADDITIONAL RUNWAY INFORMATION

RWY		USABLE LENGTHS LANDING BEYOND Threshold	Glide Slope	TAKE-OFF	WIDTH
01	HIRL (60m) CL (15m) HIALS PAPI-L (3.0°) RVR		9128′ 2782m	❶	148′ 45m
19	HIRL (60m) CL (15m) HIALS-II TDZ PAPI-L (3.0°) RVR				

❶ TAKE-OFF RUN AVAILABLE

RWY 01:		RWY 19:	
From rwy head	10,039′ (3060m)	From rwy head	10,039′ (3060m)
twy B8 int	9429′ (2874m)	twy B2 int	8409′ (2563m)
rwy 06/24 int	8091′ (2466m)	twy B4 int	7713′ (2351m)
main twy M2 int	7119′ (2170m)		
twy M1 int	6850′ (2088m)		

RWY		Threshold	Glide Slope	TAKE-OFF	WIDTH
06 / 24	HIRL (60m) CL (15m) HIALS-II TDZ PAPI-L (3.0°) RVR		10,511′ 3204m / 10,389′ 3167m	❷	197′ 60m

❷ TAKE-OFF RUN AVAILABLE

RWY 06:		RWY 24:	
From rwy head	11,483′ (3500m)	From rwy head	11,483′ (3500m)
twy A11 int	9843′ (3000m)	twy A3 int	9708′ (2959m)
twy A10 int	8327′ (2538m)	twy A5 int	8209′ (2502m)

Std TAKE-OFF

HIRL & CL (spacing 15m or less) & relevant RVR	RL & CL & relevant RVR	RL & CL	RL & RCLM DAY	RL or CL NIGHT	RL or RCLM DAY	Adequate Vis Ref DAY	Adequate Vis Ref NIGHT
TDZ R125m Mid R125m Rollout R125m	TDZ R150m Mid R150m Rollout R150m	R200m	R300m	R300m	R400m	R/V500m	NA

CHANGES: Apron.

HOT SPOTS

For information only, not to be construed as ATC instructions.

HS1 Intersection of Rwy 06/24 and Rwy 01/19

During simultaneous flight operations or simultaneous taxiing on two Rwys, flight crews must maintain an active lookout and a listening watch, strictly observe ATS unit instructions.

HS2 Intersection of Twy M1, Main Twy M2 and Rwy 01/19

ACFT taxiing from Vnukovo 3 apron must keep a lookout for runway-holding position marking on Rwy 01/19. Crossing runway-holding positions at Rwy 01/19 and taxiing onto Main Twy M2 shall be carried out strictly by ATS unit instruction.

HS3 Intersection of Twys M1, M3, A6 and A7

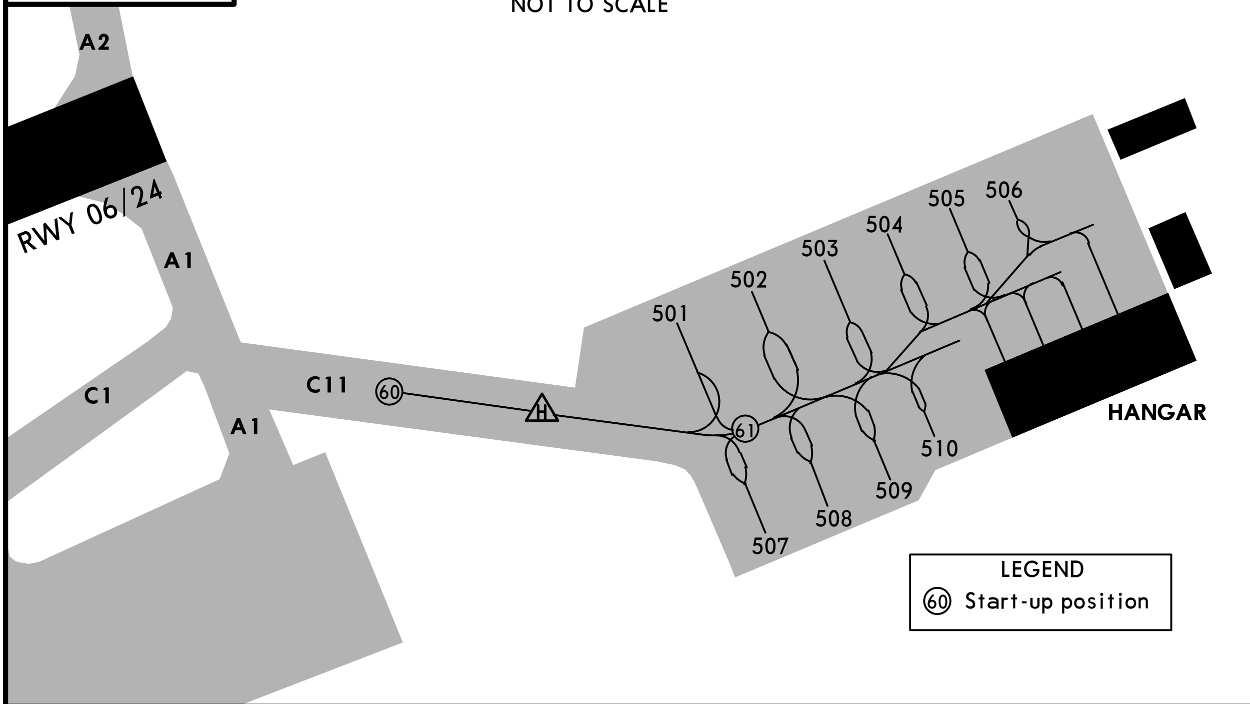
Flight crews must exercise extreme caution during taxiing, follow the marking and ATS unit instructions. ACFT exiting Rwy 06 via TWY A7 after landing must keep speed under control to avoid missing the turn onto Twy A7.

HS4 Intersection of Twys M4 and B5

ACFT exiting Rwy 01 via Twy B5 after landing must keep speed under control to avoid missing the RIGHT turn onto Vnukovo 1 apron. ACFT taxiing to Vnukovo 1 via Twy M4 must avoid missing the LEFT turn.

VNUKOVO 5

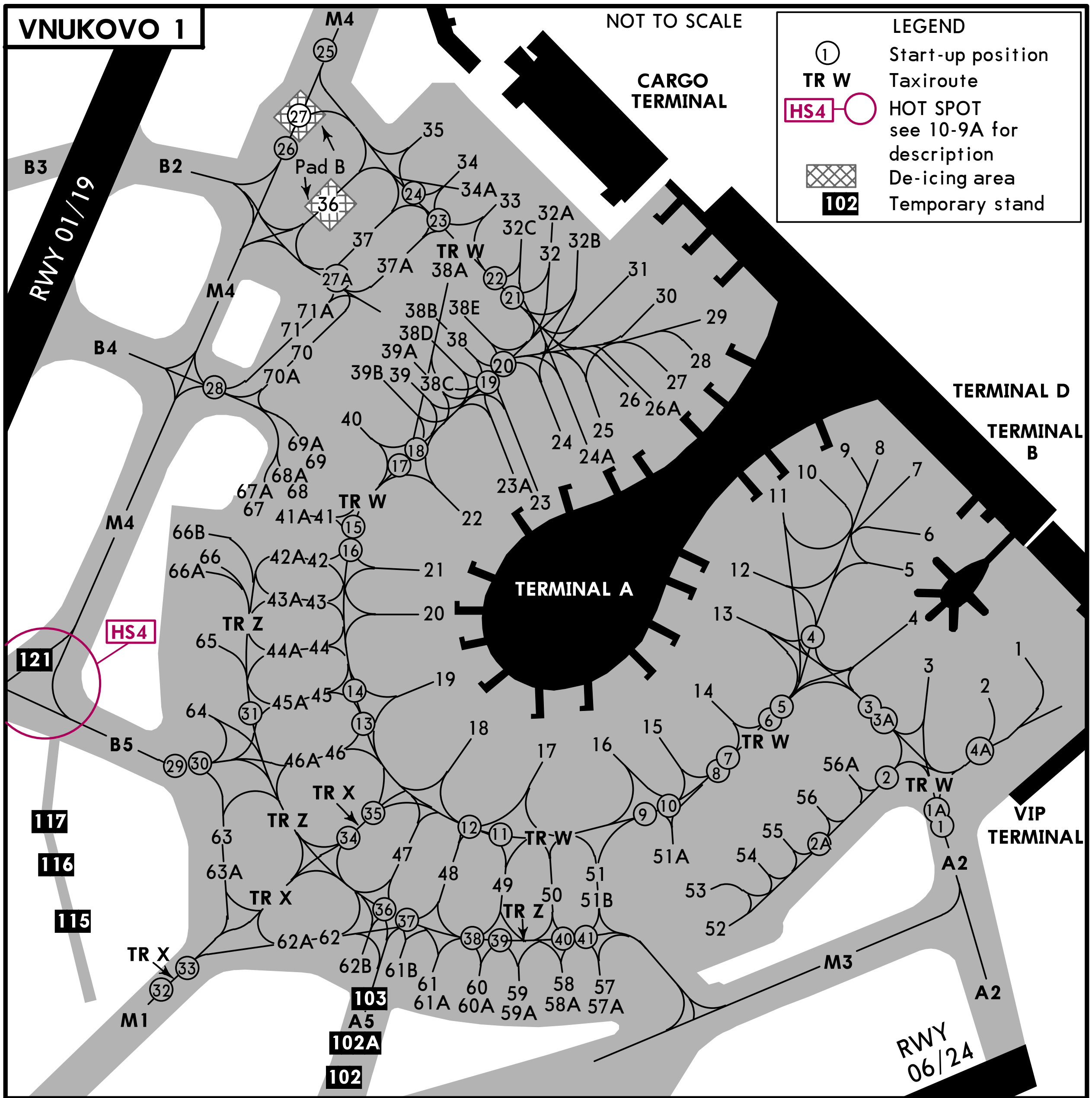
NOT TO SCALE



INS COORDINATES

STAND No.	COORDINATES	STAND No.	COORDINATES
501	N55 35.8 E037 17.7	507, 508	N55 35.8 E037 17.8
502	N55 35.8 E037 17.8	509, 510	N55 35.8 E037 17.9
503	N55 35.9 E037 17.8		
504, 505	N55 35.9 E037 17.9		
506	N55 35.9 E037 18.0		

CHANGES: None.



CHANGES: Stands, start-up postions, temporary stand 102A.

INS COORDINATES VNUKOVO 1

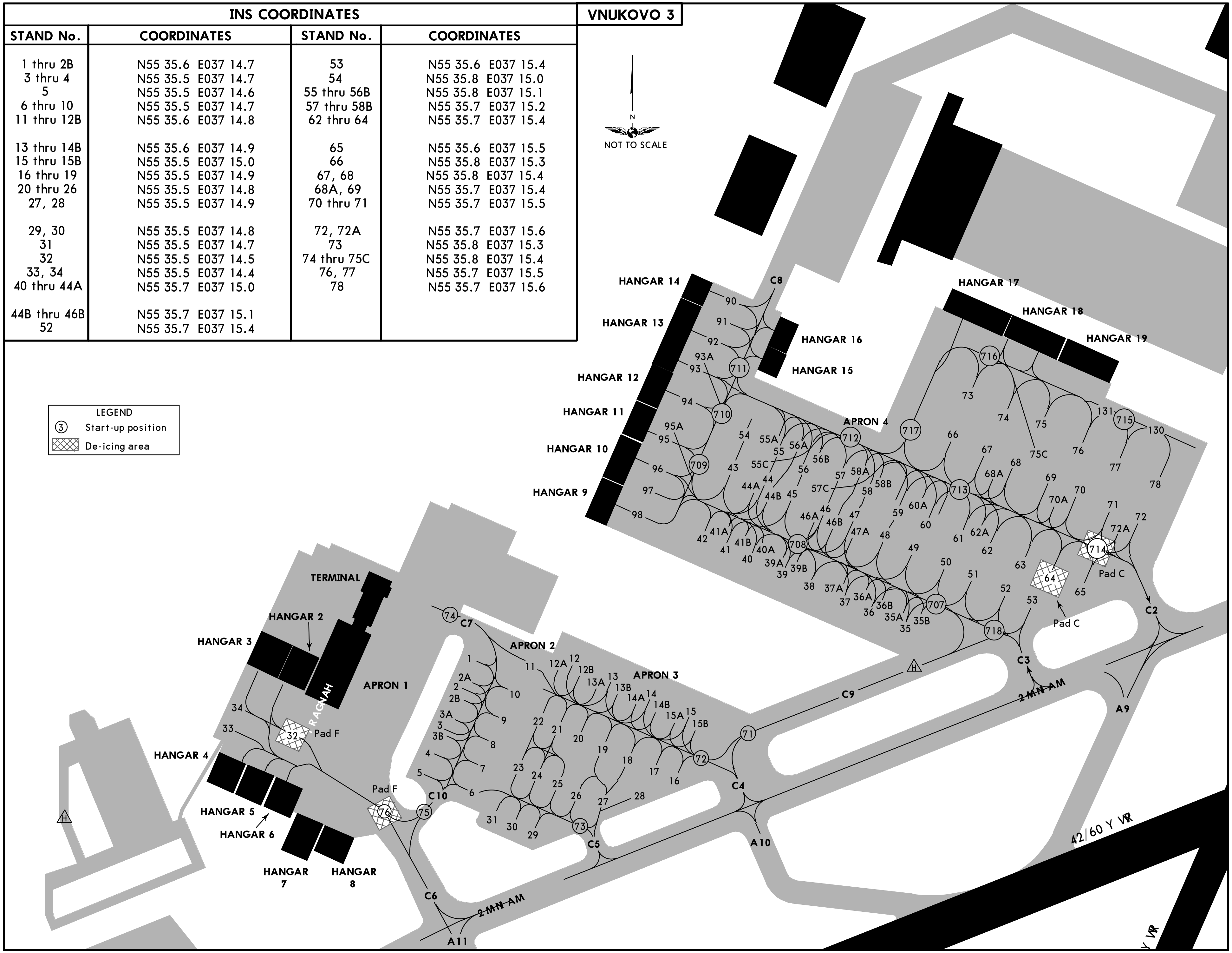
STAND No.	COORDINATES	STAND No.	COORDINATES
1	N55 36.2 E037 17.4	42A thru 43A	N55 36.2 E037 16.6
2, 3	N55 36.1 E037 17.3	44 thru 45A	N55 36.1 E037 16.6
4 thru 7	N55 36.2 E037 17.3	46	N55 36.1 E037 16.7
8	N55 36.3 E037 17.3	46A	N55 36.1 E037 16.6
9	N55 36.3 E037 17.2	47, 48	N55 36.0 E037 16.8
10	N55 36.2 E037 17.2	49, 50	N55 36.0 E037 16.9
11 thru 13	N55 36.2 E037 17.1	51 thru 51B	N55 36.0 E037 17.0
14	N55 36.1 E037 17.0	52 thru 55	N55 36.0 E037 17.1
15	N55 36.1 E037 17.0	56, 56A	N55 36.1 E037 17.2
16, 17	N55 36.1 E037 16.9	57, 57A	N55 35.9 E037 17.0
18, 19	N55 36.1 E037 16.8	58 thru 59A	N55 35.9 E037 16.9
20 thru 22	N55 36.2 E037 16.8	60 thru 61A	N55 35.9 E037 16.8
23 thru 24	N55 36.2 E037 16.9	61B, 62	N55 36.0 E037 16.7
24A	N55 36.2 E037 17.0	62A	N55 36.0 E037 16.6
25 thru 27	N55 36.3 E037 17.0	62B	N55 36.0 E037 16.7
28, 29	N55 36.3 E037 17.1	63, 63A	N55 36.0 E037 16.6
30, 31	N55 36.4 E037 17.0	64, 65	N55 36.1 E037 16.5
32 thru 33	N55 36.4 E037 16.9	66 thru 66B	N55 36.2 E037 16.5
34	N55 36.5 E037 16.8	67 thru 68	N55 36.2 E037 16.6
34A	N55 36.4 E037 16.8	68A	N55 36.3 E037 16.6
35	N55 36.5 E037 16.8	69	N55 36.3 E037 16.7
36, 37	N55 36.4 E037 16.7	69A	N55 36.3 E037 16.6
37A	N55 36.4 E037 16.8	70	N55 36.3 E037 16.7
38	N55 36.3 E037 16.8	70A	N55 36.3 E037 16.6
38A, 38B	N55 36.4 E037 16.8	71, 71A	N55 36.3 E037 16.7
38C thru 39	N55 36.3 E037 16.8		
39A thru 40	N55 36.3 E037 16.7		
41	N55 36.2 E037 16.7		
41A	N55 36.2 E037 16.6		
42	N55 36.2 E037 16.7		

CHANGES: Stands, coordinates.

INS COORDINATES

STAND No.	COORDINATES	STAND No.	COORDINATES
1 thru 2B	N55 35.6 E037 14.7	53	N55 35.6 E037 15.4
3 thru 4	N55 35.5 E037 14.7	54	N55 35.8 E037 15.0
5	N55 35.5 E037 14.6	55 thru 56B	N55 35.8 E037 15.1
6 thru 10	N55 35.5 E037 14.7	57 thru 58B	N55 35.7 E037 15.2
11 thru 12B	N55 35.6 E037 14.8	62 thru 64	N55 35.7 E037 15.4
13 thru 14B	N55 35.6 E037 14.9	65	N55 35.6 E037 15.5
15 thru 15B	N55 35.5 E037 15.0	66	N55 35.8 E037 15.3
16 thru 19	N55 35.5 E037 14.9	67, 68	N55 35.8 E037 15.4
20 thru 26	N55 35.5 E037 14.8	68A, 69	N55 35.7 E037 15.4
27, 28	N55 35.5 E037 14.9	70 thru 71	N55 35.7 E037 15.5
29, 30	N55 35.5 E037 14.8	72, 72A	N55 35.7 E037 15.6
31	N55 35.5 E037 14.7	73	N55 35.8 E037 15.3
32	N55 35.5 E037 14.5	74 thru 75C	N55 35.8 E037 15.4
33, 34	N55 35.5 E037 14.4	76, 77	N55 35.7 E037 15.5
40 thru 44A	N55 35.7 E037 15.0	78	N55 35.7 E037 15.6
44B thru 46B	N55 35.7 E037 15.1		
52	N55 35.7 E037 15.4		

VNUKOVO 3



STRAIGHT-IN RWY		A	B	C	D
01	ILS	**833'**(200') ❶ R550m	**833'**(200') ❶ R550m	**833'**(200') ❶ R550m	**833'**(200') ❶ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	GLS	**833'**(200') ❶ R550m	**833'**(200') ❶ R550m	**833'**(200') ❶ R550m	**833'**(200') ❶ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	RNP LNAV/VNAV	**958'**(325') R800m	**971'**(338') R800m	**979'**(346') R900m	**989'**(356') R900m
	ALS out	R1500m	R1500m	R1600m	R1600m
	❷RNP LNAV	**1010'**(377') R1000m	**1010'**(377') R1000m	**1060'**(427') R1300m	**1060'**(427') R1300m
	ALS out	R1500m	R1500m	R2000m	R2000m
	❷ VOR	**990'**(357') R900m	**1020'**(387') R1100m	**1050'**(417') R1200m	**1070'**(437') R1300m
	ALS out	R1500m	R1500m	R1900m	R2000m
	❷ NDB	**980'**(347') R900m	**980'**(347') R900m	**980'**(347') R900m	**980'**(347') R900m
	ALS out	R1500m	R1500m	R1600m	R1600m
	❷ NDB Y	**1160'**(527') R1500m	**1160'**(527') R1500m	**1160'**(527') R1700m	**1160'**(527') R1700m
	ALS out	R1500m	R1500m	R2400m	R2400m
06	CAT 2 ILS	**729'**(100') **RA104'** R300m	**729'**(100') **RA104'** R300m	**729'**(100') **RA104'** R300m	**729'**(100') **RA104'** ❸ R300m
	ILS	**829'**(200') R550m	**829'**(200') R550m	**829'**(200') R550m	**829'**(200') R550m
	TDZ or CL out	❶ R550m	❶ R550m	❶ R550m	❶ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	GLS	**829'**(200') R550m	**829'**(200') R550m	**829'**(200') R550m	**829'**(200') R550m
	TDZ or CL out	❶ R550m	❶ R550m	❶ R550m	❶ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	RNP LNAV/VNAV	**950'**(321') R800m	**963'**(334') R800m	**971'**(342') R900m	**981'**(352') R900m
	ALS out	R1500m	R1500m	R1600m	R1600m
	❷RNP LNAV	**1000'**(371') R1000m	**1000'**(371') R1000m	**1000'**(371') R1000m	**1000'**(371') R1000m
	ALS out	R1500m	R1500m	R1700m	R1700m
	❷ VOR	**1100'**(471') R1500m	**1100'**(471') R1500m	**1100'**(471') R1500m	**1100'**(471') R1500m
	ALS out	R1500m	R1500m	R2200m	R2200m
	❷ NDB	**1100'**(471') R1500m	**1100'**(471') R1500m	**1100'**(471') R1500m	**1100'**(471') R1500m
	ALS out	R1500m	R1500m	R2200m	R2200m
	❷ NDB Y	**1150'**(521') R1500m	**1150'**(521') R1500m	**1150'**(521') R1700m	**1150'**(521') R1700m
	ALS out	R1500m	R1500m	R2400m	R2400m

❶ R750m when a Flight Director or Autopilot or HUDLS to DA is not used.
❷ Continuous Descent Final Approach.
❸ without autoland: R350m.

CHANGES: None.

STRAIGHT-IN RWY		A	B	C	D
19	CAT 2 ILS	**734′**(100′) **RA105′** R300m	**734′**(100′) **RA105′** R300m	**734′**(100′) **RA105′** R300m	**739′**(105′) **RA111′** ❶ R300m
	ILS	**834′**(200′) R550m	**834′**(200′) R550m	**834′**(200′) R550m	**834′**(200′) R550m
	TDZ or CL out	❷ R550m	❷ R550m	❷ R550m	❷ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	GLS	**834′**(200′) R550m	**834′**(200′) R550m	**834′**(200′) R550m	**834′**(200′) R550m
	TDZ or CL out	❷ R550m	❷ R550m	❷ R550m	❷ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	RNP LNAV/VNAV	**978′**(344′) R900m	**991′**(357′) R900m	**999′**(365′) R1000m	**1009′**(375′) R1000m
	ALS out	R1500m	R1500m	R1700m	R1700m
	❸ RNP LNAV	**1130′**(496′) R1500m	**1130′**(496′) R1500m	**1130′**(496′) R1500m	**1130′**(496′) R1500m
	ALS out	R1500m	R1500m	R2300m	R2300m
	❸ VOR	**1120′**(486′) R1500m	**1120′**(486′) R1500m	**1120′**(486′) R1500m	**1120′**(486′) R1500m
	ALS out	R1500m	R1500m	R2300m	R2300m
	❸ NDB	**1120′**(486′) R1500m	**1120′**(486′) R1500m	**1120′**(486′) R1500m	**1120′**(486′) R1500m
	ALS out	R1500m	R1500m	R2300m	R2300m
	❸ NDB Y	**1270′**(636′) R1500m	**1270′**(636′) R1500m	**1270′**(636′) R2200m	**1270′**(636′) R2200m
	ALS out	R1500m	R1500m	R2400m	R2400m
24	CAT 2 ILS	**785′**(100′) **RA103′** R300m	**785′**(100′) **RA103′** R300m	**785′**(100′) **RA103′** R300m	**785′**(100′) **RA103′** ❶ R300m
	ILS	**885′**(200′) R550m	**885′**(200′) R550m	**885′**(200′) R550m	**885′**(200′) R550m
	TDZ or CL out	❷ R550m	❷ R550m	❷ R550m	❷ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	GLS	**885′**(200′) R550m	**885′**(200′) R550m	**885′**(200′) R550m	**885′**(200′) R550m
	TDZ or CL out	❷ R550m	❷ R550m	❷ R550m	❷ R550m
	ALS out	R1200m	R1200m	R1200m	R1200m
	RNP LNAV/VNAV	**949′**(264′) ❹ R750m	**961′**(276′) ❹ R750m	**999′**(314′) ❺ R750m	**1009′**(324′) R800m
	ALS out	R1300m	R1300m	R1400m	R1500m
	❸ RNP LNAV	**1070′**(385′) R1100m	**1070′**(385′) R1100m	**1070′**(385′) R1100m	**1070′**(385′) R1100m
	ALS out	R1500m	R1500m	R1800m	R1800m
	❸ VOR	**1060′**(375′) R1000m	**1060′**(375′) R1000m	**1060′**(375′) R1000m	**1060′**(375′) R1000m
	ALS out	R1500m	R1500m	R1700m	R1700m
	❸ NDB	**1150′**(465′) R1500m	**1150′**(465′) R1500m	**1150′**(465′) R1500m	**1150′**(465′) R1500m
	ALS out	R1500m	R1500m	R2200m	R2200m
	❸ NDB Y	**1280′**(595′) R1500m	**1280′**(595′) R1500m	**1280′**(595′) R2000m	**1280′**(595′) R2000m
	ALS out	R1500m	R1500m	R2400m	R2400m

❶ without autoland: R350m.
❷ R750m when a Flight Director or Autopilot or HUDLS to DA is not used.
❸ Continuous Descent Final Approach.
❹ With TDZ & CL & HUD: R600m
❺ With TDZ & CL & HUD: R700m

CHANGES: CAT 2 ILS Rwy 19 minimums added.

TAKE-OFF

Low Visibility Procedures required				RCLM or RL or CL	RL or CL	Adequate Vis Ref	
Approval for Low Visibility Take-off required							
RCLM & RL & CL(spacing 15m or less) & RVR	RCLM & RL & CL & RVR	RCLM & RL & RVR	RCLM & RVR & RL or CL	DAY	NIGHT	DAY	NIGHT
		DAY	NIGHT				
R125m	R150m	R300m		R/V400m		R/V500m	NA

MOSCOW, RUSSIA
ILS Rwy 01

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

LOC IWM	Final Apch Crs	WW151 MANDATORY	DA(H)	Apt Elev 685' Rwy 633'
111.7	012°	2000' (1367')	833' (200')	

MSA ARP: 3800. MSA ARP is computed for surface air temperature at apt -29.0°C

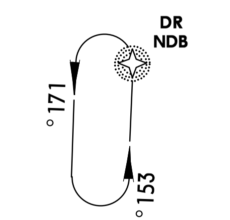
MISSED APCH: Climb STRAIGHT AHEAD to 1100' or above (MAX 190 KT), then turn LEFT to AMRED (MAX 190 KT) climbing to 3500', then proceed to BOPUS (MANDATORY 220 KT), then to SODLE (MANDATORY 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000', then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000'

RNAV 1 for initial, intermediate and missed approach.

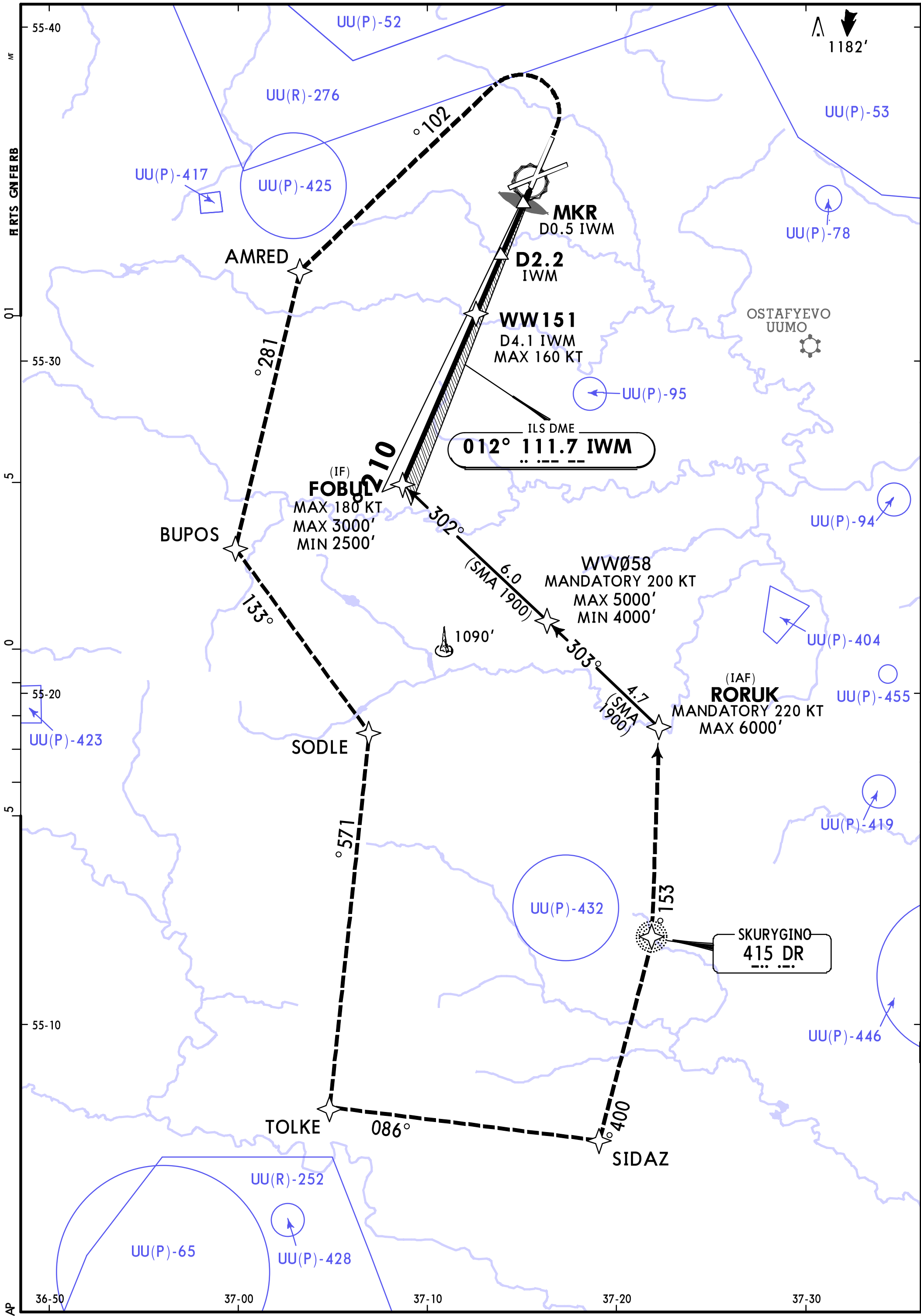
1. GNSS or DME/DME required. 2. ILS DME reads zero at rwy 01 threshold.

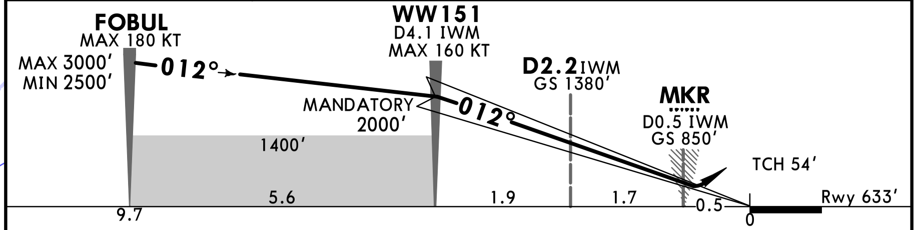
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
3000	(725)
2500	(570)
2000	(420)
1900	(390)
1400	(235)
1380	(230)
1100	(145)
850	(70)
833	(60)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
GS	3.00°	372	478	531	637	743	849

HIALS PAPI — MIN 1100' — AMRED LT

Std

STRAIGHT-IN LANDING

ILS

DA(H) 833' (200')

	FULL	ALS out
A	❶ R550m	R1200m
B		
C		
D		

❶ R750m when a Flight Director or Autopilot or HUD to DA is not used.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

LOC IGT	Final Apch Crs	WW502 MANDATORY	DA(H)	Apt Elev 685' Rwy 629'
108.9	057°	2000' (1371')	829' (200')	

MSA ARP 3800 — is computed for surface air temperature at apt -29.0°C

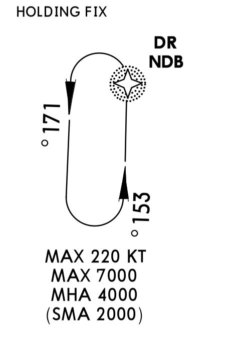
MISSED APCH: Climb STRAIGHT AHEAD to WW652 (MAX 180 KT), then turn RIGHT to WW187 (MAX 190 KT) climbing to 3500', then proceed to LO NDB (MANDATORY 200 KT) climbing to 4000', then to BEMAS (MANDATORY 200 KT) to 4000', then to TEBDI (MANDATORY 200 KT), then to TEPTA (MANDATORY 200 KT), then to SIDAZ (MANDATORY 200 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000'

RNAV 1 for initial, intermediate and missed approach.

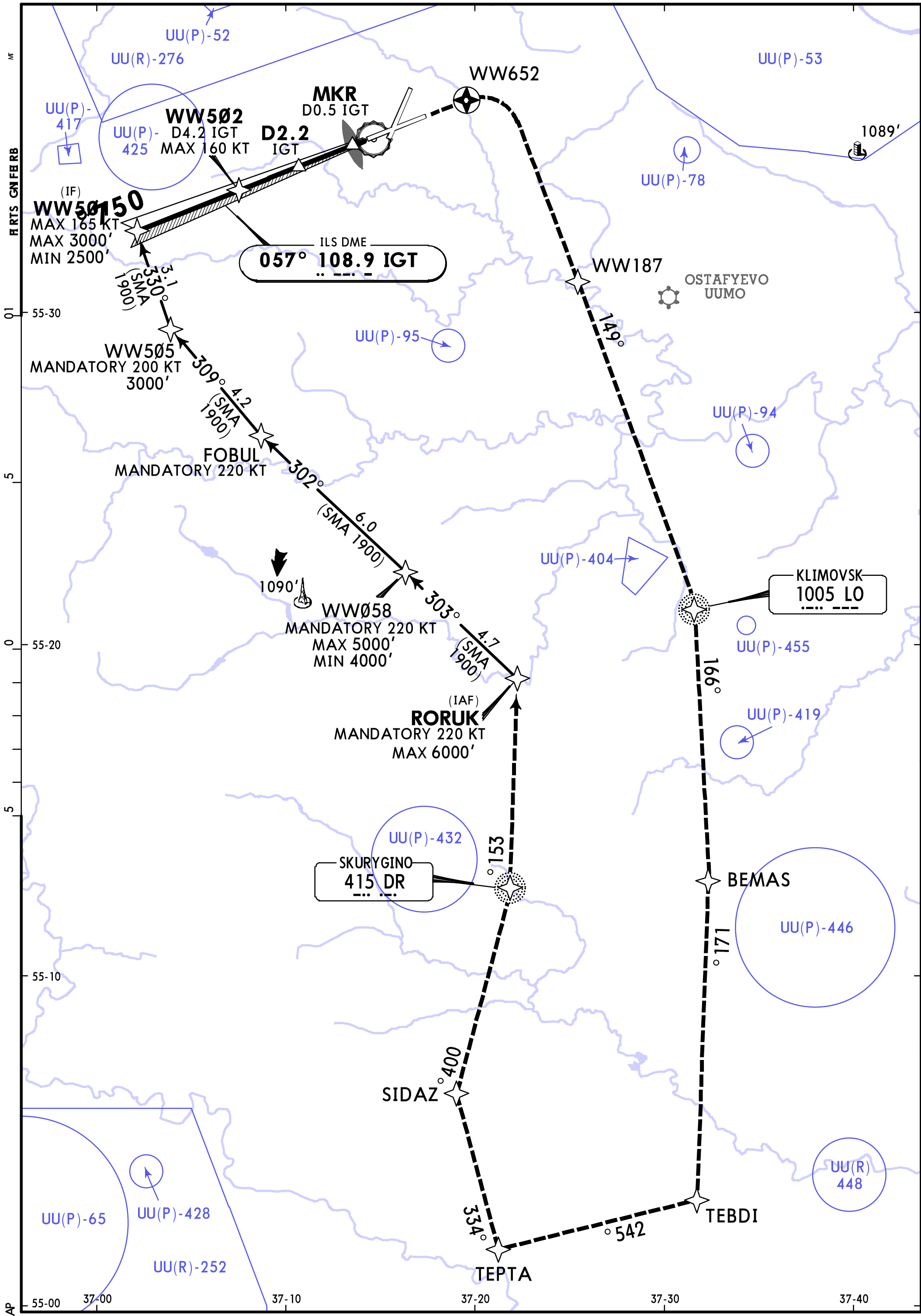
1. GNSS or DME/DME required. 2. ILS DME reads zero at rwy 06 threshold.

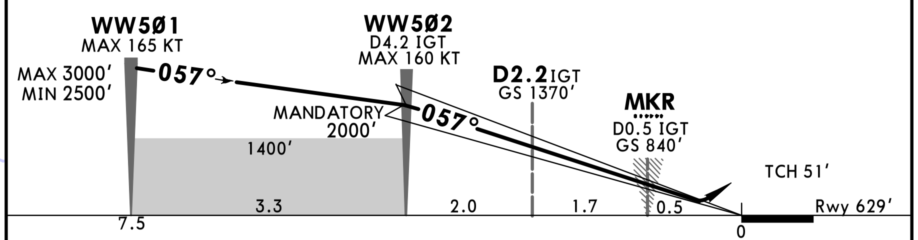
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(880)
3000	(725)
2500	(575)
2000	(420)
1900	(390)
1400	(240)
1370	(230)
840	(65)
829	(60)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts	70	90	100	120	140	160
GS 3.00°	372	478	531	637	743	849

HIALS-II
PAPI

WW652 ↑ WW187 RT

Std

STRAIGHT-IN LANDING

ILS

DA(H) 829' (200')

	FULL	TDZ or CL out	ALS out
A B C D	R550m	❶ R550m	R1200m

❶ R750m when a Flight Director or Autopilot or HUD to DA is not used.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

LOC IGT	Final Apch Crs	WW502 MANDATORY	CAT II ILS RA DA(H)	Apt Elev 685'
108.9	057°	2000' (1371')	RA 104' 729' (100')	Rwy 629'

3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to WW652 (MAX 180 KT), then turn RIGHT to WW187 (MAX 190 KT) climbing to 3500', then proceed to LO NDB (MANDATORY 200 KT) climbing to 4000', then to BEMAS (MANDATORY 200 KT) to 4000', then to TEBDI (MANDATORY 200 KT), then to TEPTA (MANDATORY 200 KT), then to SIDAZ (MANDATORY 200 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000'

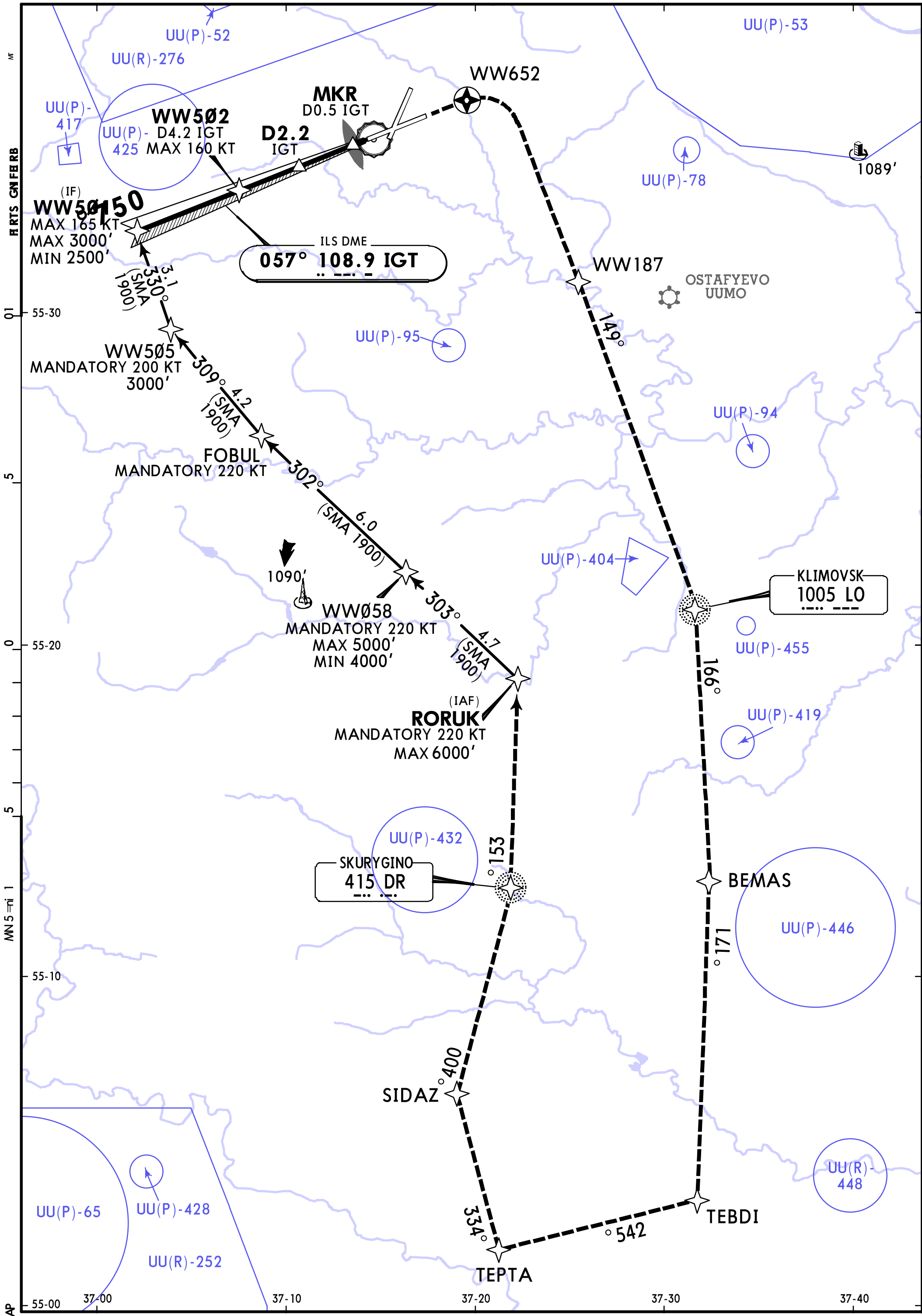
RNAV 1 for initial, intermediate and missed approach.

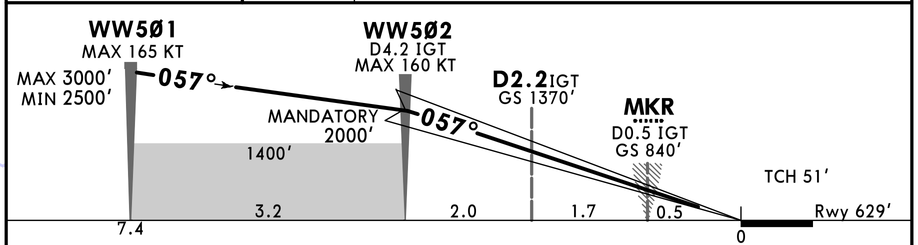
1. Special Aircrew & Acft Certification Required. 2. GNSS or DME/DME required. 3. ILS DME reads zero at rwy 06 threshold.

HOLDING FIX
DR NDB
171°
153°
MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(880)
3000	(725)
2500	(575)
2000	(420)
1900	(390)
1400	(240)
1370	(230)
840	(65)
729	(30)

❶ FL120 if pressure is less than 1013 hPa (760mm).
FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts	70	90	100	120	140	160
GS 3.00°	372	478	531	637	743	849

HIALS-II
PAPI
WW652 WW187 RT

Std

STRAIGHT-IN LANDING

CAT II ILS

RA 104'
DA(H) 729' (100')

❶ R300m

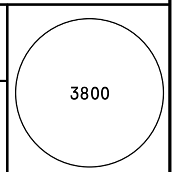
❶ CAT D without autoland: R350m.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

LOC ITA	Final Apch Crs	WW951	DA(H)	Apt Elev 685'
111.5	192°	1700' (1066')	834' (200')	Rwy 634'

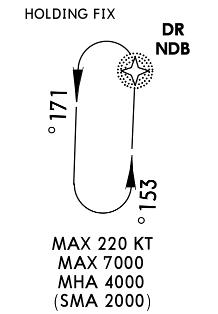


MISSED APCH: Climb STRAIGHT AHEAD to WW952, then turn LEFT to FOBUL (MANDATORY 220 KT) climbing to 3500', then proceed to SODLE (MANDATORY 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000', then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req)	Rwy Elev: 23 hPa	Trans level: FL110 ❶	Trans alt: 10000'

RNAV 1 for initial, intermediate and missed approach.

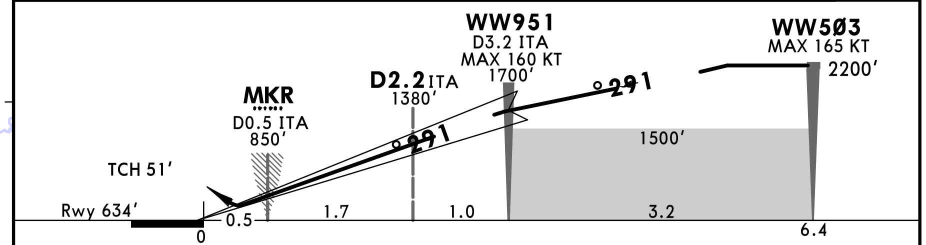
1. GNSS or DME/DME required. 2. ILS DME reads zero at rwy 19 threshold.



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
2500	(570)
2200	(480)
1700	(325)
1500	(265)
1380	(230)
1030	(125)
850	(70)
834	(60)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).

ITA DME	1.1	2.2
ALTITUDE	1030'	1380'



Gnd speed-Kts	70	90	100	120	140	160
GS 3.00°	372	478	531	637	743	849

HIALS-II
PAPI
WW952

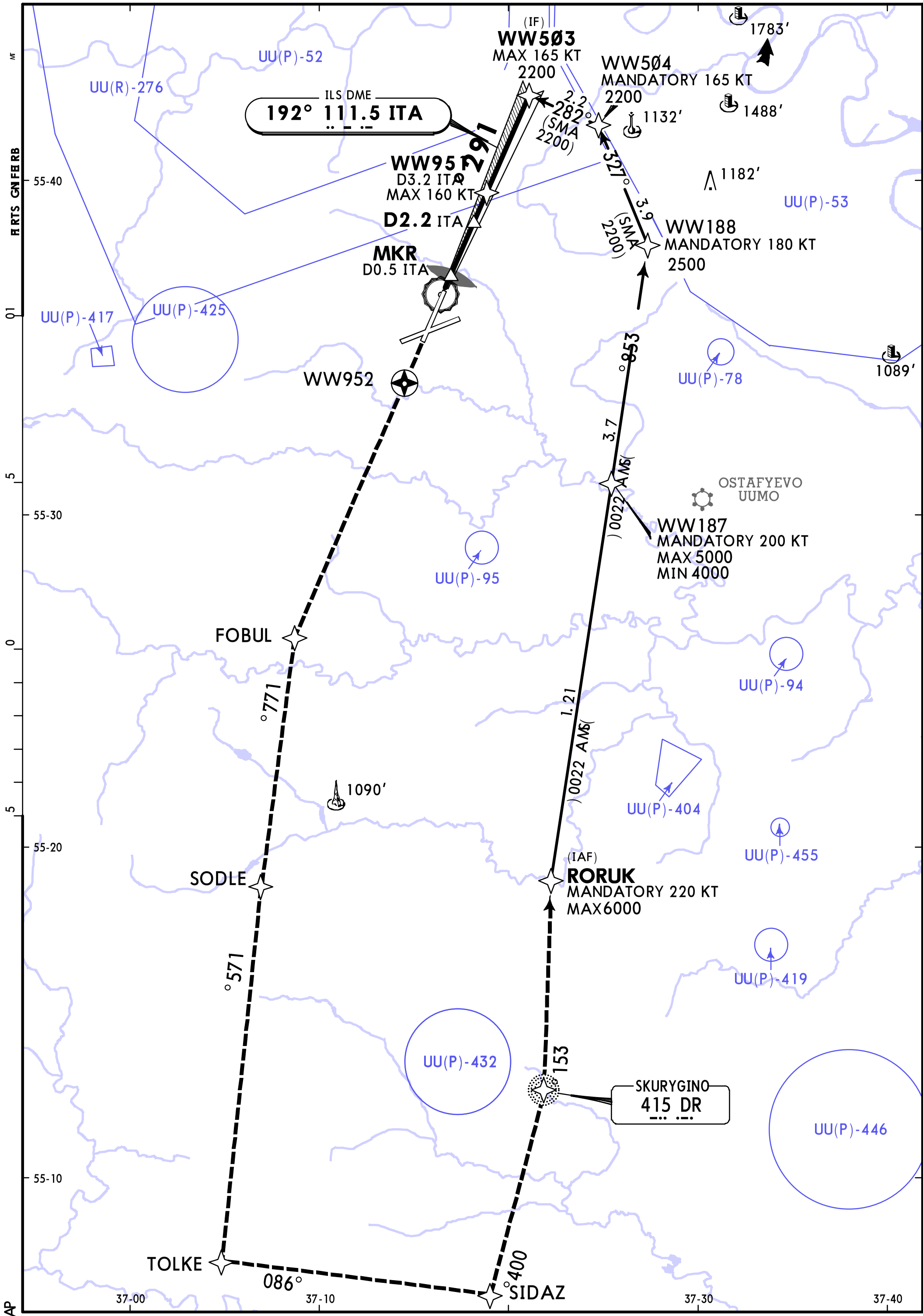
Std STRAIGHT-IN LANDING

ILS

DA(H) 834' (200')

	TDZ or CL out	ALS out
A B C D	R550m	❶ R550m	R1200m

❶ R750m when a Flight Director or Autopilot or HUD to DA is not used.



CHANGES: Recommended altitudes.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

LOC ITA	Final Apch Crs	WW951	CAT II ILS RA/DA(H)	Apt Elev 685′
111.5	192°	1700′ (1066′)	Refer to Minimums	Rwy 634′

MSA ARP: 3800. MSA ARP is computed for surface air temperature at apt -29.0°C

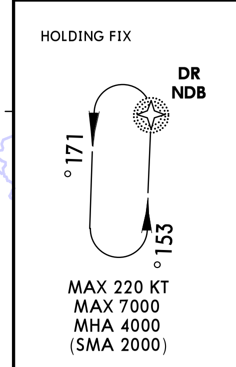
MISSED APCH: Climb STRAIGHT AHEAD to WW952, then turn LEFT to FOBUL (MANDATORY 220 KT) climbing to 3500′, then proceed to SODLE (MANDATORY 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000′, then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

Alt Set: hPa (MM on req) | Rwy Elev: 23 hPa | Trans level: FL110 ❶ | Trans alt: 10000′

RNAV 1 for initial, intermediate and missed approach.

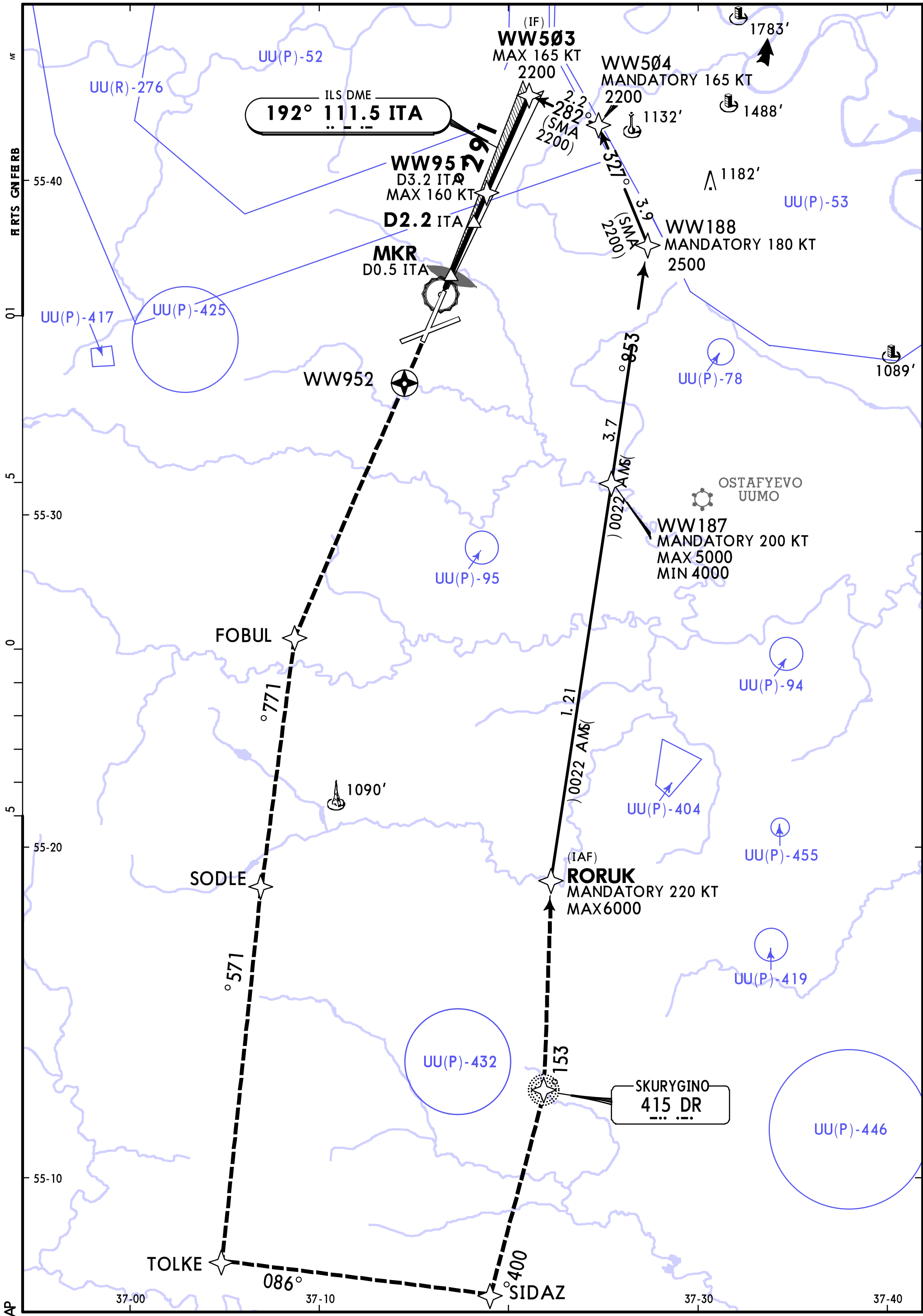
1. GNSS or DME/DME required.
2. ILS DME reads zero at rwy 19 threshold.

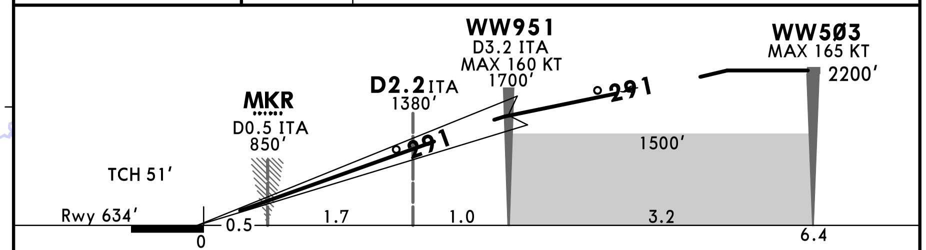
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
2500	(570)
2200	(480)
1700	(325)
1500	(265)
1380	(230)
850	(70)
739	(32)
734	(30)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
GS	3.00°	372	478	531	637	743	849

HIALS-II PAPI

WW952

Std

STRAIGHT-IN LANDING
CAT II ILS

ABC	D
RA 105′ DA(H) 734′ (100′)	RA 111′ DA(H) 739′ (105′)

❶ R300m

❶ CAT D without autoland: R350m.

CHANGES: New procedure.

UUWW/VKO
VNUKOVO

22 NOV 24 (11-4) Eff 28 Nov

JEPPESEN

MOSCOW, RUSSIA
ILS Rwy 24

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-2
118.3	118.3	121.7

LOC IOB	Final Apch Crs	WW451	DA(H)	Apt Elev 685′ Rwy 685′
111.1	237°	1700′(1015′)	885′(200′)	

3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

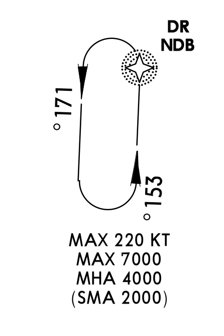
MISSED APCH: Climb STRAIGHT AHEAD to WW452 (MAX 220 KT), then turn LEFT to AMRED (MAX 220 KT) climbing to 3500′, then proceed to BUPOS (MAX 220 KT), then to SODLE (MAX 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000′, then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 25 hPa Trans level: FL110[1] Trans alt: 10000′

RNAV 1 for initial, intermediate and missed approach.

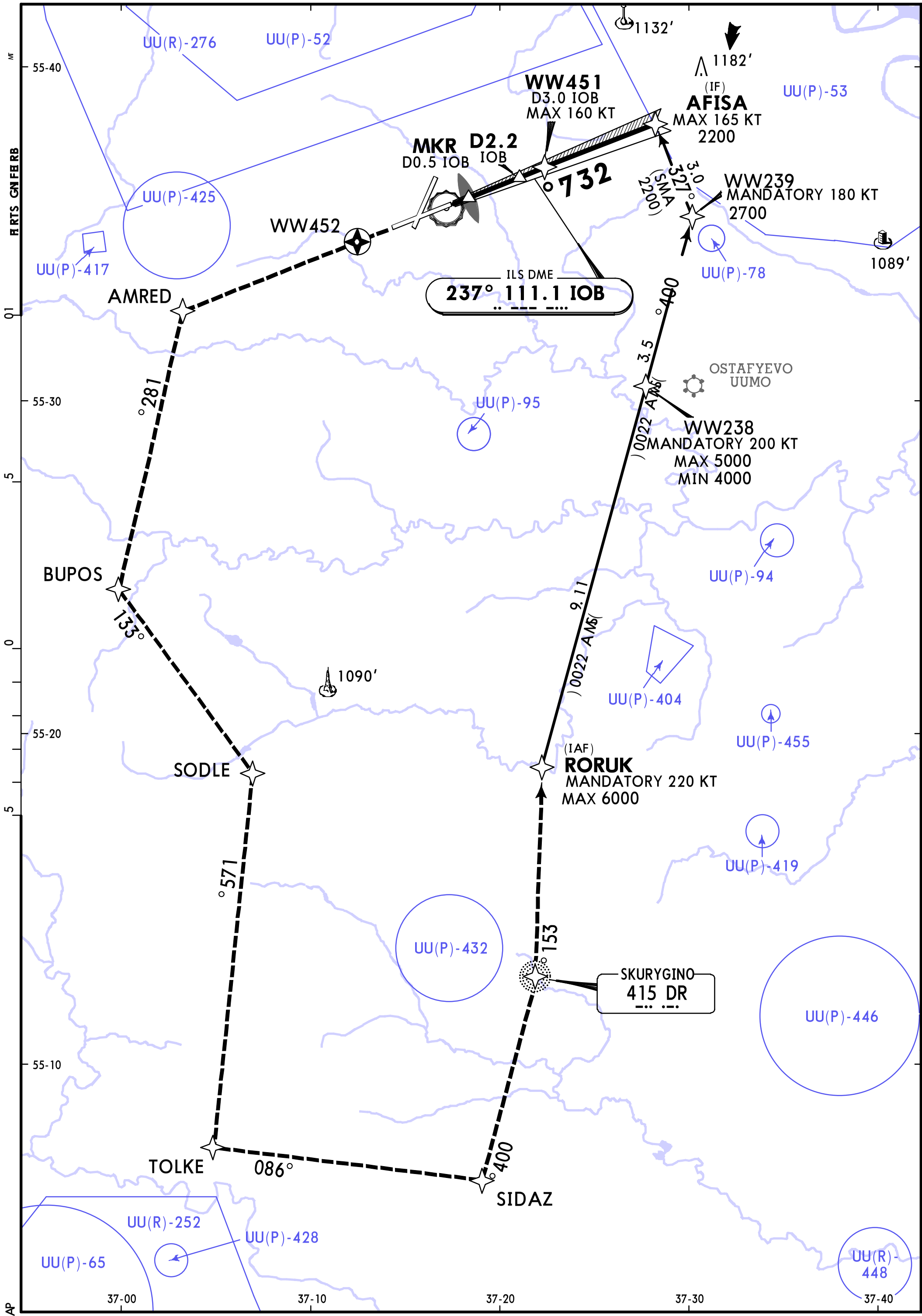
1. GNSS or DME/DME required. 2. ILS DME reads zero at rwy 24 threshold.

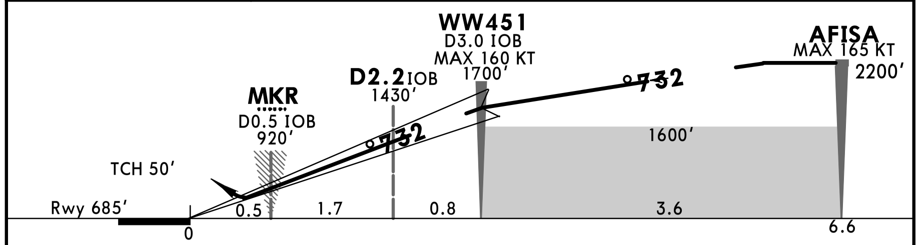
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1925)
6000	(1625)
5000	(1320)
4000	(1015)
3800	(1000)
3500	(860)
2700	(615)
2200	(465)
2000	(405)
1700	(310)
1600	(280)
1430	(230)
920	(75)
885	(60)

[1] FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
GS	3.00°	372	478	531	637	743	849

HIALS-II
PAPI

WW452

Std

STRAIGHT-IN LANDING

ILS

DA(H) 885′(200′)

		TDZ or CL out	ALS out
A B C D	R550m	[1] R550m	R1200m

[1] R750m when a Flight Director or Autopilot or HUD to DA is not used.

CHANGES: Airspace.

UUWW/VKO
VNUKOVO

JEPPESEN
22 NOV 24 (11-4A) Eff 28 Nov

MOSCOW, RUSSIA
CAT II ILS Rwy 24

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-2
118.3	118.3	121.7

LOC IOB	Final Apch Crs	WW451	CAT II ILS RA 103′ DA(H)	Apt Elev 685′
111.1	237°	1700′(1015′)	785′(100′)	Rwy 685′

MSA ARP: 3800. MSA ARP is computed for surface air temperature at apt -29.0°C

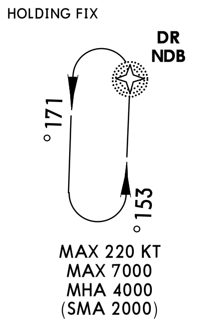
MISSED APCH: Climb STRAIGHT AHEAD to WW452 (MAX 220 KT), then turn LEFT to AMRED (MAX 220 KT) climbing to 3500′, then proceed to BUPOS (MAX 220 KT), then to SODLE (MAX 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000′, then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

Alt Set: hPa (MM on req) | Rwy Elev: 25 hPa | Trans level: FL110 ❶ | Trans alt: 10000′

RNAV 1 for initial, intermediate and missed approach.

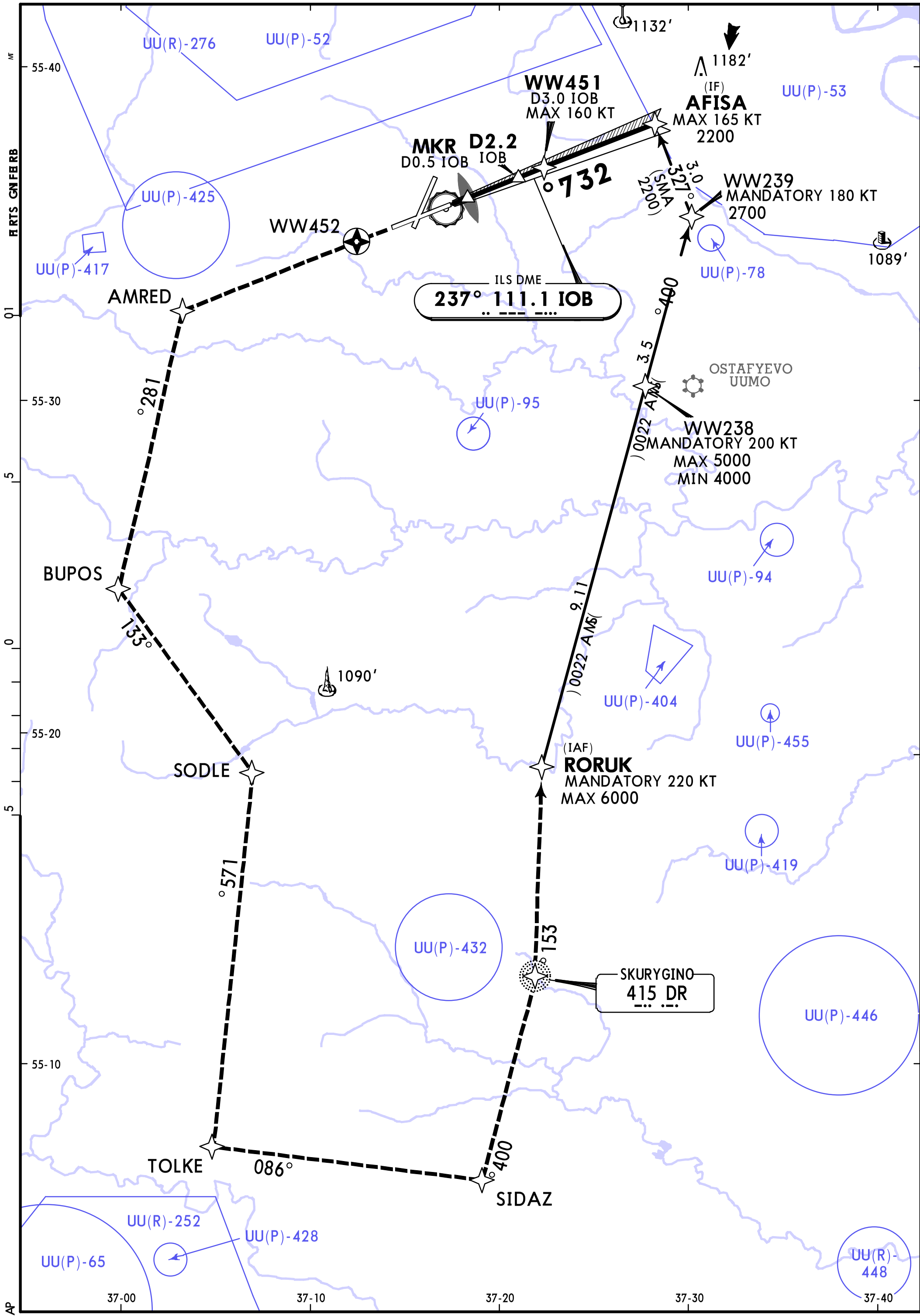
1. GNSS or DME/DME required. 2. ILS DME reads zero at rwy 24 threshold. 3. Special Aircrew & Acft Certification Required.

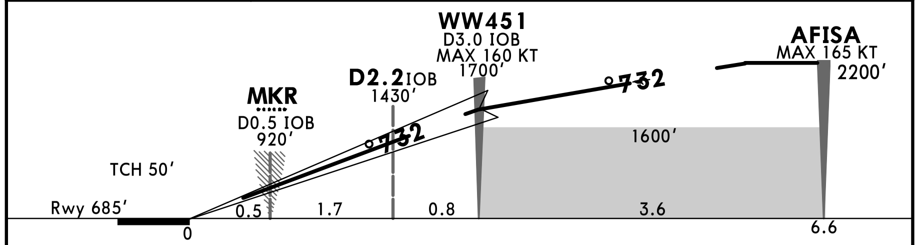
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1925)
6000	(1625)
5000	(1320)
4000	(1015)
3800	(1000)
3500	(860)
2700	(615)
2200	(465)
2000	(405)
1700	(310)
1600	(280)
1430	(230)
920	(75)
785	(30)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
GS	3.00°	372	478	531	637	743	849

HIALS-II
PAPI

WW452

Std

STRAIGHT-IN LANDING

CAT II ILS

RA 103′
DA(H) 785′(100′)

❶ R300m

❶ CAT D without autoland: R350m.

CHANGES: Airspace.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

RNAV	Final Apch Crs	WW15Ø MANDATORY	LNAV/VNAV DA(H)	Apt Elev 685'
	012°	2000' (1367')	Refer to Minimums	Rwy 633'

3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to WW152 (MAX 190 KT), then turn LEFT to AMRED (MAX 190 KT) climbing to 3500', then proceed to BOPUS (MANDATORY 220 KT), then to SODLE (MANDATORY 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000', then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) | Rwy Elev: 23 hPa | Trans level: FL110 ❶ | Trans alt: 10000'

RNP Apch | RNAV 1 for initial, intermediate and missed approach.

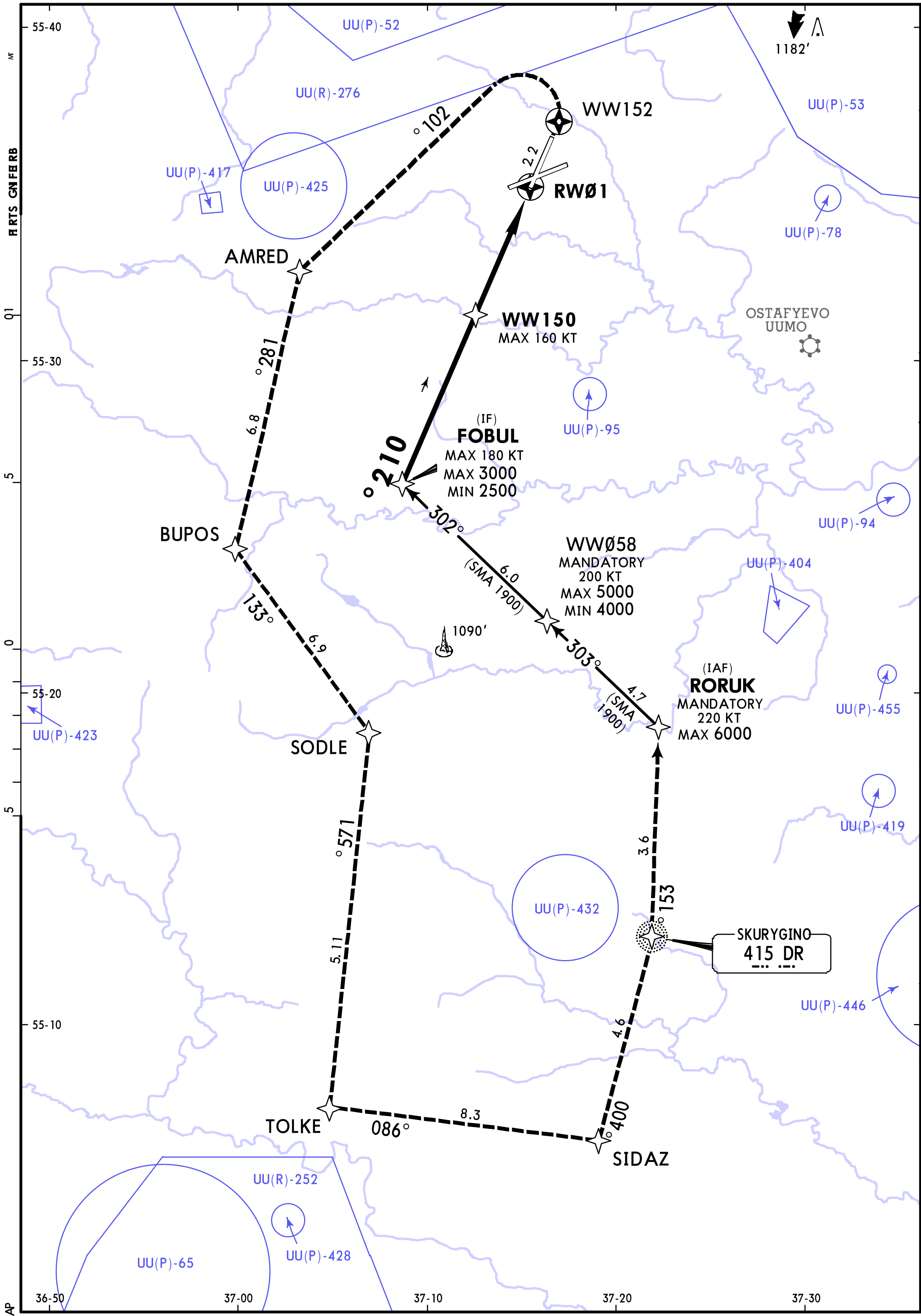
1. GNSS required. 2. Baro-VNAV not authorized below -31°C.

HOLDING FIX
DR NDB
171°
153°
MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

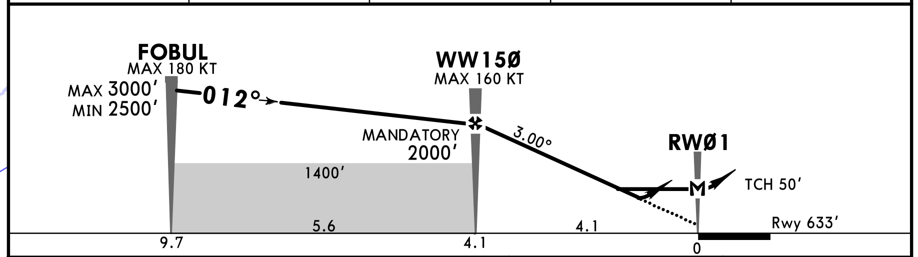
FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
3000	(725)
2500	(570)
2050	(435)
2000	(420)
1900	(390)
1710	(330)
1400	(235)
1370	(225)
1060	(135)
1030	(125)

FEET QNH	METERS (QFE)
1010	(115)
989	(109)
979	(106)
971	(104)
958	(100)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



DIST to RWØ1	4.3	3.2	2.2	1.1
ALTITUDE	2050'	1710'	1370'	1030'



Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849
MAP at RWØ1							

Std

STRAIGHT-IN LANDING

	LNAV/VNAV DA(H) A: 958' (325') B: 971' (338')	C: 979' (346') D: 989' (356')	LNAV CDFA ❶ DA/MDA(H) AB: 1010' (377')	CD: 1060' (427')
		ALS out		ALS out
A	R800m	R1500m	R1000m	R1500m
B				
C	R900m	R1600m	R1300m	R2000m
D				

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces, general note 1.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

RNAV	Final Apch Crs	WW506 MANDATORY	LNAV/VNAV DA(H)	Apt Elev 685'
	057°	2000' (1371')	Refer to Minimums	Rwy 629'

MISSED APCH: Climb STRAIGHT AHEAD to WW652 (MAX 180 KT), turn RIGHT to WW187 (MAX 190 KT) climbing to 3500', then proceed to LO NDB (MANDATORY 200 KT) climbing to 4000', then to BEMAS (MANDATORY 200 KT) to 4000', then to TEBDI (MANDATORY 200 KT), then to TEPTA (MANDATORY 200 KT), then to SIDAZ (MANDATORY 200 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

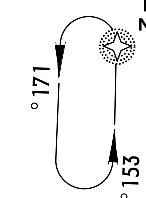
3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

Alt Set: hPa (MM on req) | Rwy Elev: 23 hPa | Trans level: FL110 (1) | Trans alt: 10000'

RNP Apch | RNAV 1 for initial, intermediate and missed approach.

1. GNSS required. 2. Baro-VNAV not authorized below -30°C.

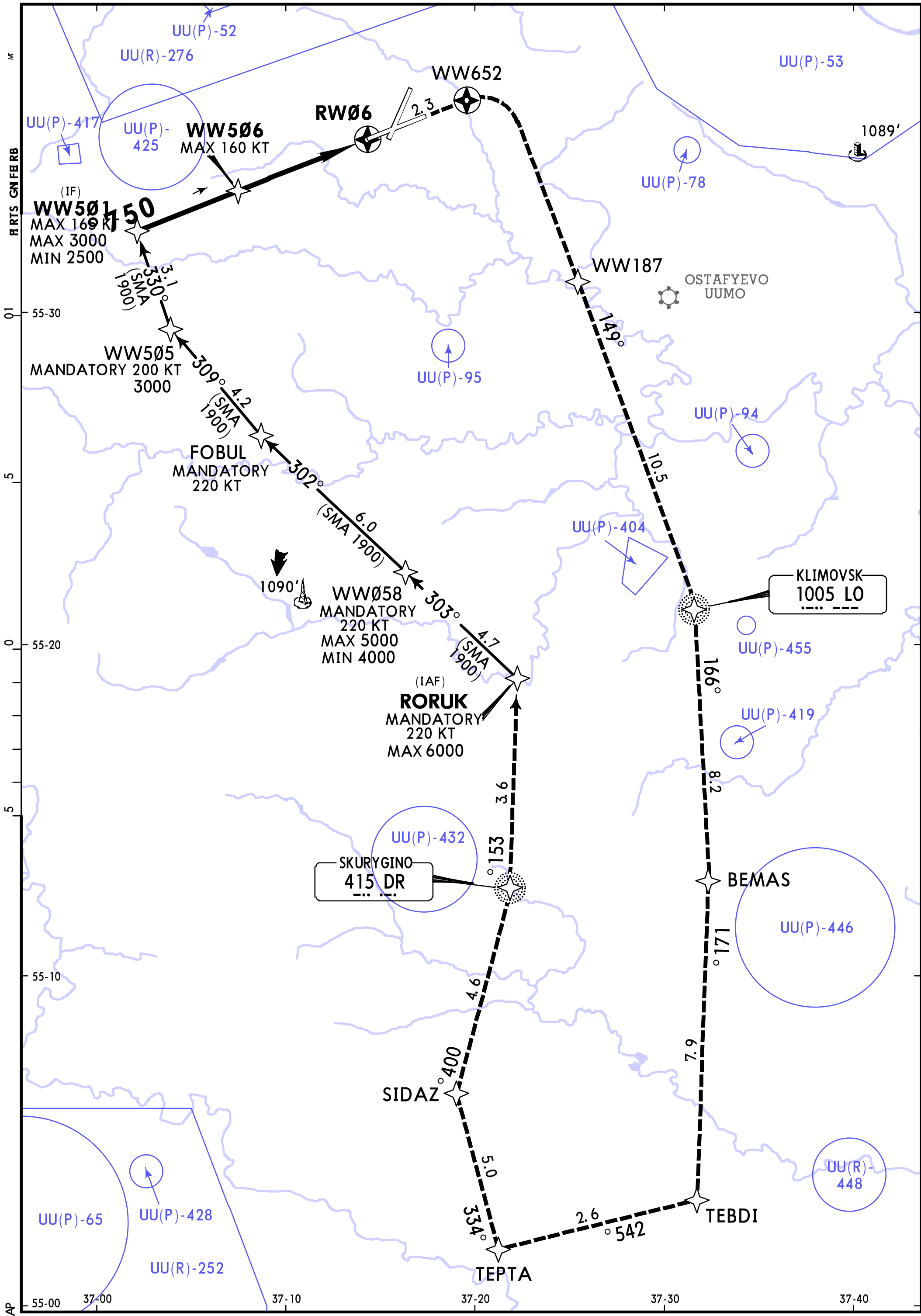
HOLDING FIX



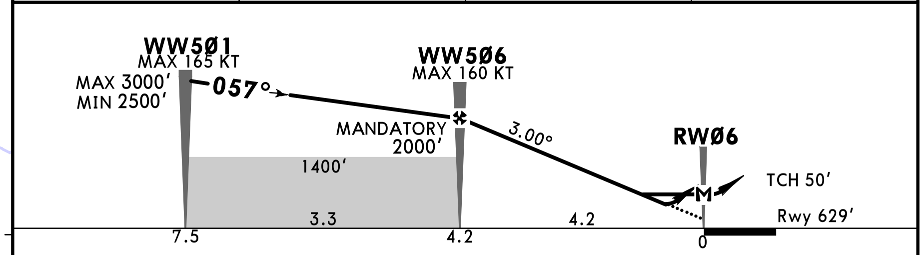
FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(880)
3000	(725)
2500	(575)
2000	(420)
1900	(390)
1710	(330)
1400	(240)
1370	(225)
1020	(120)
1000	(115)

FEET QNH	METERS (QFE)
981	(108)
971	(105)
963	(102)
950	(98)

(1) FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



DIST to RW06	3.2	2.2	1.1
ALTITUDE	1710'	1370'	1020'



Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849
MAP at RW06							

HIALS-II
PAPI
WW652 ↑ | WW187 RT

Std STRAIGHT-IN LANDING

	LNAV/VNAV DA(H) A: 950' (321') C: 971' (342') B: 963' (334') D: 981' (352')		LNAV CDFA (1) DA/MDA(H) 1000' (371')	
		ALS out		ALS out
A	R800m	R1500m	R1000m	R1500m
B	R800m	R1500m	R1000m	R1500m
C	R900m	R1600m	R1000m	R1700m
D	R900m	R1600m	R1000m	R1700m

(1) VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces, general note 1.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

RNAV	Final Apch Crs	WW95Ø MANDATORY	LNAV/VNAV DA(H)	Apt Elev 685'
	192°	1700'(1066')	Refer to Minimums	Rwy 634'

MSA ARP: 3800
MSA ARP is computed for surface air temperature at apt -29.0°C

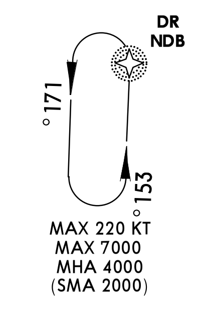
MISSED APCH: Climb STRAIGHT AHEAD to WW952, then turn LEFT to FOBUL (MANDATORY 220 KT) climbing to 3500', then proceed to SODLE (MANDATORY 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000', then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) | Rwy Elev: 23 hPa | Trans level: FL110 ❶ | Trans alt: 10000'

RNP Apch | RNAV 1 for initial, intermediate and missed approach.

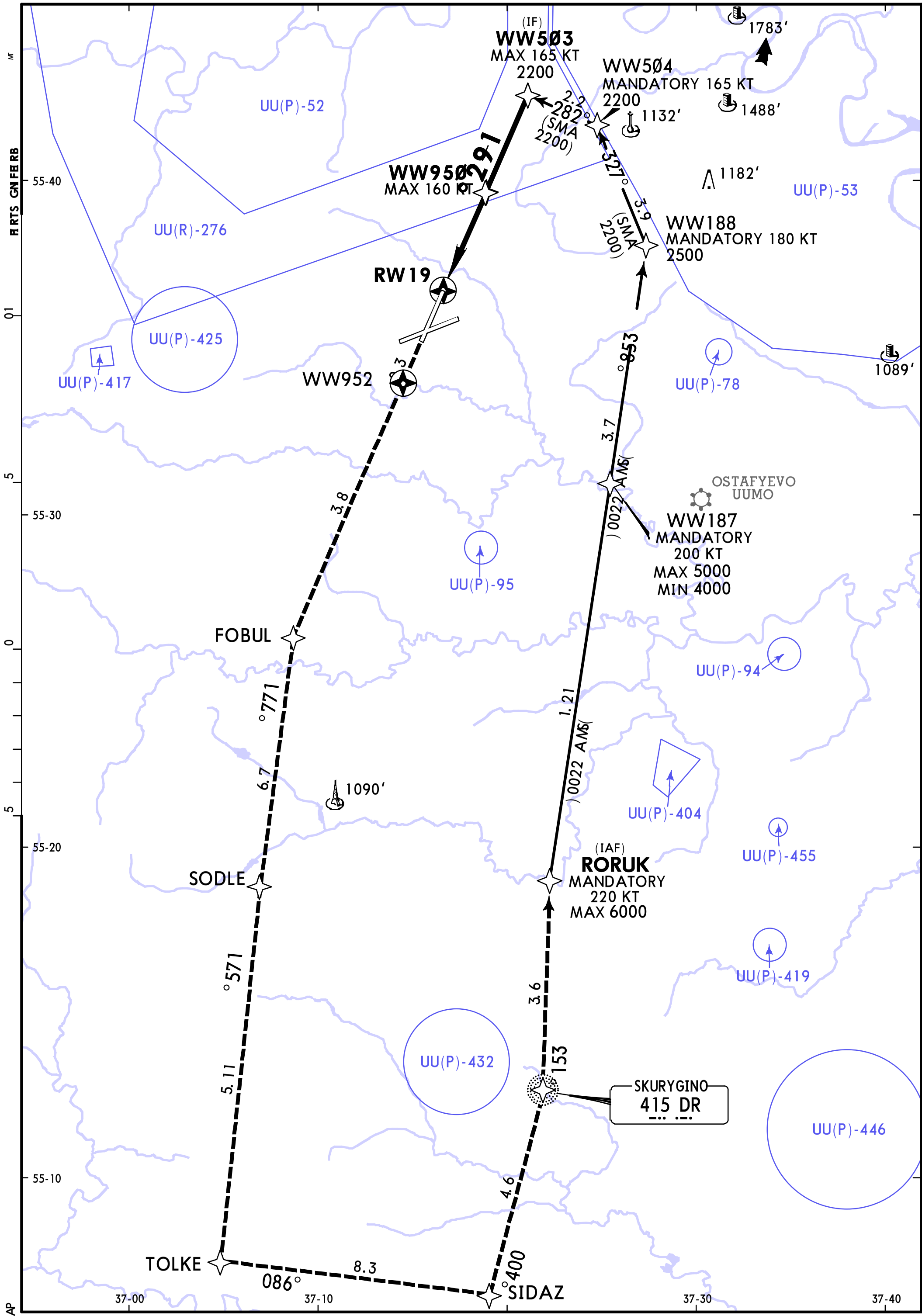
1. GNSS required. 2. Baro-VNAV not authorized below -30°C.

HOLDING FIX

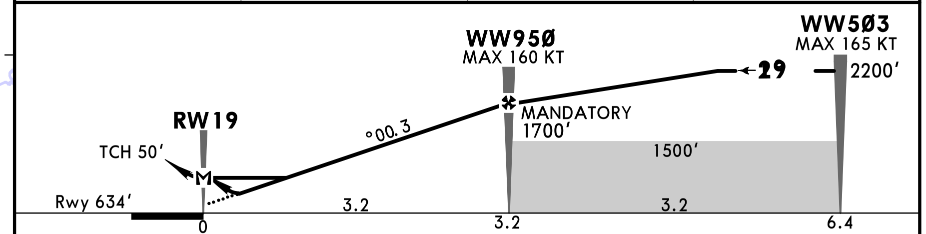


FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
2500	(570)
2200	(480)
1710	(330)
1700	(325)
1500	(265)
1370	(225)
1130	(155)
1030	(125)
1009	(115)
999	(112)
991	(109)
978	(105)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



DIST to RW19	1.1	2.2	3.2
ALTITUDE	1030'	1370'	1710'



Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849

MAP at RW19

HIALS-II
PAPI

WW952

Std STRAIGHT-IN LANDING

	LNAV/VNAV DA(H) A: 978'(344') C: 999'(365') B: 991'(357') D: 1009'(375')	ALS out	LNAV CDFA ❶ DA/MDA(H) 1130'(496')	ALS out
A	R900m	R1500m	R1500m	R1500m
B	R900m	R1500m	R1500m	R1500m
C	R1000m	R1700m	R1500m	R2300m
D	R1000m	R1700m	R1500m	R2300m

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces, general note 1.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-2
118.3	118.3	121.7

RNAV	Final Apch Crs	WW45Ø MANDATORY	LNAV/VNAV DA(H)	Apt Elev 685'
	237°	1700'(1015')	Refer to Minimums	

3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

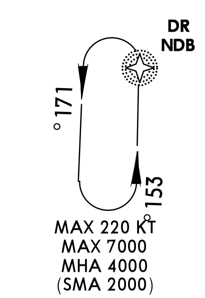
MISSED APCH: Climb STRAIGHT AHEAD to WW452 (MAX 220 KT), then turn LEFT to AMRED (MAX 220 KT) climbing to 3500', proceed to BUPOS (MAX 220 KT), then to SODLE (MAX 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000', then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) | Apt Elev: 25 hPa | Trans level: FL110 1 | Trans alt: 10000'

RNP Apch | RNAV 1 for initial, intermediate and missed approach.

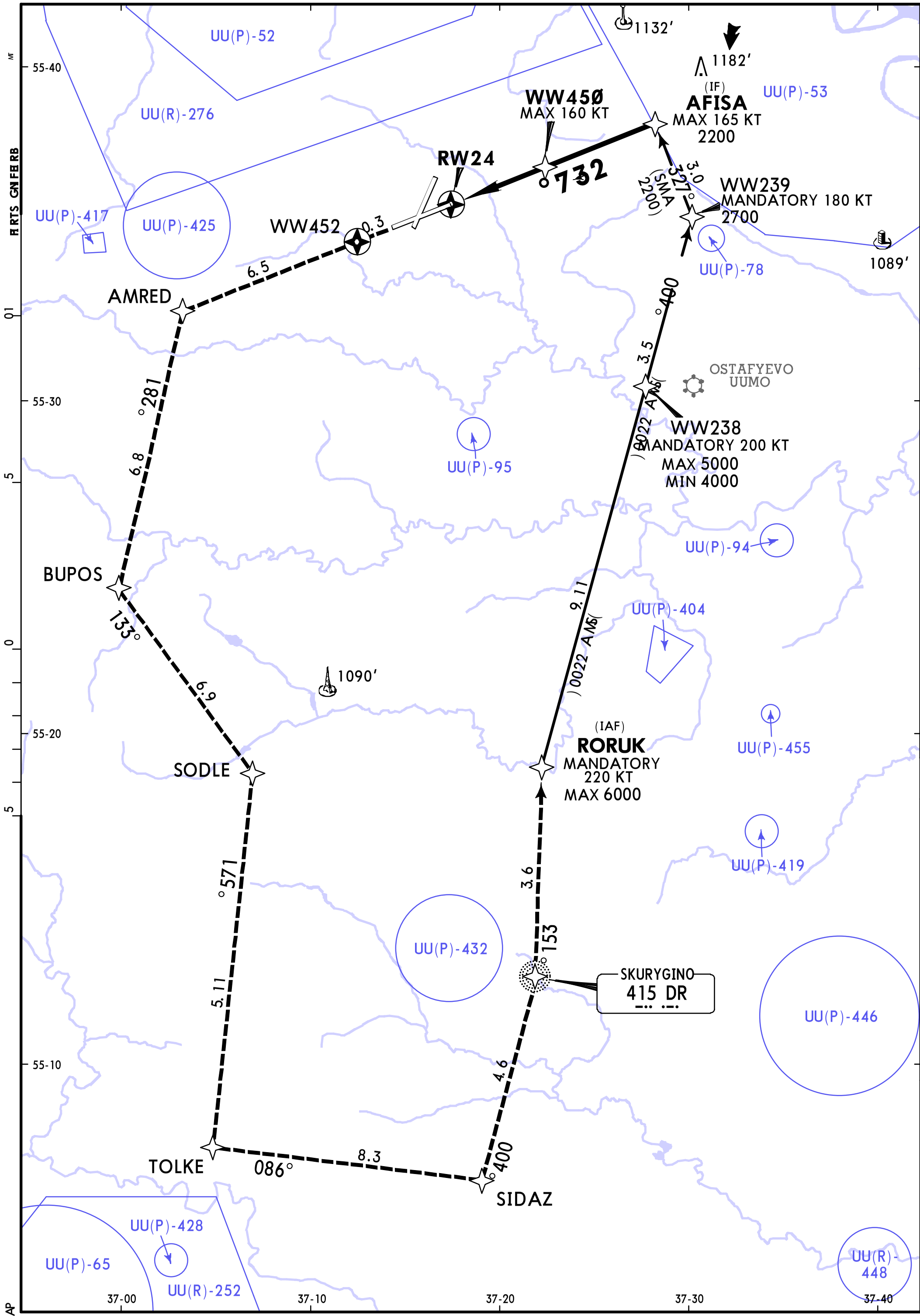
1. GNSS required. 2. Baro-VNAV not authorized below -30°C.

HOLDING FIX

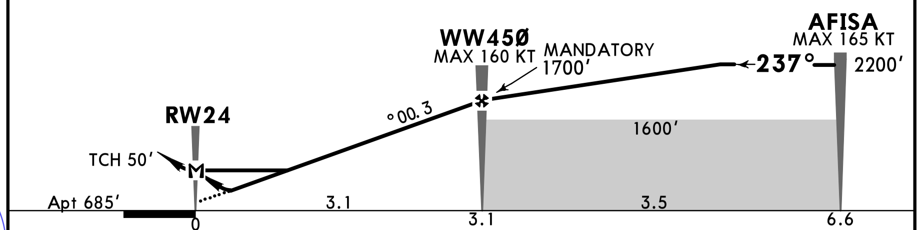


FEET QNH	METERS (QFE)
10000	(2840)
7000	(1925)
6000	(1625)
5000	(1320)
4000	(1015)
3800	(1000)
3500	(860)
2700	(615)
2200	(465)
2000	(405)
1760	(330)
1700	(310)
1600	(280)
1420	(225)
1080	(125)
1070	(120)
1009	(99)
999	(96)
961	(85)
949	(81)

1 FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



DIST to RW24	1.1	2.2	3.2
ALTITUDE	1080'	1420'	1760'



Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849

MAP at RW24

HIALS-II
PAPI

WW452

Std STRAIGHT-IN LANDING

	LNAV/VNAV DA(H) A: 949'(264') C: 999'(314') B: 961'(276') D: 1009'(324')		LNAV CDFA 1 DA/MDA(H) 1070'(385')	
		ALS out		ALS out
A	R750m	R1300m	R1100m	R1500m
B				
C	R800m	R1400m		R1800m
D		R1500m		

1 VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces, general note 1.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

GBAS	Final Apch Crs	WW151 MANDATORY	GLS DA(H)	Apt Elev 685′ Rwy 633′
Ch 20431 G01A	012°	2000′ (1367′)	833′ (200′)	

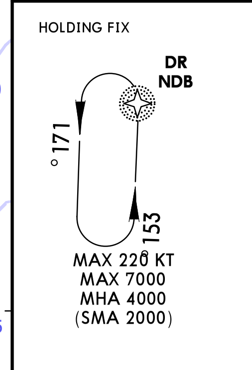
MSA ARP: 3800. MSA ARP is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to 1100′ or above (MAX 190 KT), then turn LEFT to AMRED (MAX 190 KT) climbing to 3500′, then proceed to BOPUS (MANDATORY 220 KT), then to SODLE (MANDATORY 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000′, then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

Alt Set: hPa (MM on req) | Rwy Elev: 23 hPa | Trans level: FL110 ❶ | Trans alt: 10000′

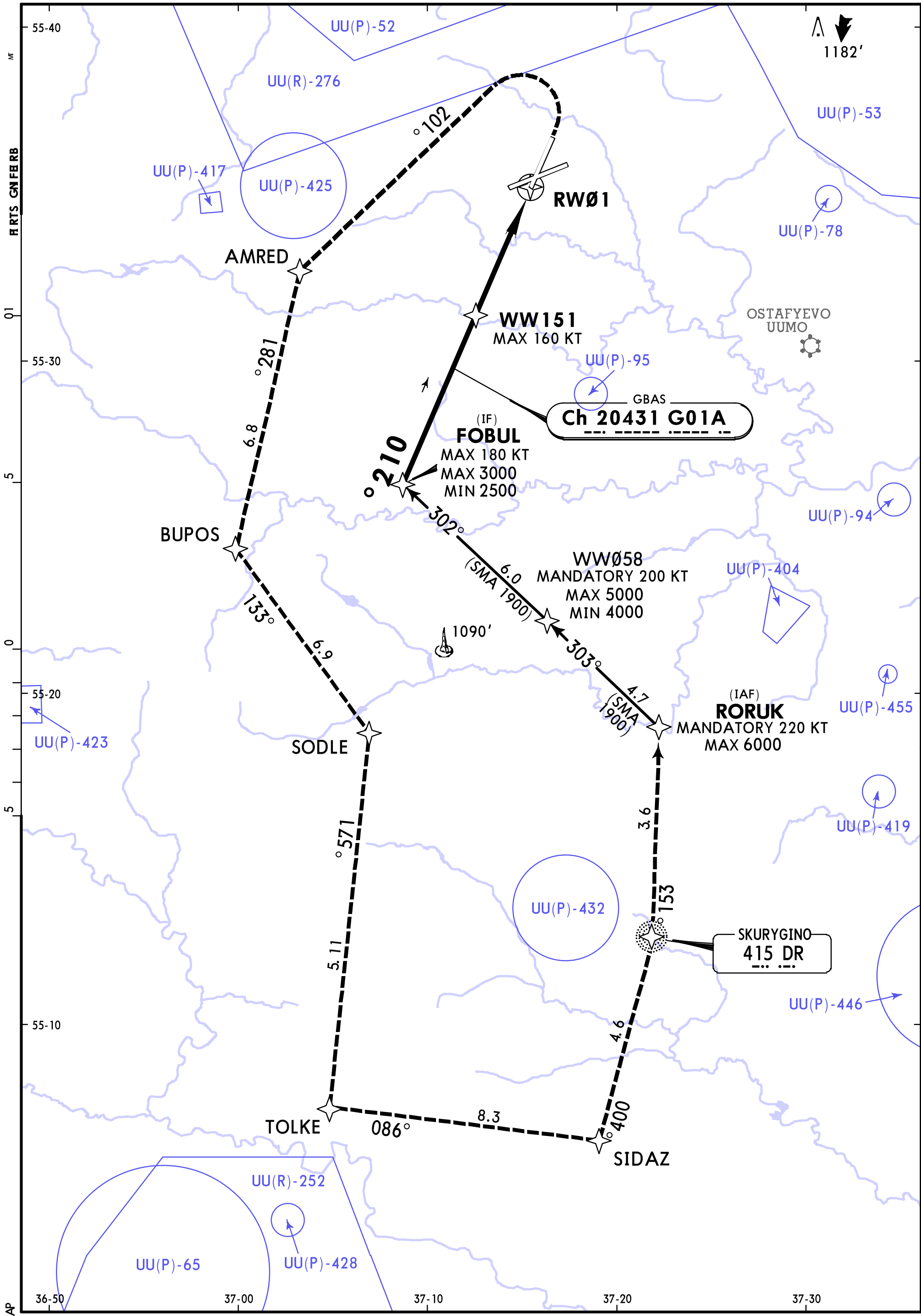
RNAV 1 for initial, intermediate and missed approach.

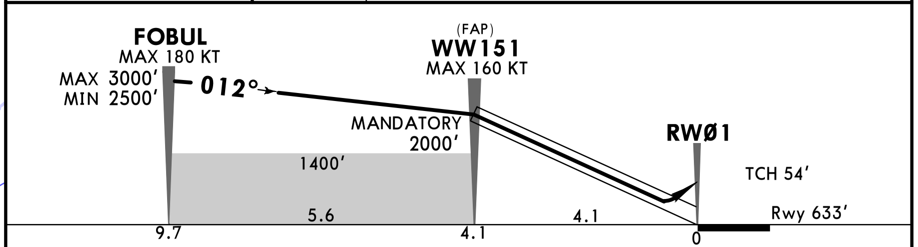
GNSS or DME/DME required.



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
3000	(725)
2500	(570)
2000	(420)
1900	(390)
1400	(235)
1100	(145)
833	(60)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849

HIALS PAPI | MIN 1100′ | AMRED LT

Std

	STRAIGHT-IN LANDING GLS DA(H) 833′ (200′)	
		ALS out
A B C D	❶ R550m	R1200m

❶ R750m when a Flight Director or Autopilot or HUD to DA is not used.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

GBAS	Final Apch Crs	WW502	GLS DA(H)	
Ch 21253 G06A	057°	MANDATORY 2000′(1371′)	829′(200′)	Apt Elev 685′ Rwy 629′

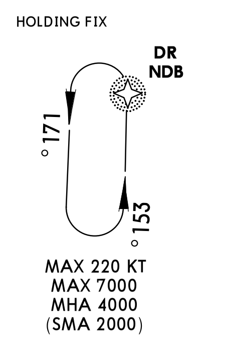
3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to WW652 (MAX 180 KT), then turn RIGHT to WW187 (MAX 190 KT) climbing to 3500′, then proceed to LO NDB (MANDATORY 200 KT) climbing to 4000′, then to BEMAS (MANDATORY 200 KT) to 4000′, then to TEBDI (MANDATORY 200 KT), then to TEPTA (MANDATORY 200 KT), then to SIDAZ (MANDATORY 200 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000′

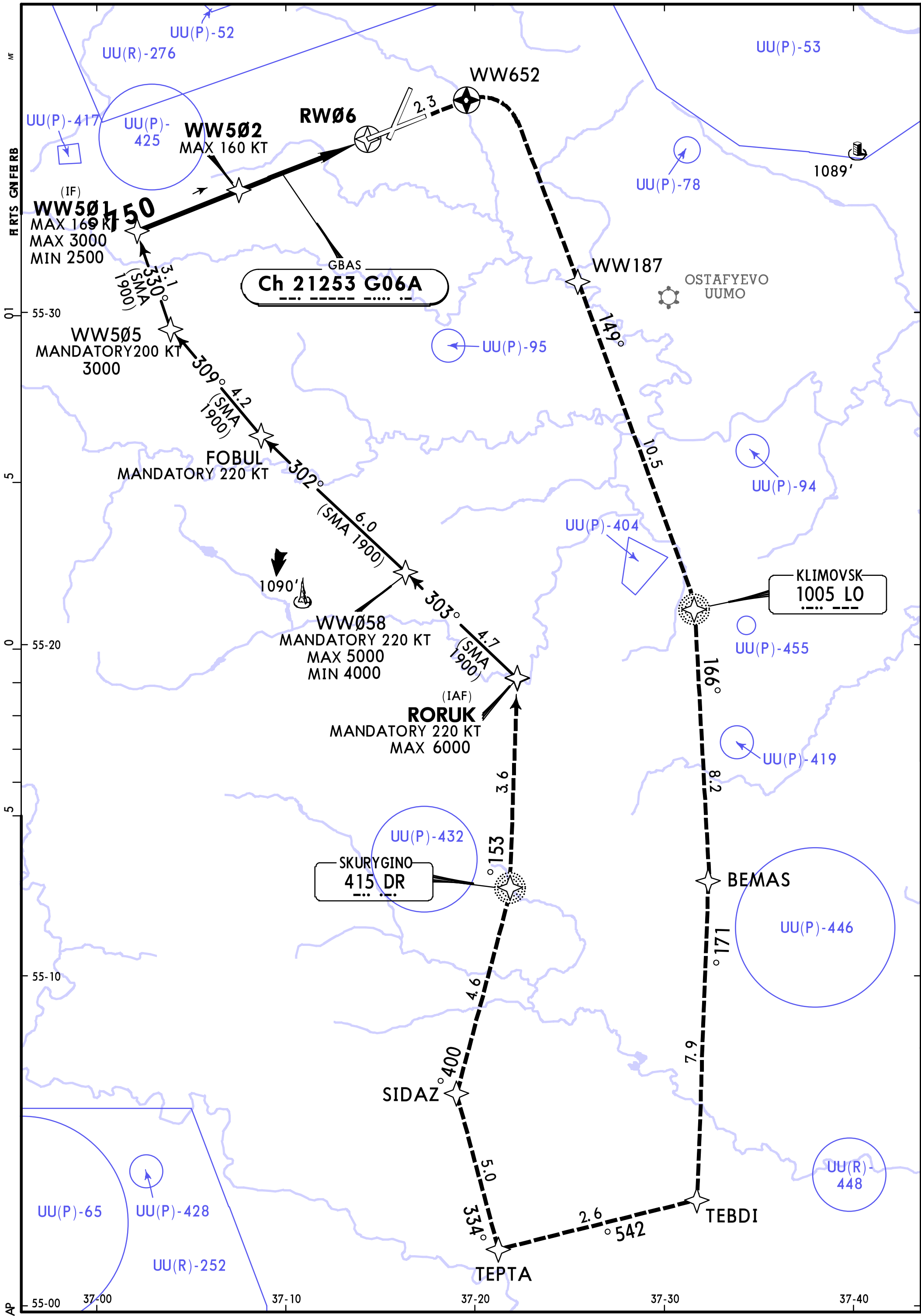
RNAV 1 for initial, intermediate and missed approach.

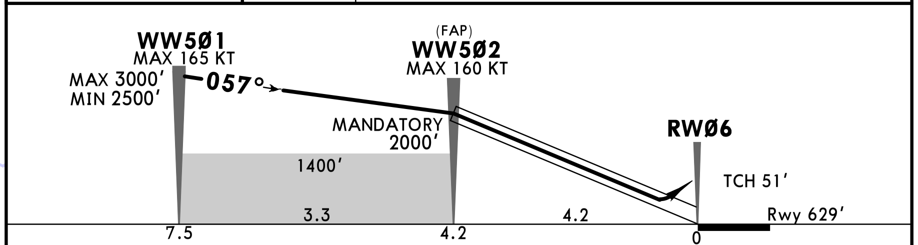
GNSS or DME/DME required.



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(880)
3000	(725)
2500	(575)
2000	(420)
1900	(390)
1400	(240)
829	(60)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849

HIALS-II
PAPI

WW652 WW187 RT

Std

STRAIGHT-IN LANDING

GLS

DA(H) 829′(200′)

	TDZ or CL out	ALS out
A B	R550m	❶ R550m	R1200m
C D			

❶ R750m when a Flight Director or Autopilot or HUD to DA is not used.

CHANGES: Airspaces.

MOSCOW, RUSSIA
GLS Rwy 19

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian 125.875)	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

GBAS	Final Apch Crs	WW951 MANDATORY	GLS DA(H)	Apt Elev 685'
Ch 20842 G19A	192°	1700' (1066')	834' (200')	Rwy 634'

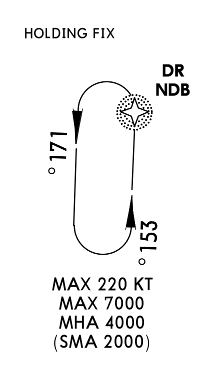
MSA ARP: 3800. MSA ARP is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to WW952, then turn LEFT to FOBUL (MANDATORY 220 KT) climbing to 3500', then proceed to SODLE (MANDATORY 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000', then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 (1) Trans alt: 10000'

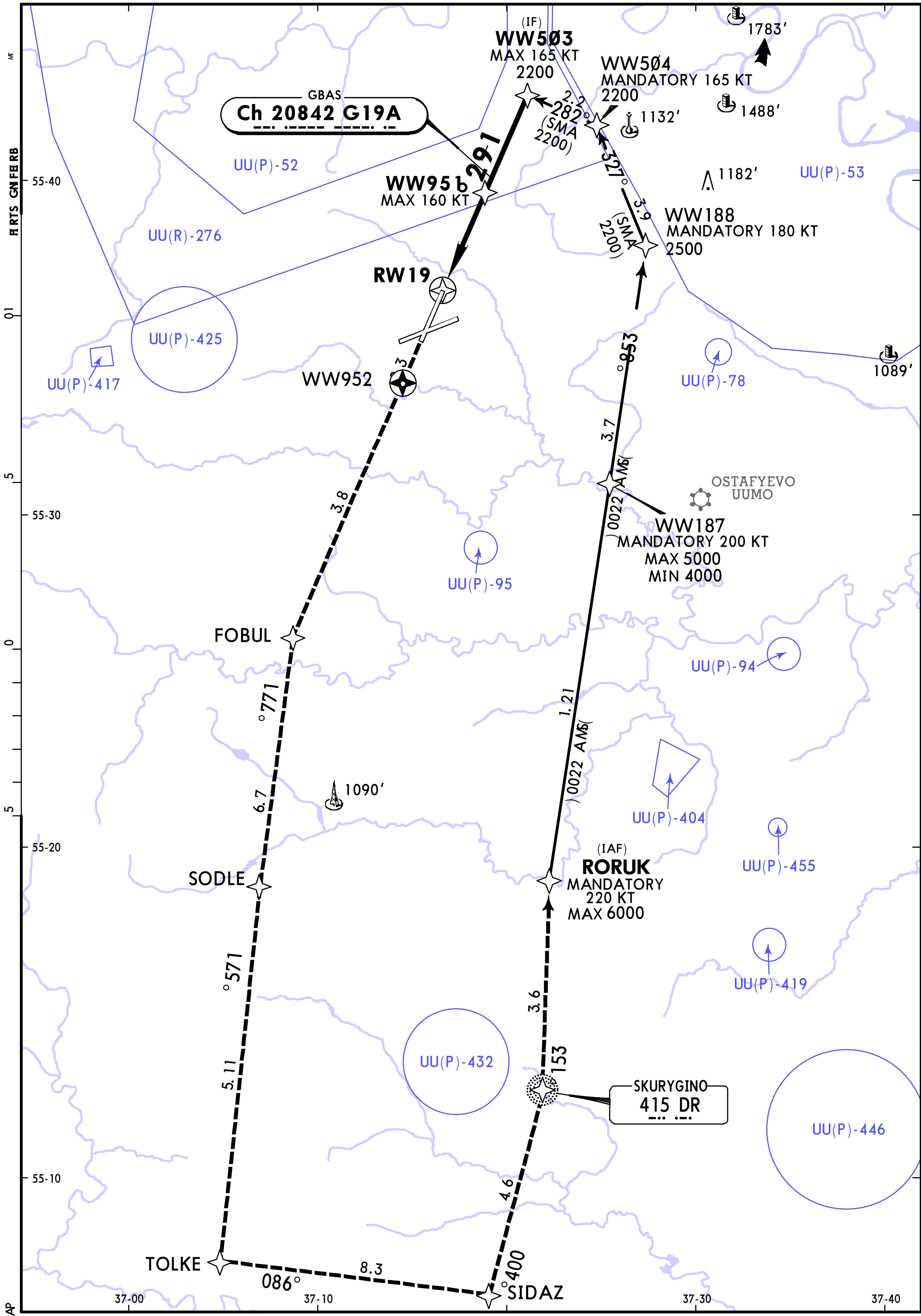
RNAV 1 for initial, intermediate and missed approach.

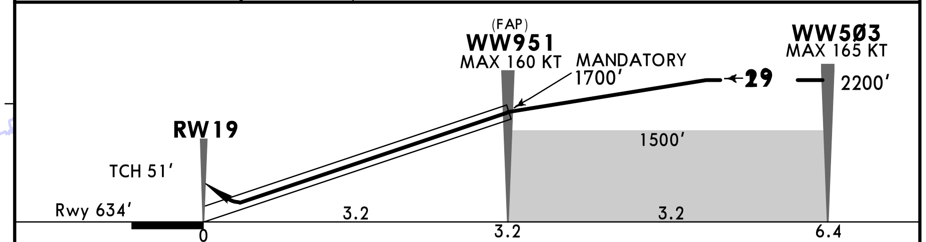
GNSS or DME/DME required.



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
2500	(570)
2200	(480)
1700	(325)
1500	(265)
834	(60)

(1) FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849

HIALS-II
PAPI
WW952

Std

STRAIGHT-IN LANDING
GLS

DA(H) 834' (200')

	TDZ or CL out		ALS out
A B C D	R550m	(1) R550m	R1200m

(1) R750m when a Flight Director or Autopilot or HUD to DA is not used.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-2
118.3	118.3	121.7

GBAS	Final Apch Crs	WW451 MANDATORY	GLS DA(H)	Apt Elev 685' Rwy 685'
Ch 21664 G24A	237°	1700'(1015')	885'(200')	

3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

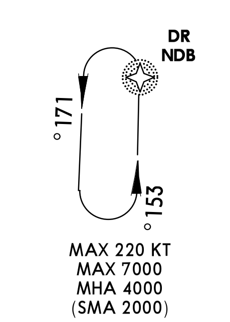
MISSED APCH: Climb STRAIGHT AHEAD to WW452 (MAX 220 KT), then turn LEFT to AMRED (MAX 220 KT) climbing to 3500', then proceed to BUPOS (MAX 220 KT), then to SODLE (MAX 220 KT), then to TOLKE (MANDATORY 220 KT) climbing to 4000', then to SIDAZ (MANDATORY 220 KT), then to DR NDB (MANDATORY 220 KT), then to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 25 hPa Trans level: FL110 ❶ Trans alt: 10000'

RNAV 1 for initial, intermediate and missed approach.

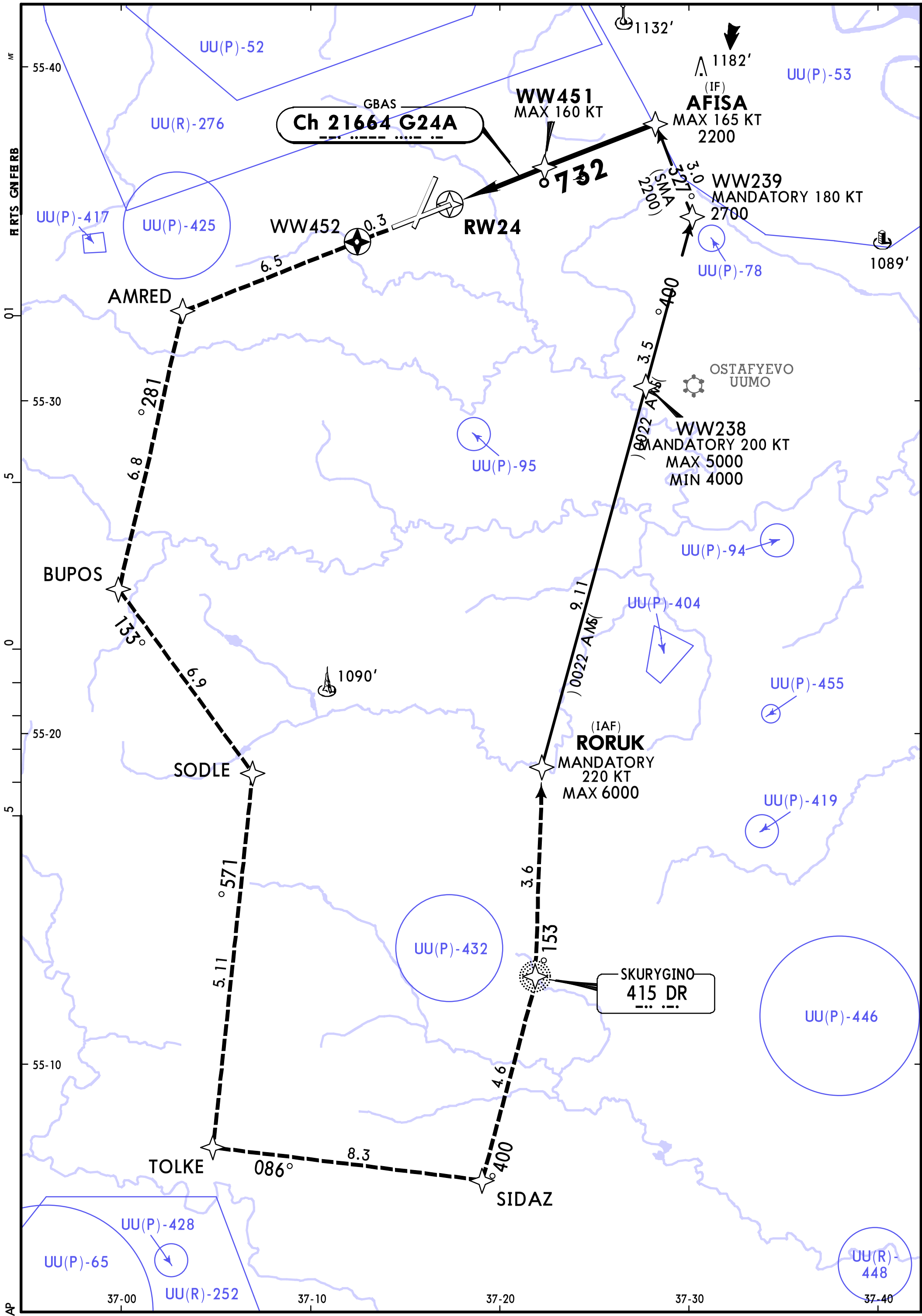
GNSS or DME/DME required.

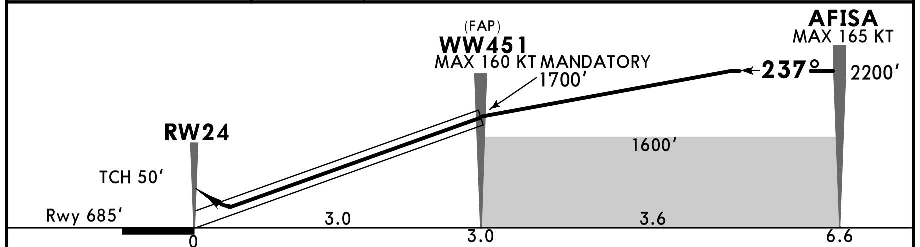
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1925)
6000	(1625)
5000	(1320)
4000	(1015)
3800	(1000)
3500	(860)
2700	(615)
2200	(465)
2000	(405)
1700	(310)
1600	(280)
885	(60)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849

HIALS-II
PAPI
WW452

Std

STRAIGHT-IN LANDING
GLS
DA(H) 885'(200')

		TDZ or CL out	ALS out
A	R550m	❶ R550m	R1200m
B			
C			
D			

❶ R750m when a Flight Director or Autopilot or HUD to DA is not used.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

VOR WNK	Final Apch Crs	D4.2 MANDATORY	DA/MDA(H)	Apt Elev 685′
113.7	002°	2000′(1367′)	Refer to Minimums	Rwy 633′

MISSED APCH: Climb STRAIGHT AHEAD to D2.1, then turn RIGHT onto 154° LO NDB to LO NDB (MANDATORY 200 KT) climbing to 4000′, then onto 346° LO NDB and proceed to BEMAS (MANDATORY 200KT) to 4000′, then turn RIGHT to DR NDB (MANDATORY 220 KT), then onto 171° DR NDB proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

3800

MSA ARP is computed for surface air temperature at apt -29.0°C

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000′

RNAV 1 for initial approach. 1. GNSS or DME/DME required. 2. Final aprroach track offset bz 10° from runway centerline.

HOLDING FIX

DR NDB

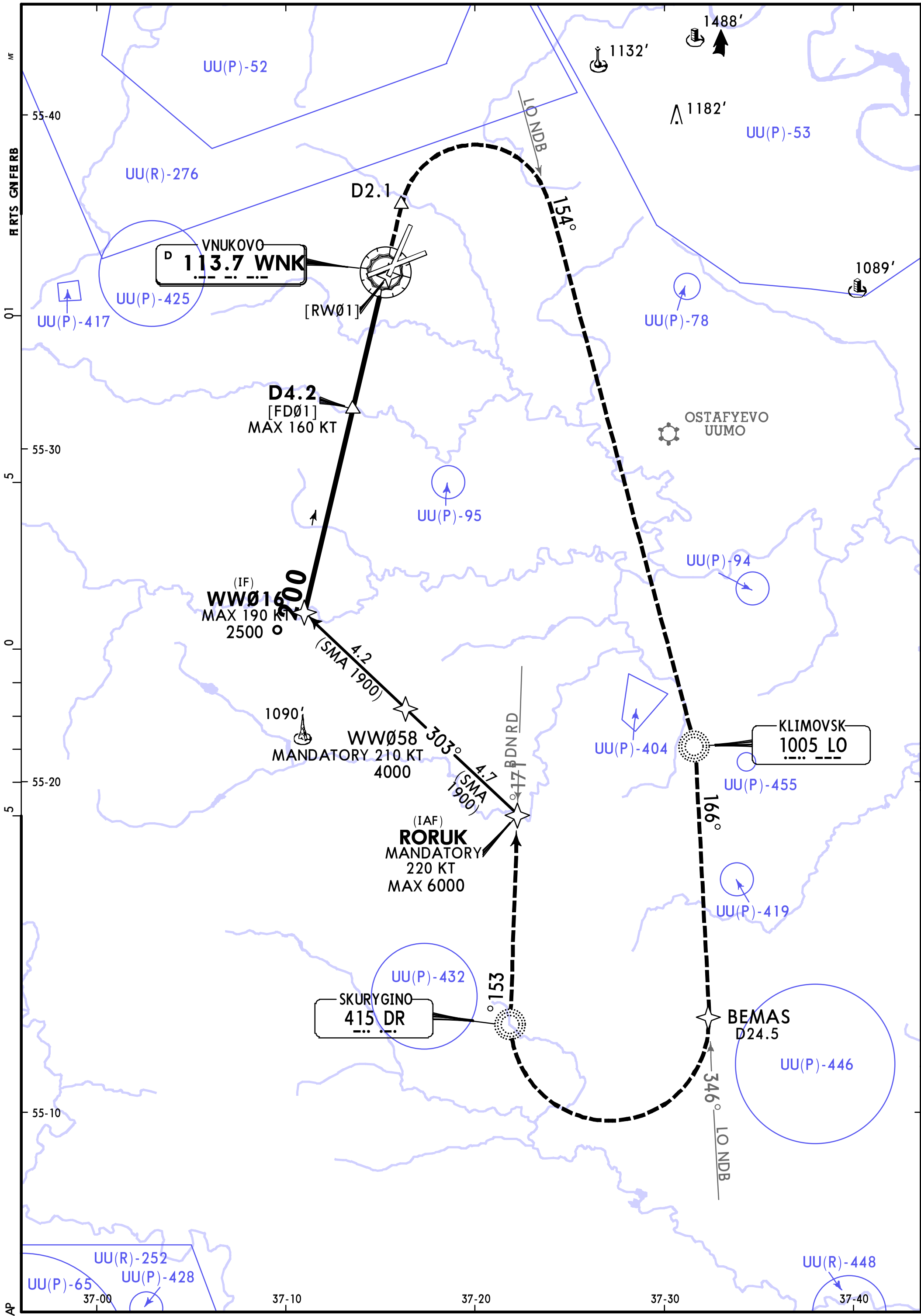
171°

153°

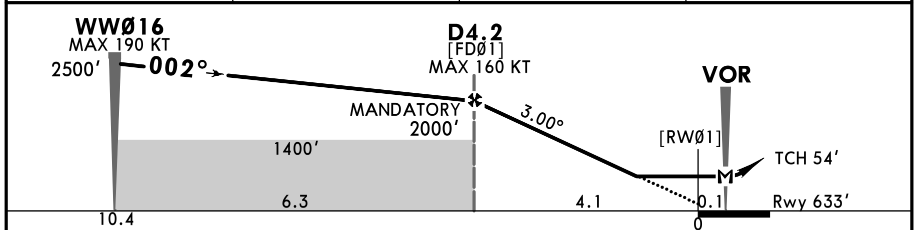
MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
4000	(1030)
3800	(1000)
2500	(570)
2000	(420)
1900	(390)
1690	(325)
1400	(235)
1350	(220)
1070	(135)
1050	(130)
1020	(120)
990	(110)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



WNK DME	4.2	3.2	2.2
ALTITUDE	2000′	1690′	1350′



Gnd speed-Kts		70	90	100	120	140	160
Descent Angle	3.00°	372	478	531	637	743	849
MAP at VOR							

HIALS PAPI

D2.1 LO 1005 RT onto LO 1005 154°

Std

STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) A: 990′(357′) C: 1050′(417′)
B: 1020′(387′) D: 1070′(437′)

		ALS out
A	R900m	R1500m
B	R1100m	
C	R1200m	R1900m
D	R1300m	R2000m

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

VOR WNK	Final Apch Crs	D4.7 MANDATORY	DA/MDA(H)	
113.7	059°	2000′(1371′)	1100′(471′)	Apt Elev 685′ Rwy 629′

3800

MSA ARP is computed for surface air temperature at apt -29.0°C

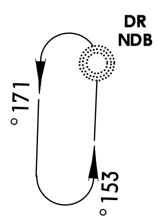
MISSED APCH: Climb STRAIGHT AHEAD to D2.5, then turn RIGHT onto 142° LO NDB to LO NDB (MANDATORY 200 KT) climbing to 4000′, then onto 346° LO NDB and proceed to BEMAS (MANDATORY 200 KT) to 4000, then turn RIGHT to DR NDB (MANDATORY 220 KT), then onto 171° DR NDB proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

Alt Set: hPa (MM on req) | Rwy Elev: 23 hPa | Trans level: FL110 ❶ | Trans alt: 10000′

RNAV 1 for initial approach.

1. GNSS or DME/DME required. 2. Final approach offset by 2° from runway centerline.

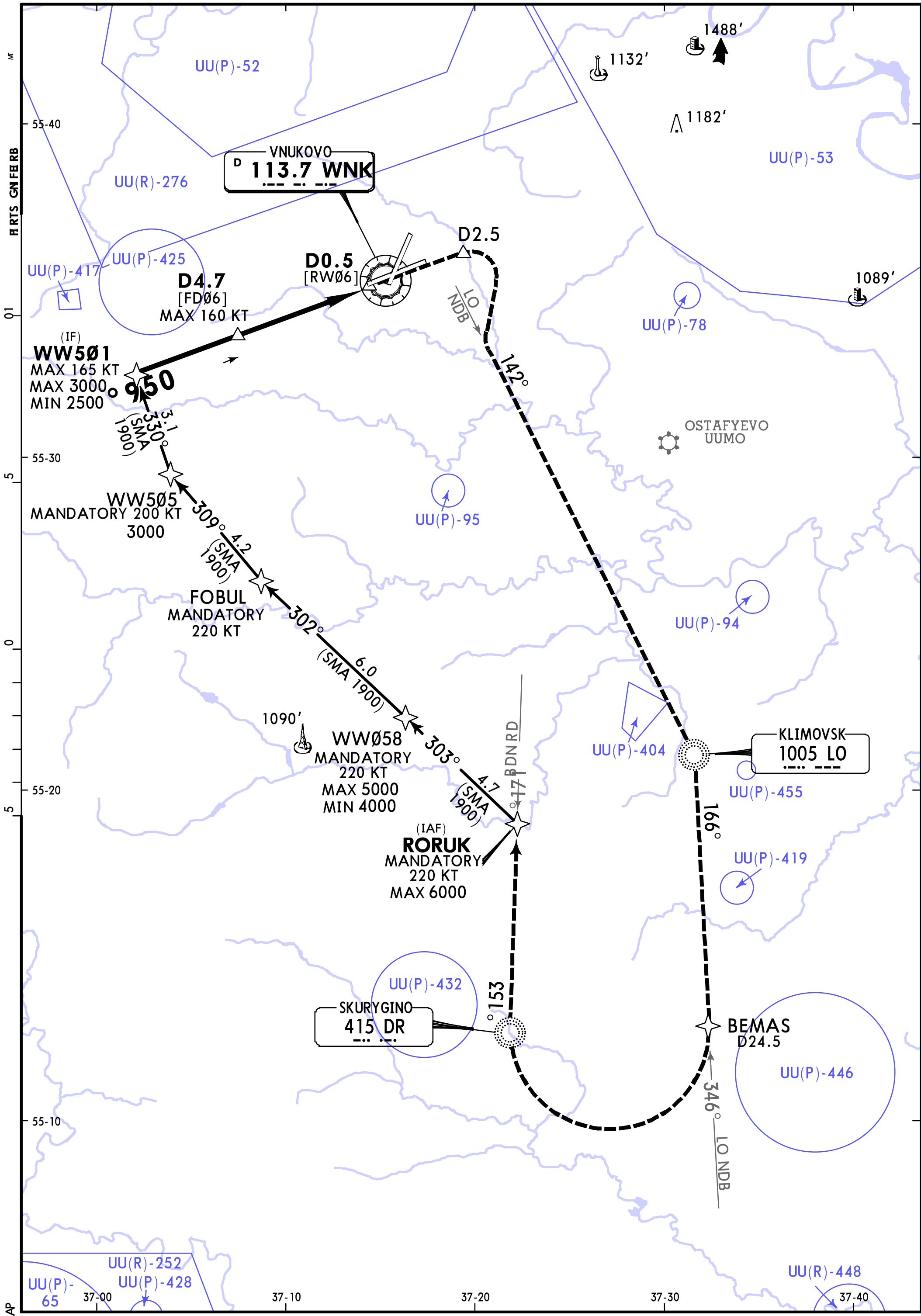
HOLDING FIX



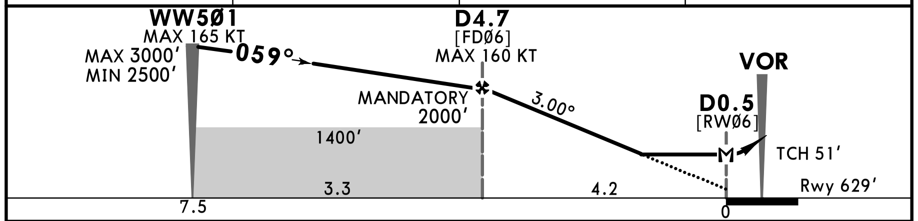
MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

FEET	METERS QNH (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
4000	(1030)
3800	(1000)
3000	(725)
2500	(575)
2000	(420)
1900	(390)
1700	(330)
1540	(280)
1400	(240)
1200	(175)
1100	(145)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



WNK DME	3.8	3.2	2.2
ALTITUDE	1700′	1540′	1200′



Gnd speed-Kts		70	90	100	120	140	160
Descent Angle	3.00°	372	478	531	637	743	849
MAP at D0.5							

HIALS-II
PAPI
D2.5
LO 1005 RT
onto
LO 1005 142°

Std

STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1100′(471′)

		ALS out
A	R1500m	R1500m
B		
C		R2200m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

VOR WNK	Final Apch Crs	D4.8 MANDATORY	DA/MDA(H)	Apt Elev 685′ Rwy 634′
113.7	194°	1700′(1066′)	1120′(486′)	

MSA ARP 3800
MSA ARP is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to D1.5, then turn LEFT onto 191° WZ NDB to WZ NDB climbing to 3500′, then onto 082° DR NDB to DR NDB (MANDATORY 220 KT), then turn LEFT onto 171° DR NDB and proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000′

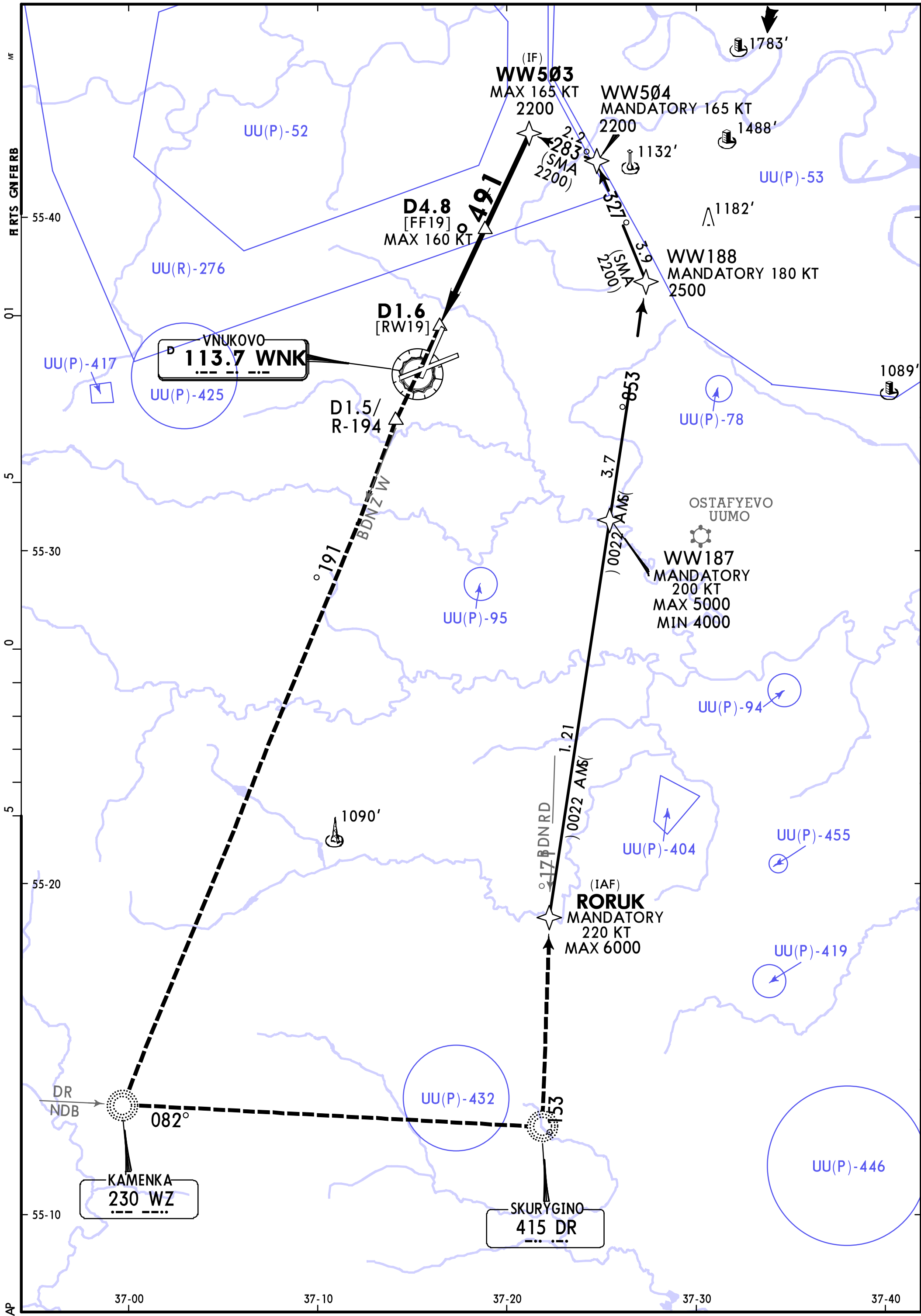
RNAV 1 for initial approach.

1. GNSS or DME/DME required. 2. Final approach offset by 2° from runway centerline.

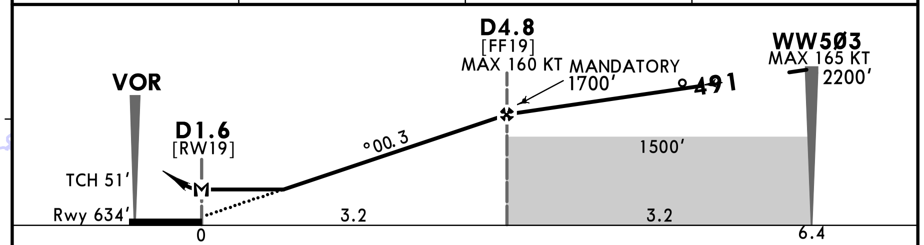
HOLDING FIX
DR NDB
171° 153°
MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
2500	(570)
2200	(480)
2000	(420)
1700	(325)
1540	(280)
1500	(265)
1200	(175)
1120	(150)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



WNK DME	3.2	4.3	4.8
ALTITUDE	1200′	1540′	1700′



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at D1.6						

HIALS-II
PAPI
D1.5

Std STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1120′(486′)

		ALS out
A	R1500m	R1500m
B		
C		R2300m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-2
118.3	118.3	121.7

VOR WNK	Final Apch Crs	D4.4 MANDATORY	DA/MDA(H)	Apt Elev 685'
113.7	235°	1700' (1015')	1060' (375')	

3800

MSA ARP is computed for surface air temperature at apt -29.0°C

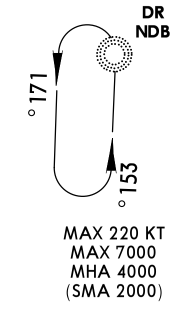
MISSED APCH: Climb STRAIGHT AHEAD to D1.7, then turn LEFT onto 186° WZ NDB to WZ NDB climbing to 3500', then onto 082° DR NDB to DR NDB (MANDATORY 220 KT), then turn LEFT onto 171° DR NDB and proceed to proceed to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.

Alt Set: hPa (MM on req) Apt Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000'

RNAV 1 for initial approach.

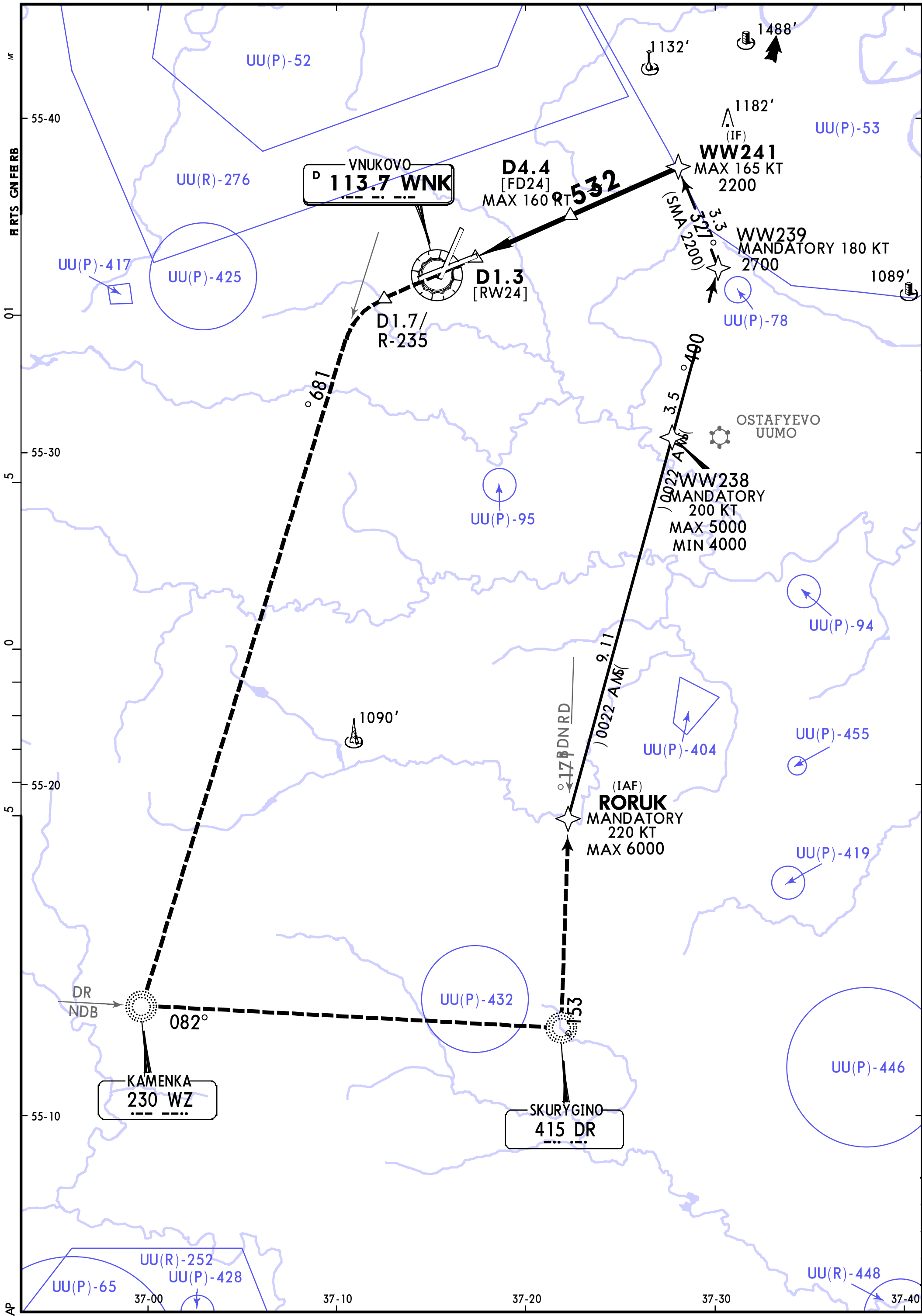
1. GNSS or DME/DME required. 2. Final approach offset by 2° from runway centerline.

HOLDING FIX

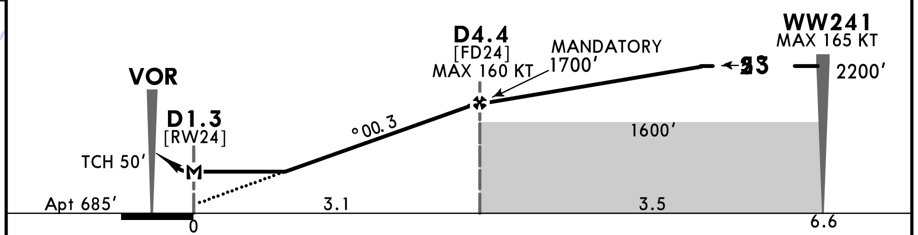


FEET QNH	METERS (QFE)
10000	(2900)
7000	(1925)
6000	(1625)
5000	(1320)
4000	(1015)
3800	(1000)
3500	(860)
2700	(615)
2200	(465)
2000	(405)
1700	(310)
1670	(305)
1600	(280)
1330	(200)
1060	(115)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



WNK DME	3.2	4.3	4.4
ALTITUDE	1330'	1670'	1700'



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at D1.3						

HIALS-II
PAPI
D1.7

Std

STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1060' (375')

		ALS out
A	R1000m	R1500m
B		
C		R1700m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Airspaces.

MOSCOW, RUSSIA
NDB Rwy 01

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

NDB OE 949	Final Apch Crs 012°	4.2 NM RADAR FIX MANDATORY 2000′ (1367′)	DA/MDA(H) 980′ (347′)	Apt Elev 685′ Rwy 633′

MISSED APCH: Climb on 012° OE NDB to 1300′ or above, then turn RIGHT onto 154° LO NDB to LO NDB (MANDATORY 200 KT) climbing to 4000′, then onto 346° LO NDB and proceed to BEMAS (MANDATORY 200 KT) to 4000′, then turn RIGHT to DR NDB (MANDATORY 220 KT), then onto 171° DR NDB proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed. Turn before MAP is PROHIBITED.

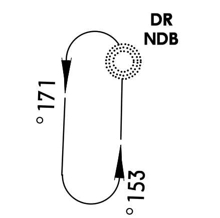
3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 1 Trans alt: 10000′

RNAV 1 for initial approach.

GNSS or DME/DME required.

HOLDING FIX

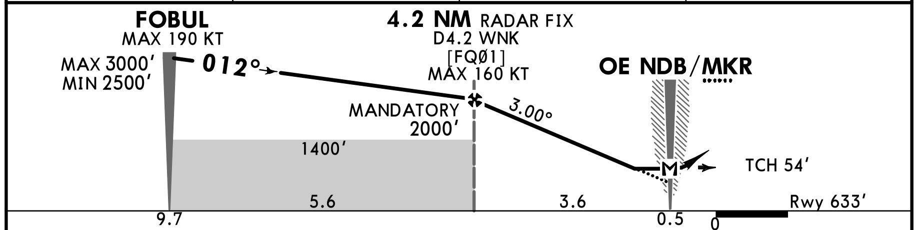


MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3000	(725)
2500	(570)
2000	(420)
1900	(390)
1690	(325)
1400	(235)
1350	(220)
1300	(205)
980	(110)

1 FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).

WNK DME	4.2	3.2	2.2
ALTITUDE	2000′	1690′	1350′



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at OE NDB/MKR						
Timing not authorized for defining MAP.						

HIALS PAPI | OE 949 012° | MIN 1300′ | LO 1005 onto RT | LO 1005 154°

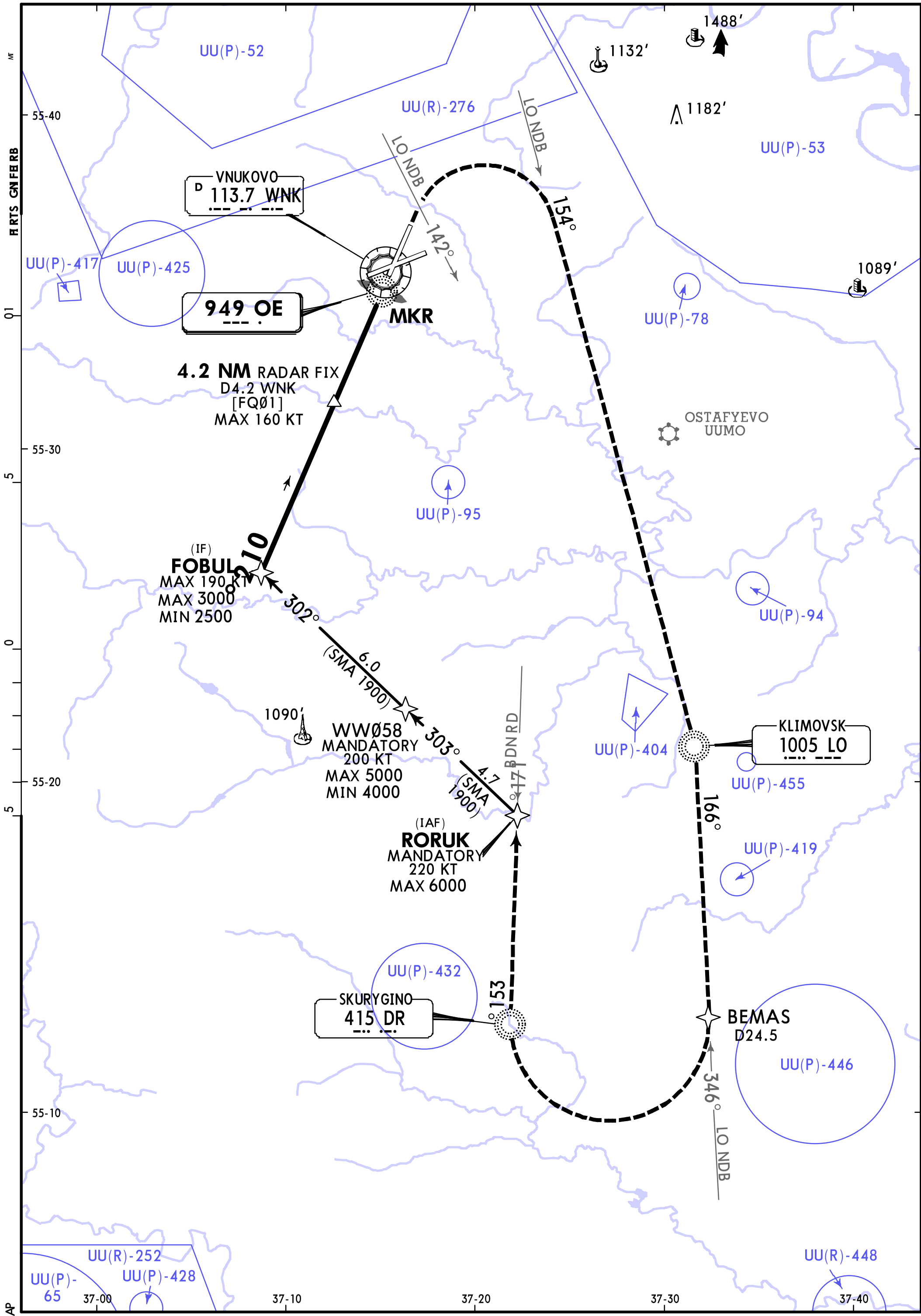
Std STRAIGHT-IN LANDING

CDFA

1 DA/MDA(H) 980′ (347′)

		ALS out
A	R900m	R1500m
B		
C		R1600m
D		

1 VNAV DA(H) in lieu of MDA(H) depends on operator policy.



CHANGES: FAF, missed approach.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

NDB	Final Apch Crs	4.2 NM RADAR FIX MANDATORY	DA/MDA(H)	Apt Elev 685'
GT 294	057°	2000'(1371')	1100'(471')	Rwy 629'

MSA ARP 3800 — MSA ARP is computed for surface air temperature at apt -29.0°C

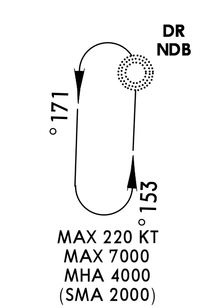
MISSED APCH: Climb STRAIGHT AHEAD to 1500' or above, then turn RIGHT onto 142° LO NDB to LO NDB (MANDATORY 200 KT) climbing to 4000', then onto 346° LO NDB and proceed to BEMAS (MANDATORY 200 KT) to 4000', then turn RIGHT to DR NDB (MANDATORY 220 KT), then onto 171° DR NDB proceed to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed. Turn before MAP is PROHIBITED.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000'

RNAV 1 for initial approach.

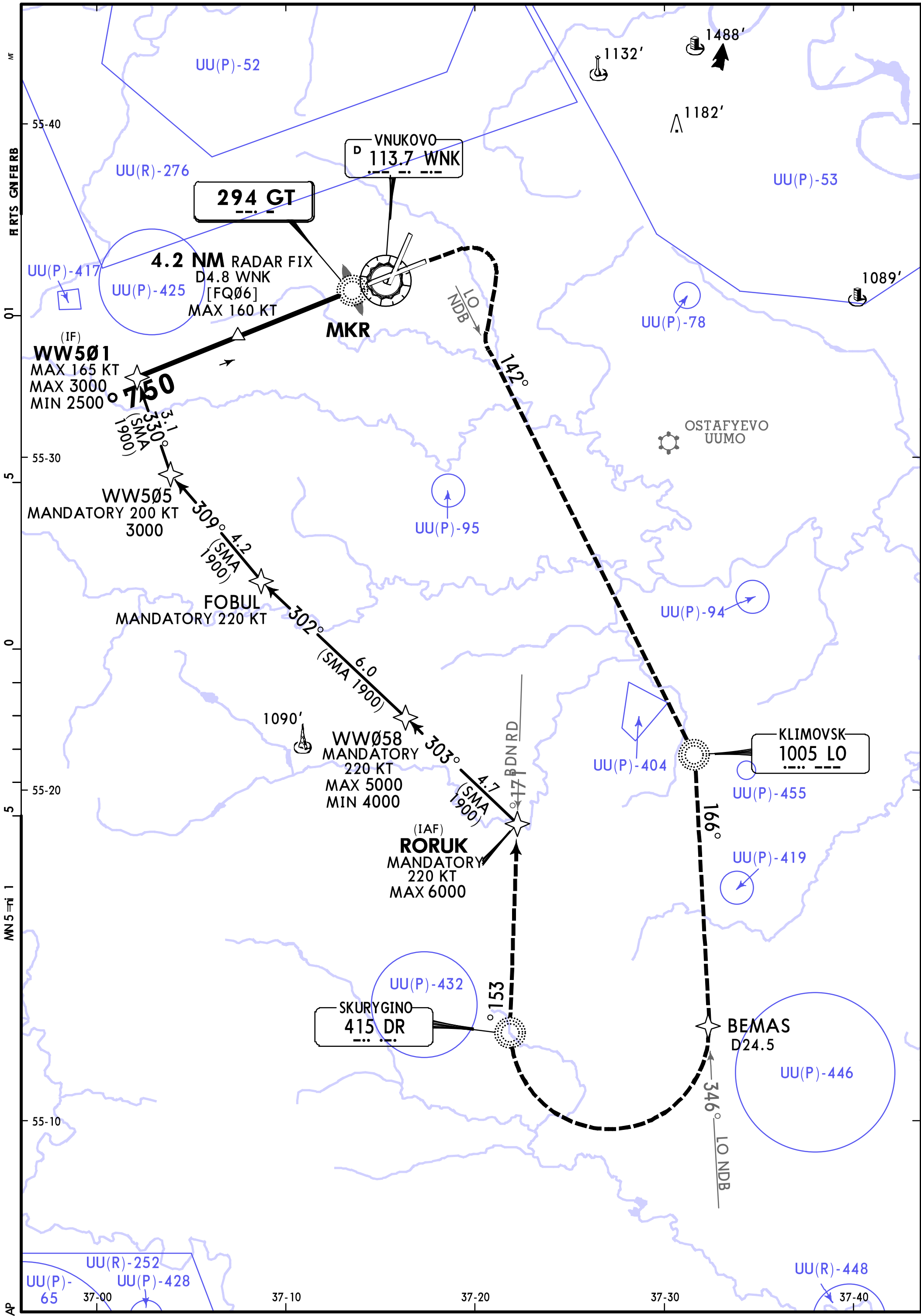
GNSS or DME/DME required.

HOLDING FIX

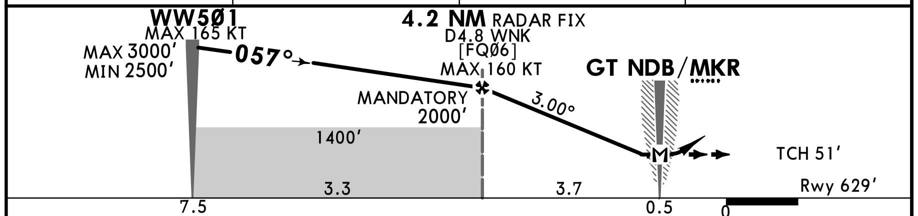


FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3000	(725)
2500	(575)
2000	(420)
1900	(390)
1700	(330)
1540	(280)
1500	(270)
1400	(240)
1200	(175)
1100	(145)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).



WNK DME	3.8	3.2	2.2
ALTITUDE	1700'	1540'	1200'



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at GT NDB/MKR						
Timing not authorized for defining MAP.						

HIALS-II PAPI — MIN 1500' — LO 1005 RT — LO 1005 onto 142°

Std STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1100'(471')

	ALS out	
A	R1500m	
B		
C	R1500m	R2200m
D		

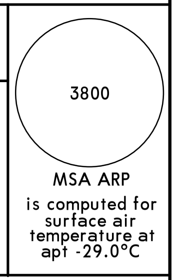
❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: FAF, missed approach.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

NDB	Final Apch Crs	3.2 NM RADAR FIX MANDATORY	DA/MDA(H)	Apt Elev 685′
SX 914	192°	1700′(1066′)	1120′(486′)	Rwy 634′



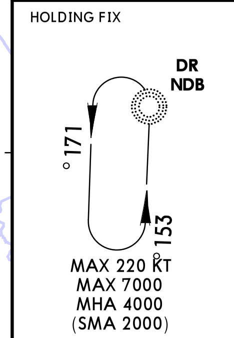
is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to 1500′ or above, then turn LEFT onto 191° WZ NDB to WZ NDB climbing to 3500′, then onto 082° DR NDB to DR NDB (MANDATORY 220 KT), then turn LEFT onto 171° DR NDB and proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.
Turn before MAP is PROHIBITED.

Alt Set: hPa (MM on req) | Rwy Elev: 23 hPa | Trans level: FL110 ❶ | Trans alt: 10000′

RNAV 1 for initial approach.
GNSS or DME/DME required.

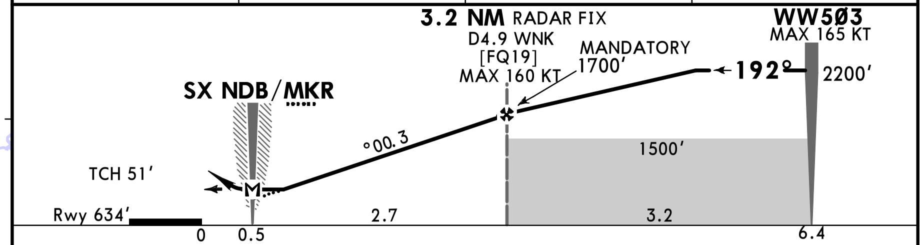
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
2500	(570)
2200	(480)
2000	(420)
1700	(325)
1540	(280)
1500	(265)
1500	(250)
1200	(175)
1120	(150)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).

WNK DME	3.2	4.3	4.8
ALTITUDE	1200′	1540′	1700′



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at SX NDB/MKR						
Timing not authorized for defining MAP.						

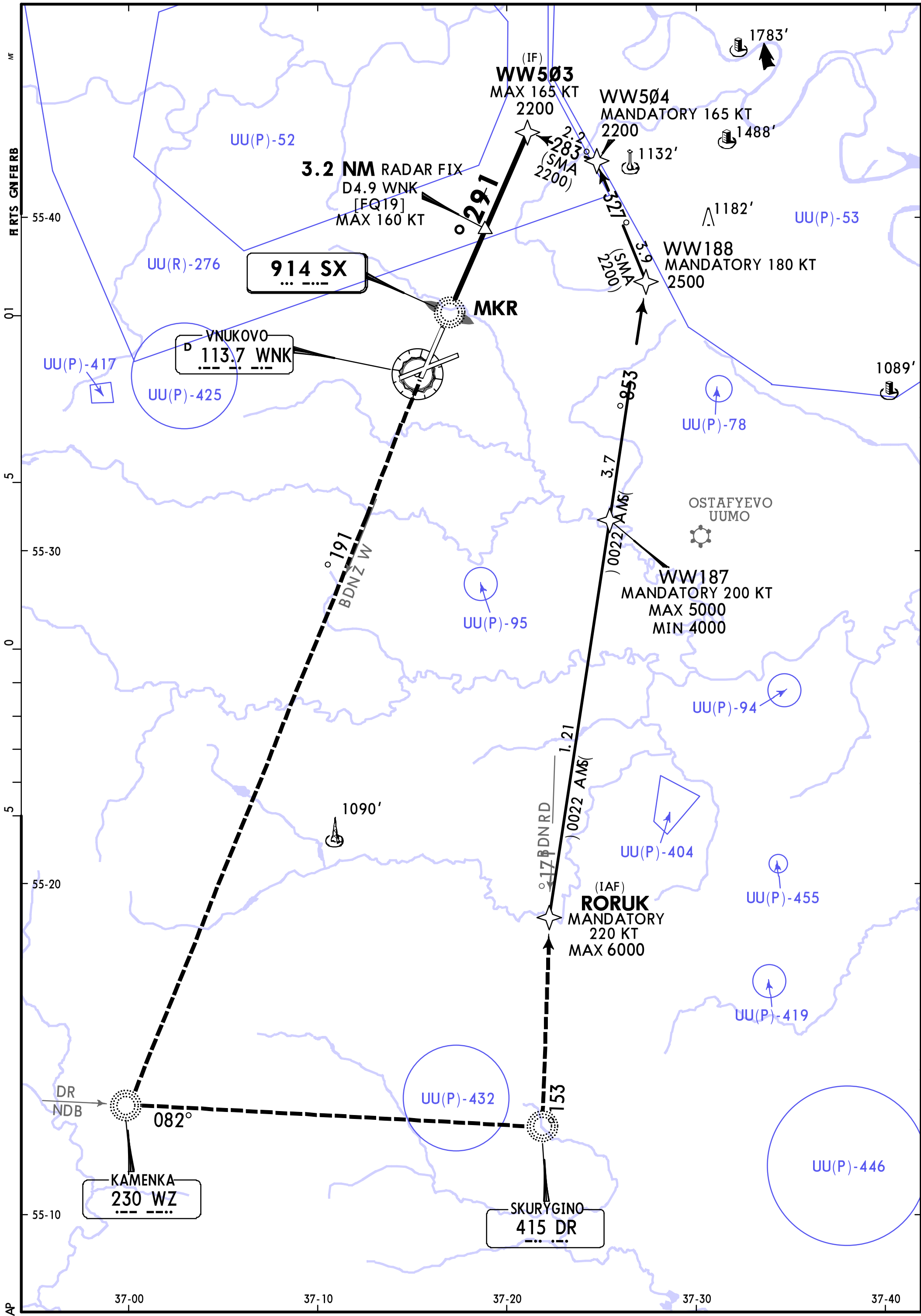
HIALS-II
PAPI
MIN 1500′

Std

STRAIGHT-IN LANDING
CDFA
❶ DA/MDA(H) 1120′(486′)

	ALS out	
A	R1500m	
B		
C	R1500m	R2300m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.



CHANGES: FAF, missed approach.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-2
118.3	118.3	121.7

NDB OB	Final Apch Crs	3.0 NM RADAR FIX MANDATORY	DA/MDA(H)	Apt Elev 685'
852	237°	1700'(1015')	1150'(465')	

3800

MSA ARP is computed for surface air temperature at apt -29.0°C

MISSED APCH: Climb STRAIGHT AHEAD to 1500' or above, then turn LEFT onto 186° WZ NDB to WZ NDB climbing to 3500', then onto 082° DR NDB to DR NDB (MANDATORY 220 KT), then turn LEFT onto 171° DR NDB and proceed to RORUK (MANDATORY 220 KT) climbing to 6000' or below, or as directed.
Turn before MAP is PROHIBITED.

Alt Set: hPa (MM on req) Apt Elev: 23 hPa Trans level: FL110❶ Trans alt: 10000'

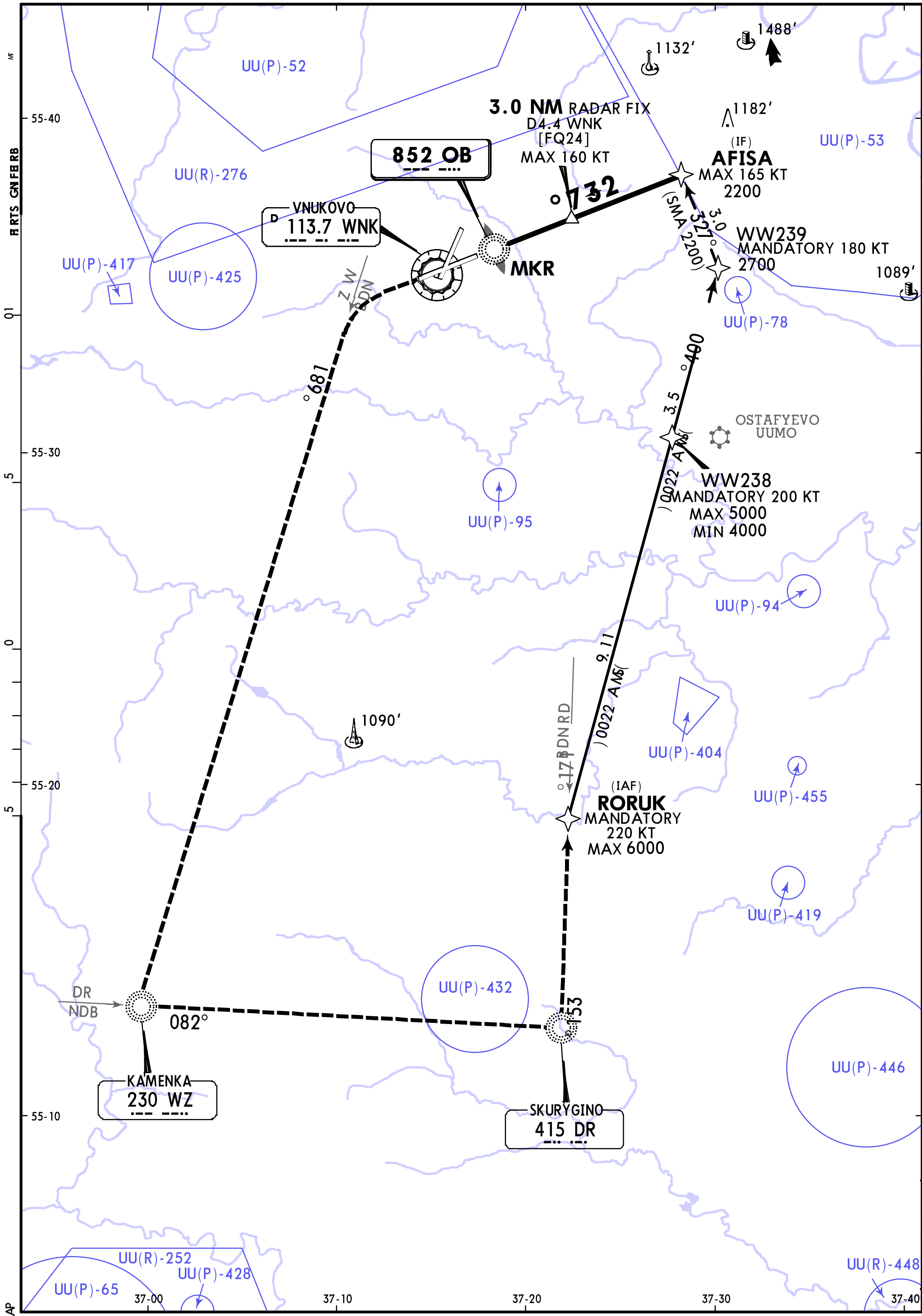
RNAV 1 for initial approach.

GNSS or DME/DME required.

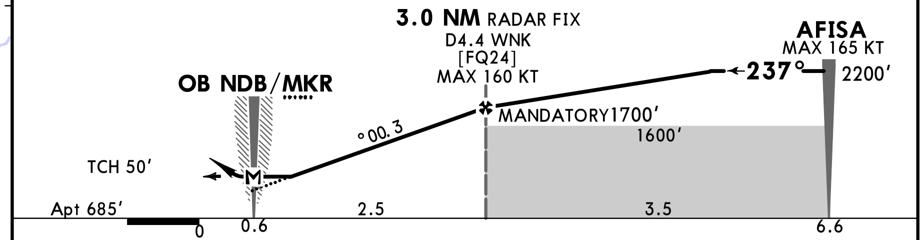
HOLDING FIX
DR NDB
171°
153°
MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

FEET QNH	METERS (QFE)
10000	(2900)
7000	(1925)
6000	(1625)
5000	(1320)
4000	(1015)
3800	(1000)
3500	(860)
2700	(615)
2200	(465)
2000	(405)
1700	(310)
1670	(305)
1600	(280)
1500	(250)
1330	(200)
1150	(140)

❶ FL120 if pressure is less than 1013 hPa (760mm).
FL130 if pressure is less than 977 hPa (733mm).



WNK DME	3.2	4.3	4.4
ALTITUDE	1330'	1670'	1700'



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at OB NDB/MKR						
Timing not authorized for defining MAP.						

HIALS-II PAPI

MIN 1500'

Std

STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1150'(465')

	ALS out	
A	R1500m	
B		
C	R1500m	R2200m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: FAF, missed approach, minimums.

MOSCOW, RUSSIA
NDB Y Rwy 01

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

NDB OE	Final Apch Crs	FOBUL	DA/MDA(H)	Apt Elev 685′ Rwy 633′
949	012°	2500′(1867′)	1160′(527′)	

3800
MSA ARP is computed for surface air temperature at apt -29.0°C

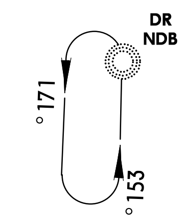
MISSED APCH: Climb STRAIGHT AHEAD to 1400′ or above, then turn RIGHT onto 154° LO NDB to LO NDB (MANDATORY 200 KT) climbing to 4000′ then onto 346° LO NDB and proceed to BEMAS (MANDATORY 200 KT) to 4000′, then turn RIGHT to DR NDB (MANDATORY 220 KT), then onto 171° DR NDB proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed. Turn before MAP is PROHIBITED.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 1 Trans alt: 10000′

RNAV 1 for initial approach.

GNSS or DME/DME required.

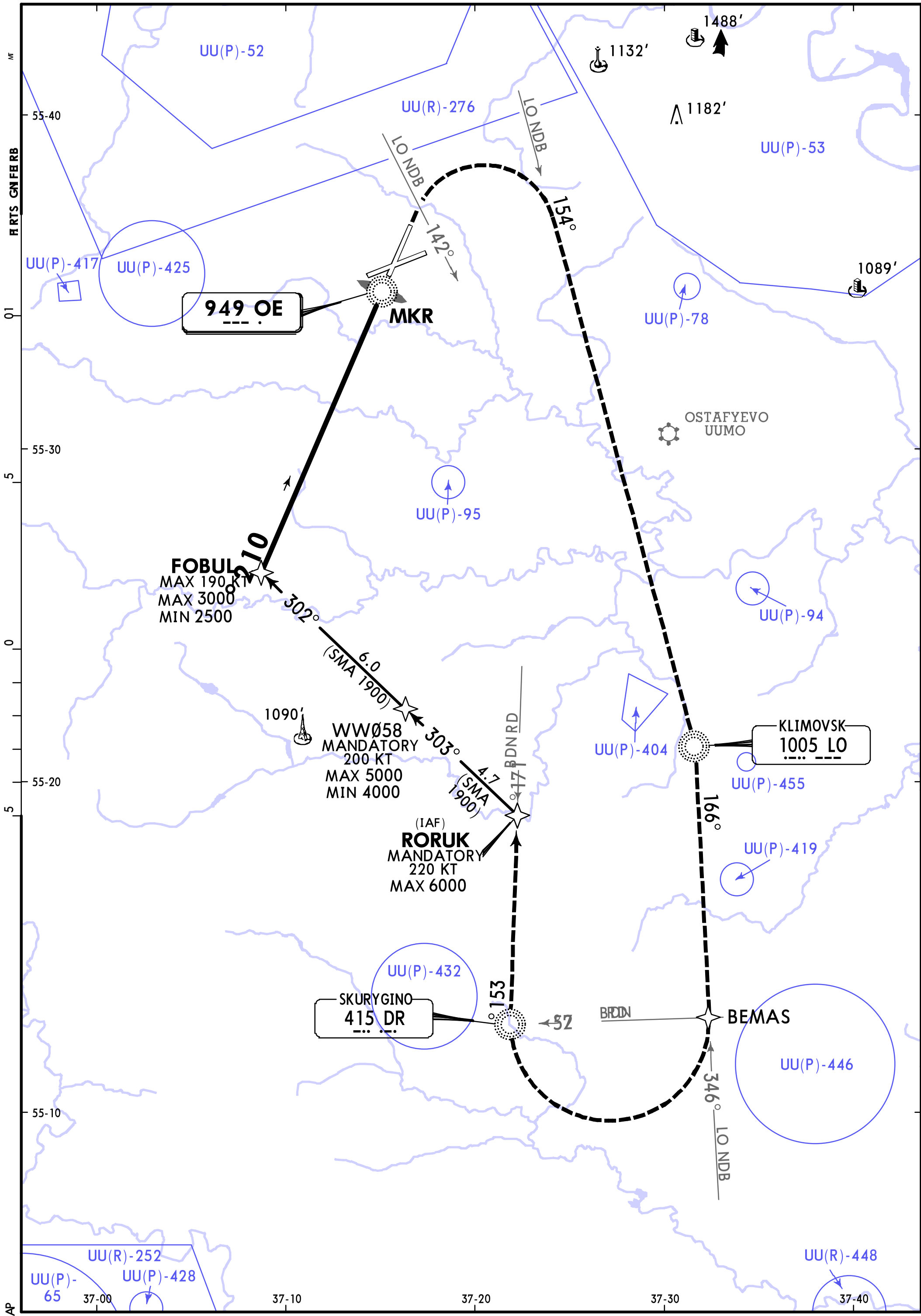
HOLDING FIX

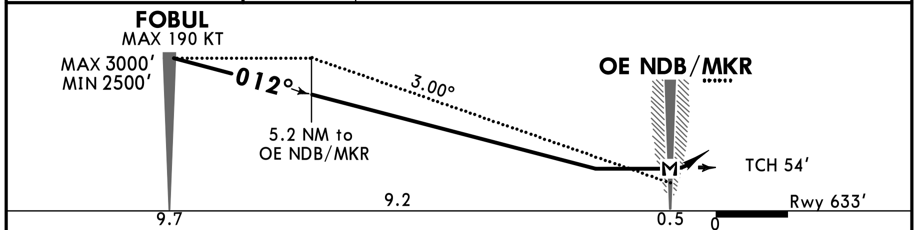


MAX 220 KT
MAX 7000
MHA 4000
(SMA 2000)

FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3000	(725)
2500	(570)
2000	(420)
1900	(390)
1400	(235)
1160	(160)

1 FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at OE NDB/MKR						

HIALS PAPI | MIN 1400′ | LO 1005 onto RT | LO 1005 154°

Std

STRAIGHT-IN LANDING

CDFA

1 DA/MDA(H) 1160′(527′)

		ALS out
A	R1500m	
B		
C	R1700m	R2400m
D		

1 VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: New procedure.

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian) 125.875	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

NDB GT	Final Apch Crs	WW5Ø1	DA/MDA(H)	Apt Elev 685′ Rwy 629′
294	057°	2500′ (1871′)	1150′ (521′)	

3800
MSA ARP
is computed for surface air temperature at apt -29.0°C

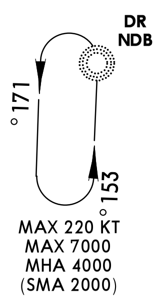
MISSED APCH: Climb STRAIGHT AHEAD to 1500′ or above, then turn RIGHT onto 142° LO NDB to LO NDB (MANDATORY 200 KT) climbing to 4000′, then onto 346° LO NDB and proceed to BEMAS (MANDATORY 200 KT) to 4000′, then turn RIGHT to DR NDB (MANDATORY 220 KT), then onto 171° DR NDB proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed. Turn before MAP is PROHIBITED.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000′

RNAV 1 for initial approach.

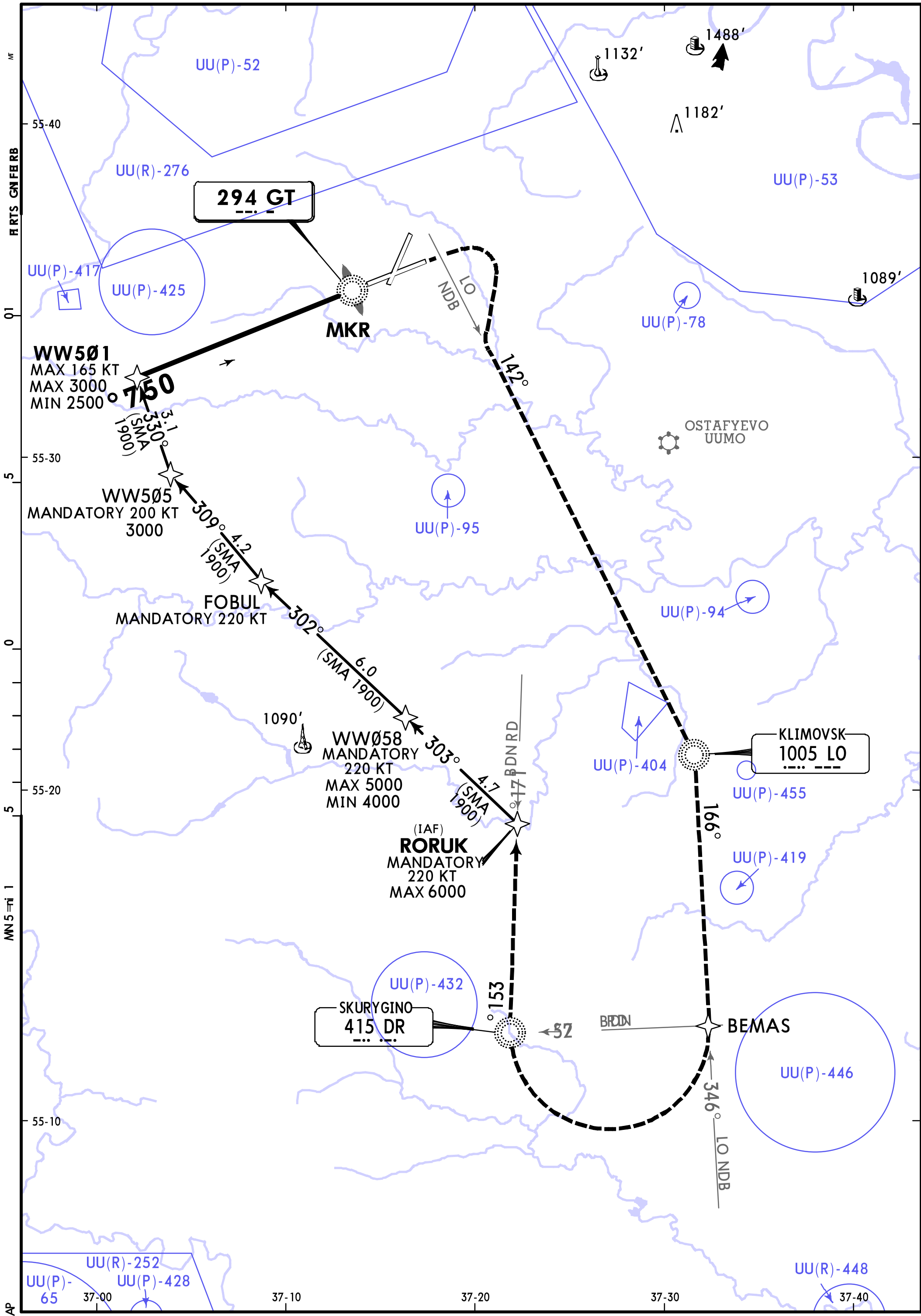
GNSS or DME/DME required.

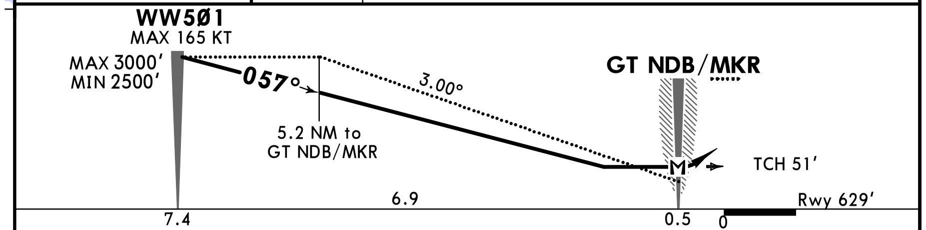
HOLDING FIX



FEET	METERS
QNH	(QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3000	(725)
2500	(575)
1900	(390)
1500	(270)
1150	(160)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at GT NDB/MKR						

HIALS-II
PAPI

MIN 1500′ | LO 1005 RT onto LO 1005 142°

Std

STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1150′(521′)

	ALS out	
A	R1500m	
B		
C	R1700m	R2400m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: New procedure.

UUWW/VKO
VNUKOVO

JEPPESEN
11 APR 25 16-7 Eff 17 Apr

MOSCOW, RUSSIA
NDB Y Rwy 19

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian 125.875)	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-1
118.3	118.3	120.450

NDB	Final Apch Crs	WW5Ø3	DA/MDA(H)	Apt Elev 685′
SX 914	192°	2200′ (1566′)	1270′ (636′)	Rwy 634′

3800

MSA ARP is computed for surface air temperature at apt -29.0°C

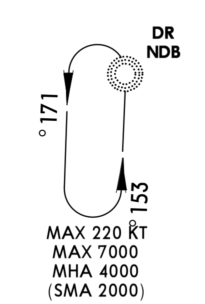
MISSED APCH: Climb STRAIGHT AHEAD to 1500′ or above, then turn LEFT onto 191° WZ NDB to WZ NDB climbing to 3500′, then onto 082° DR NDB to DR NDB (MANDATORY 220 KT), then turn LEFT onto 171° DR NDB and proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.
Turn before MAP is PROHIBITED.

Alt Set: hPa (MM on req) Rwy Elev: 23 hPa Trans level: FL110 ❶ Trans alt: 10000′

RNAV 1 for initial approach.

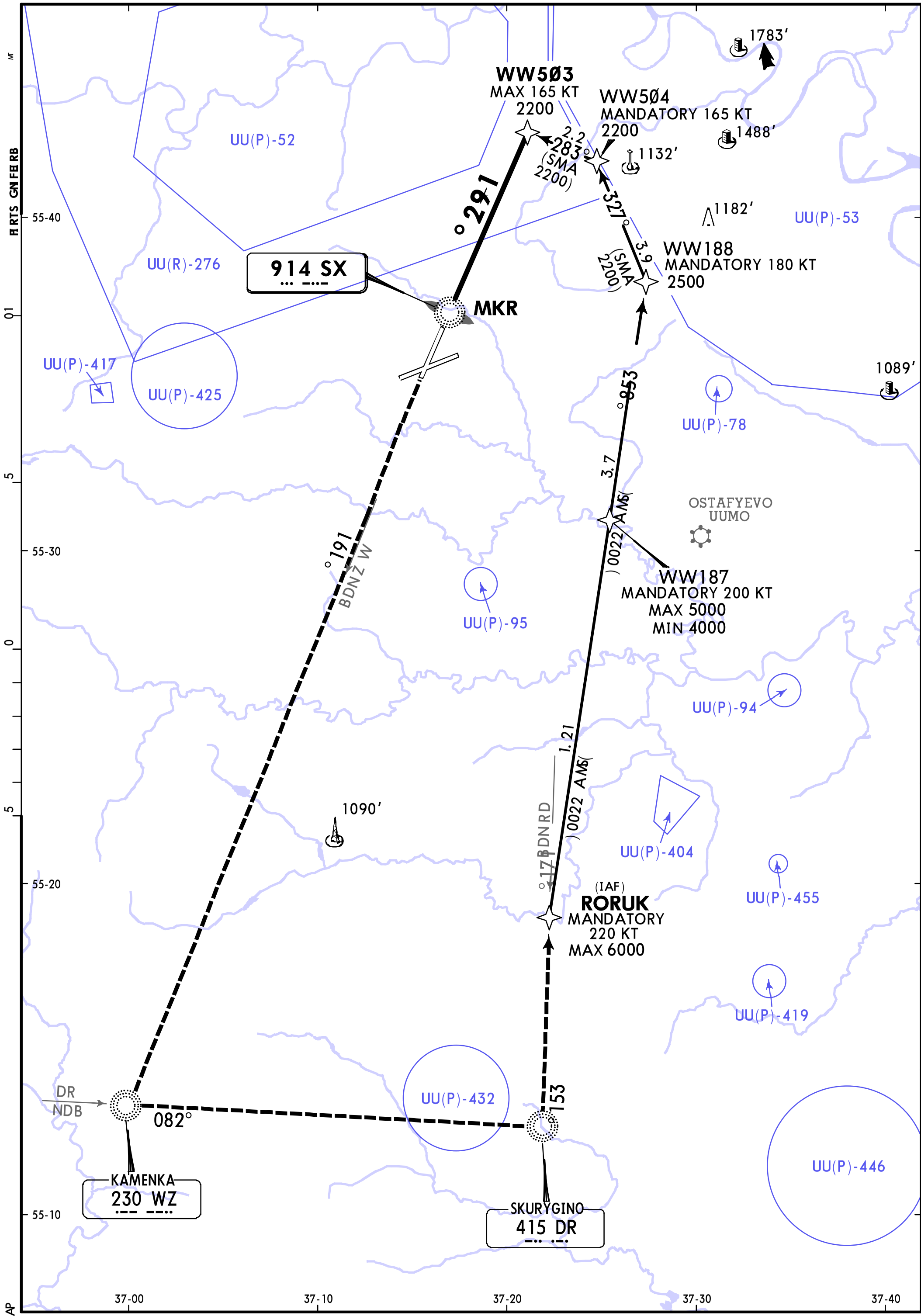
GNSS or DME/DME required.

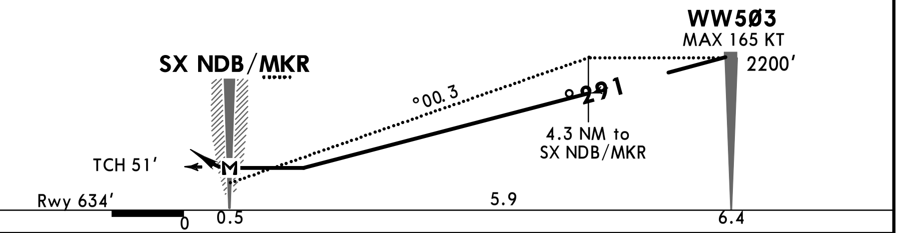
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1945)
6000	(1640)
5000	(1335)
4000	(1030)
3800	(1000)
3500	(875)
2500	(570)
2200	(480)
2000	(420)
1500	(265)
1270	(195)

❶ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at SX NDB/MKR						

HIALS-II
PAPI

MIN 1500′

Std

STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1270′ (636′)

	ALS out
A	R1500m
B	R1500m
C	R2200m / R2400m
D	R2200m / R2400m

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: New procedure.

UUWW/VKO
VNUKOVO

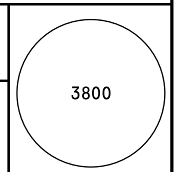
JEPPESEN
11 APR 25 (16-8) Eff 17 Apr

MOSCOW, RUSSIA
NDB Y Rwy 24

ATIS		VNUKOVO Radar (TWR)		
131.850	(Russian 125.875)	123.4	126.0	135.9

VNUKOVO Precision (TWR)	VNUKOVO Tower	Ground-2
118.3	118.3	121.7

NDB OB 852	Final Apch Crs 237°	AFISA 2200′(1515′)	DA/MDA(H) 1280′(595′)	Apt Elev 685′



MSA ARP is computed for surface air temperature at apt -29.0°C

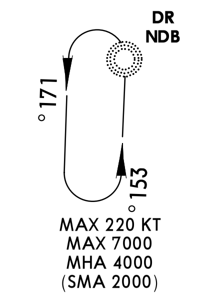
MISSED APCH: Climb STRAIGHT AHEAD to 1600′ or above, then turn LEFT onto 186° WZ NDB to WZ NDB climbing to 3500′, then onto 082° DR NDB to DR NDB (MANDATORY 220 KT), then turn LEFT onto 171° DR NDB and proceed to RORUK (MANDATORY 220 KT) climbing to 6000′ or below, or as directed.
Turn before MAP is PROHIBITED.

Alt Set: hPa (MM on req) Apt Elev: 23 hPa Trans level: FL110❶ Trans alt: 10000′

RNAV 1 for initial approach.

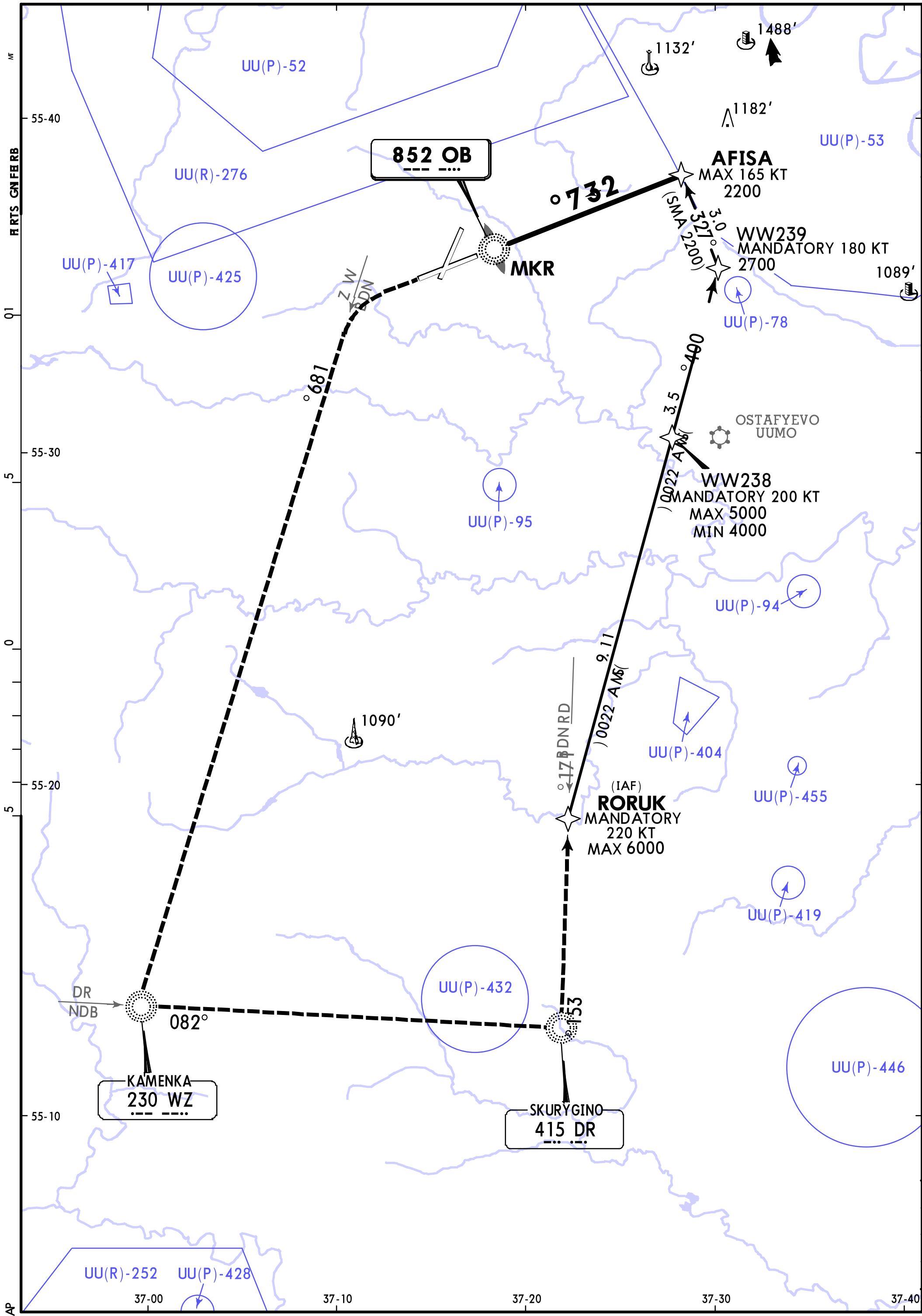
GNSS or DME/DME required.

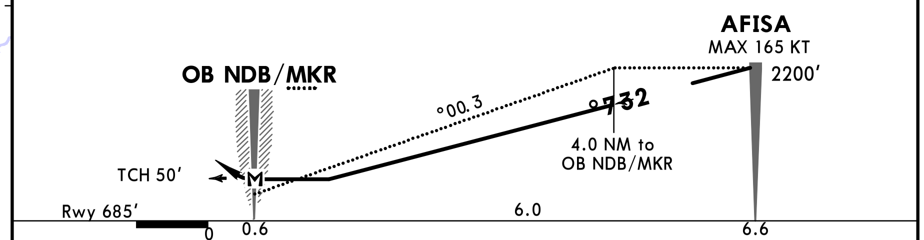
HOLDING FIX



FEET QNH	METERS (QFE)
10000	(2900)
7000	(1925)
6000	(1625)
5000	(1320)
4000	(1015)
3800	(1000)
3500	(860)
2700	(615)
2200	(465)
2000	(405)
1600	(280)
1280	(180)

❶ FL120 if pressure is less than 1013 hPa (760mm).
FL130 if pressure is less than 977 hPa (733mm).





Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at OB NDB/MKR						

HIALS-II PAPI — MIN 1600′

Std

STRAIGHT-IN LANDING

CDFA

❶ DA/MDA(H) 1280′(595′)

	ALS out	
A	R1500m	
B		
C	R2000m	R2400m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: New procedure.